Drug Delivery: Engineering and Principles

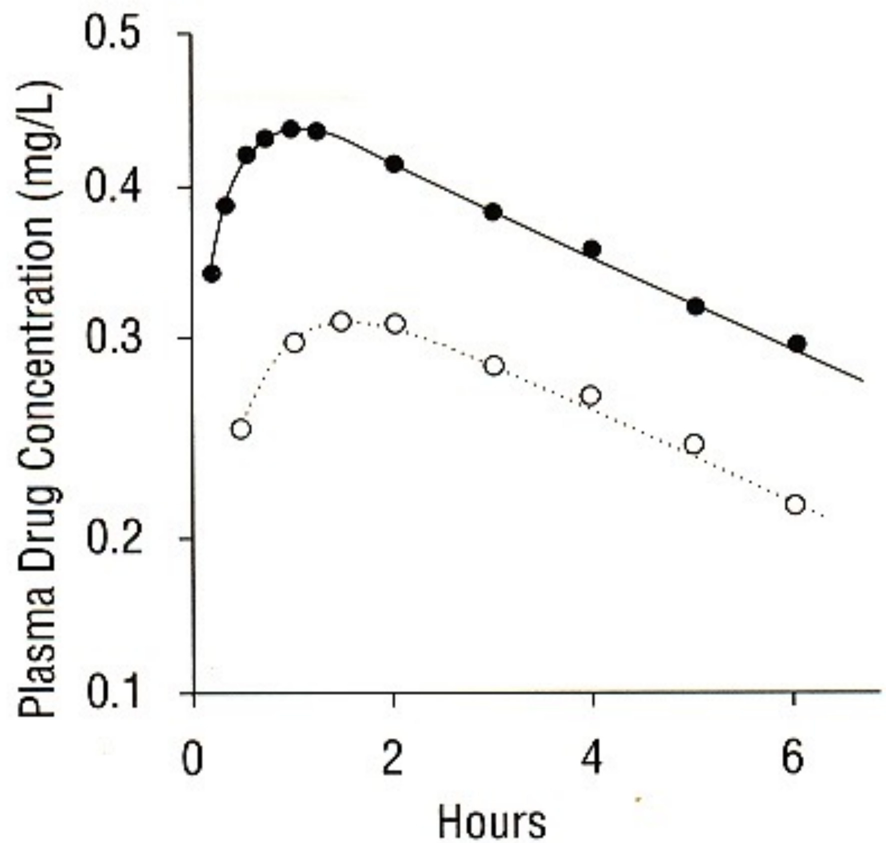
Rachit Agarwal
Associate Professor
Department of Bioengineering
Indian Institute of Science

Route specific delivery

- Depending upon application appropriate routes and specific delivery systems are chosen

- 500 mg dose given
  - intramuscularly
  - orally

**to the same subject on separate occasions



Oral Administration

- Advantages
  - Patient: Convenience, not invasive, higher compliance
  - Manufacture: well established processes, available infrastructure
- Disadvantages
  - Unconscious patients cannot take dose
  - Low solubility
  - Low permeability
  - Degradation by GI enzymes or flora
  - First pass metabolism
  - Food interactions
  - Irregular absorption

Oral Administration

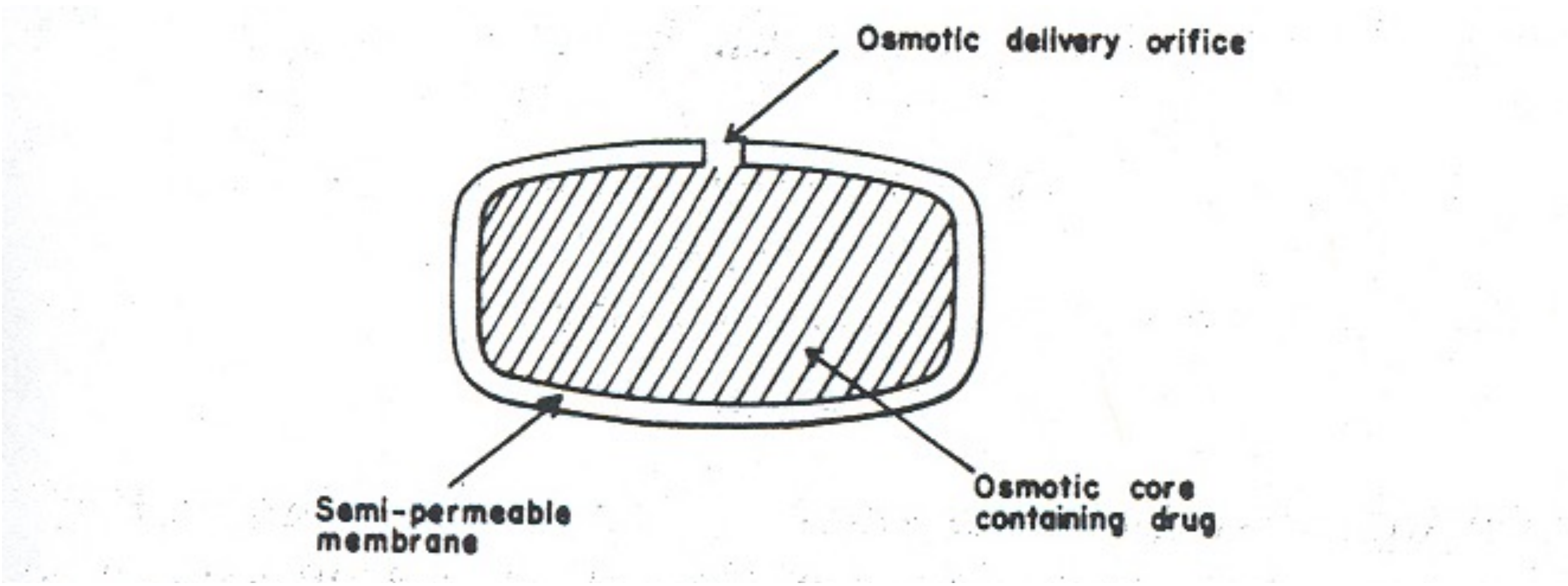
- Traditional oral delivery systems
  - Tablets
  - Capsules
  - Soft gelatin capsules
  - Suspensions

Controlled release dosage forms

Generally for highly water soluble drugs

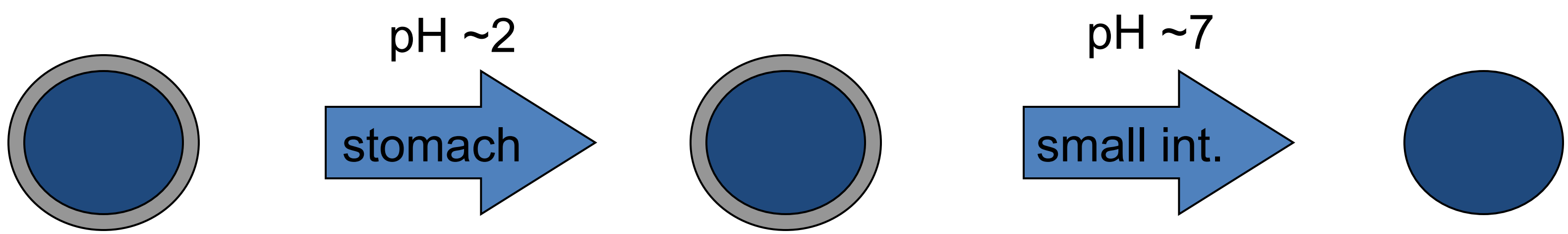
Osmotic tablets

- ***Driven by osmotic pressure***



What is an Enteric Coating?

A thin film applied to the dosage form to minimize drug dissolution in stomach



Coated with polymer containing numerous carboxylic acids

Carboxylic acids protonated (low solubility)

Carboxylic acids deprotonated (high solubility)

Subcutaneous

- Advantages
  - Patient self-administration
  - Slow, complete absorption
  - By-pass FPM
- Disadvantages
  - Invasive
  - Irritation, inflammation
  - Maximum dose volume - 2mL

Bydureon

- Glucagon like peptide
- Effective for type-2 diabetes
- Need injection twice daily
- PLGA microparticle formulation: Once a week: SC administration

Intramuscular

- Advantages
  - Patient can administer the drug himself
  - Larger volume than subcutaneous
  - By-pass first pass metabolism
- Disadvantages
  - Invasive – patient disconfort
  - Irritation, inflamation
  - May require some training

Biodegradable Polymer (PLGA), 40µm PLGA particles
Drug: luteinizing hormone-releasing hormone (LHRH) analog
Name: Decapeptyl SR
Use: Advanced prostate cancer
Dose: ~3 months
Intramuscular Injection

Moses, M.; Brem, H.; Langer, R. Cancer Cell, 2003, 4, 337

Risperdal® Consta

- First long-acting atypical antipsychotic
- IM injection every two weeks versus daily treatment

Schizophrenia: Mental Disorder that results in delusions and hallucinations
Risperdal: Antipsychotic medication
Bio-erodible microspheres: Injected every two weeks instead of daily administration
>5 million patients have taken this.

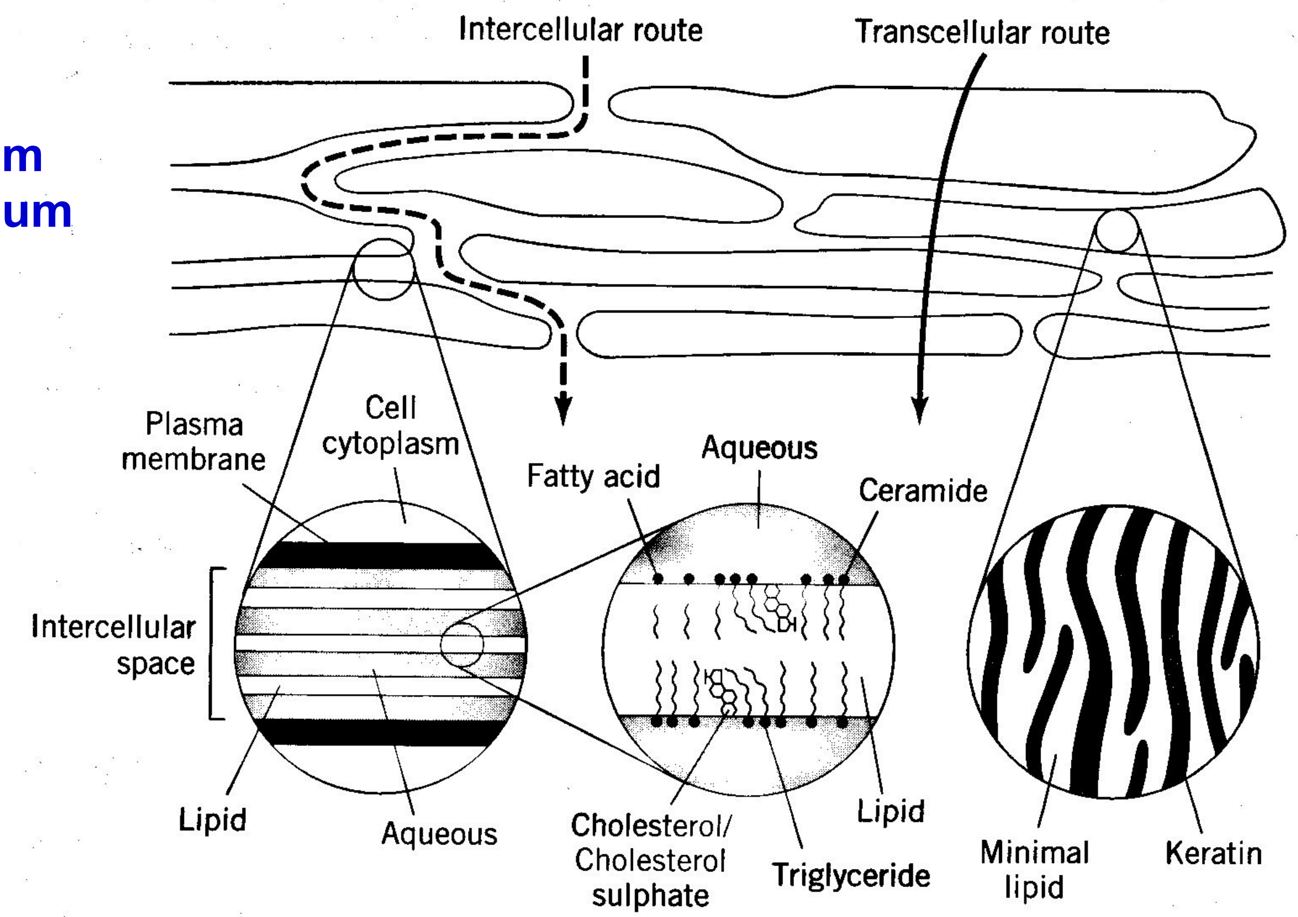
Transdermal

- Advantages
  - Local effect
  - Ease of administration
- Disadvantages
  - Low absorption for some drugs
  - May cause allergic reactions

- Requirement
  - MW< 1,000 Da

Trans-dermal delivery: Possible micro-routes

Stratum Corneum



BLOCKS OF "DEAD" KERATINOCYTES EMBEDDED IN A 'MORTAR' OF LIPIDS THESE LIPIDS ARE ARRANGED IN INTERLOCKING STRUCTURE, "BRICK AND MORTAR" STRUCTURE. THE LIPID "MORTAR' IS ARRANGED IN BILAYERS WHICH FORMS SHEETS AND CONSTITUTES THE MAJOR BARRIER

Transdermal drug delivery systems (TDDS)

- Advantages:
  - Reliable blood levels (steady levels can be achieved)
  - No first-pass metabolism
  - Very non-invasive → high patient compliance, provides simple localization → easy to apply and remove

- Disadvantages / limitations:
  - Skin is impermeable to most substances (main barrier: SC)
  - Lag time to reach steady state (generally several hours → skin loading and diffusion through SC)
  - Mostly useful for drugs with
    - High skin permeability (high partition coefficient in both water and oil) i.e. need to have some hydrophilicity and good lipophilicity
    - Low molecular weights

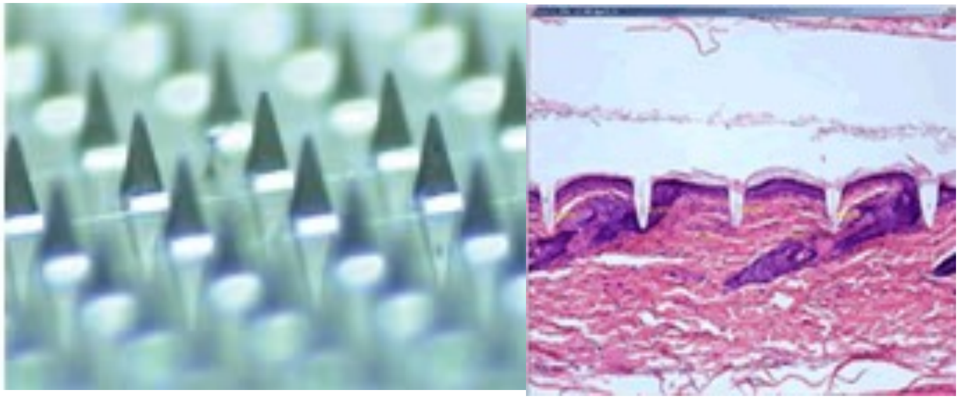
SKIN PERMEABILITY CAN BE ENHANCED USING PHYSICAL AND CHEMICAL MECHANISMS

Transdermal drug delivery systems (TDDS)

- Matrix type systems
  - FemPatch (Estradiol) → Cygnus
  - Vivelle (Estradiol) → Nova
  - Climara (Estradiol) → 3M
- Reservoir type systems:
  - Transderm-nitro (Nitroglycerin) → Angina treatment (per day)
    - ALZA-CIBA
    - Microporous membrane (EVAc)
    - Delivers 0.5 mg/cm^2/day
  - Transderm-scop (Scopolamine) → motion sickness (per 3 days)
    - Microporous polypropylene membrane
    - Delivers 0.5 mg/day
  - Nicoderm CQ (Nicotine)

Example from Industry: 3M Company

- **M**icrostructured **T**ransdermal **S**ystem: MTS
  - Microneedle system
  - Drug-in-adhesive technology platform



(a) 3M microneedle system and (b) histological section of microneedles in guinea pig skin

Micro-fabricated micro-needles for transdermal drug delivery

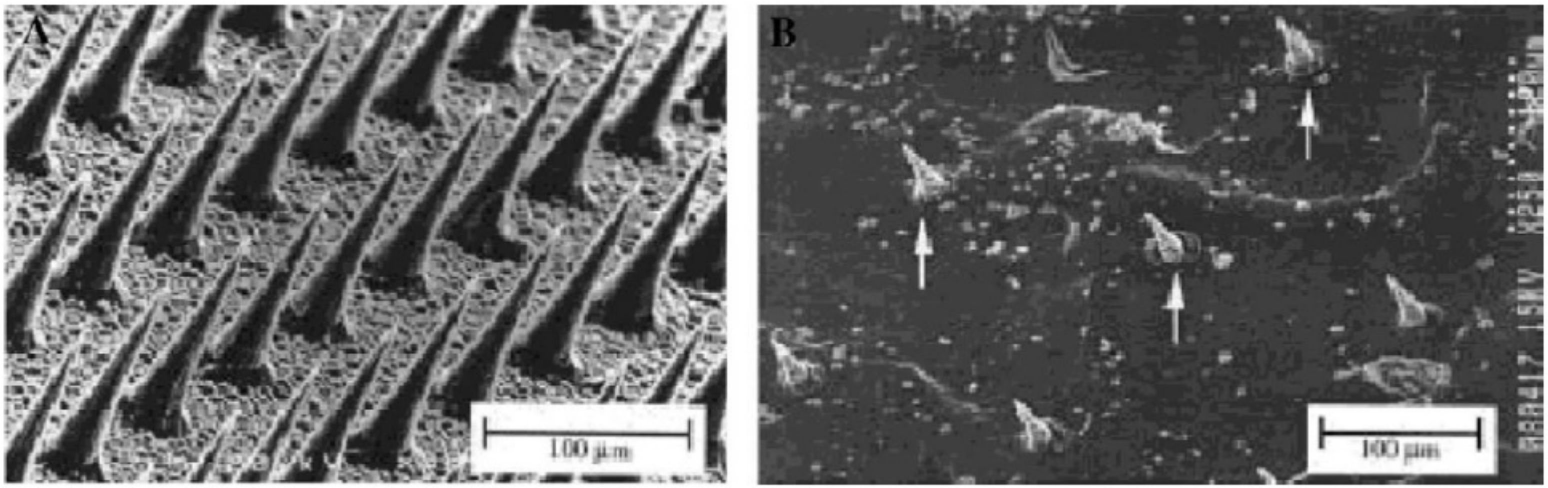


Fig. 1. (A) Scanning electron micrograph of microneedles made by reactive ion etching technique. (B) Microneedle tips inserted across epidermis. The underside of the epidermis is shown, indicating that the microneedles penetrated across the tissue and that the tips were not damaged. Arrows indicate some of the microneedle tips. Reproduced with permission from Ref. [14].

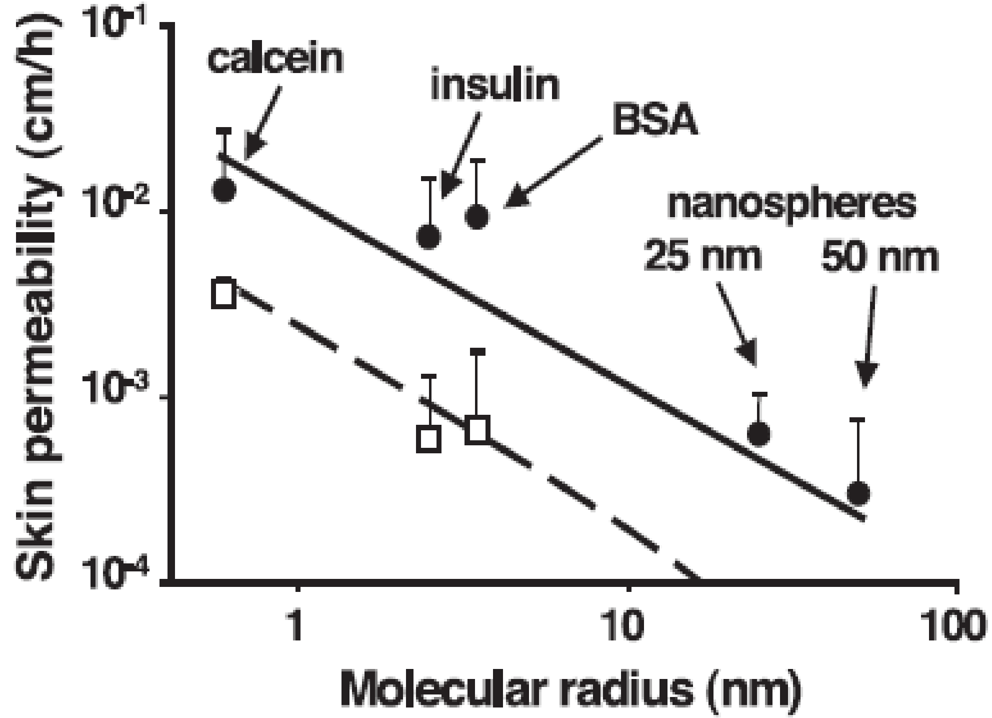


Fig. 4. Skin permeability to molecules and particles of different sizes after treatment with microneedles. The permeability of human cadaver epidermis was increased by orders of magnitude with a 400-needle array (Fig. 1*A*) inserted (□) and after the array was removed (●) for calcein, insulin, BSA, and latex nanospheres of 25 nm and 50 nm radius. Permeability to nanospheres with needles inserted was below the detection limit, on the order of 10^{-4} cm/h. In the absence of microneedles, permeability to all compounds was below their detection limits, on the order of 10^{-6} to 10^{-4} cm/h (data not shown). Mean values ± SEM are shown for at least six replicates. Predictions are shown for needles inserted (dashed line) and needles removed (solid line) by using a model requiring no adjustable parameters (Eq. 1 coupled with the Stokes–Einstein equation to interrelate molecular radius and diffusivity).

Hollow microneedles

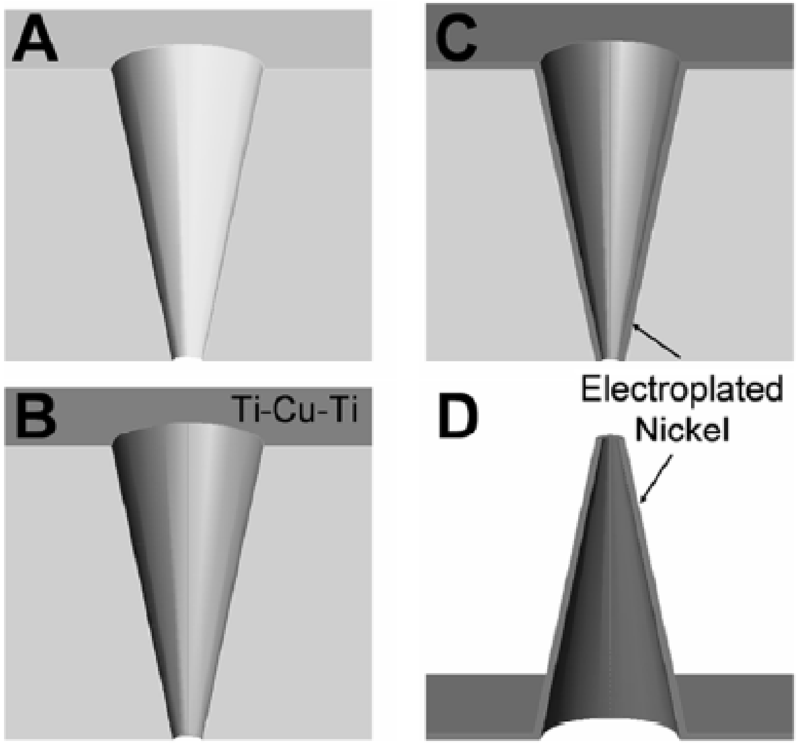


Fig. 1. Schematic sequence of the microneedle fabrication process using a polymer mold: (a) tapered holes are laser drilled through a polymer substrate, (b) a conductive seed layer is deposited on the top and sidewalls of the mold, (c) metal is electroplated onto the seed layer and (d) the mold is selectively wet etched to release the metal microneedles.

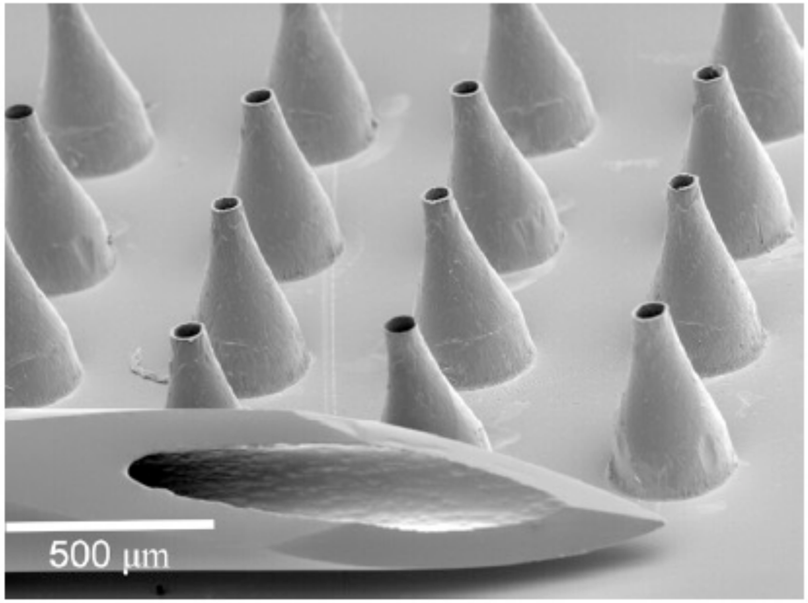


Fig. 5. An array of hollow, metal microneedles shown next to a 27-gauge hypodermic needle. The microneedles taper from a 300-μm base to a tip diameter of 75 μm over a 500-μm length and are arranged in a 4×4 array (i.e., 16 needles). Arrays of this geometry were used for insulin delivery experiments. Image from scanning electron microscopy.

IEEE TRANSACTIONS ON BIOMEDICAL ENGINEERING, VOL. 52, NO. 5, MAY 2005

Delivering insulin across skin using hollow microneedles

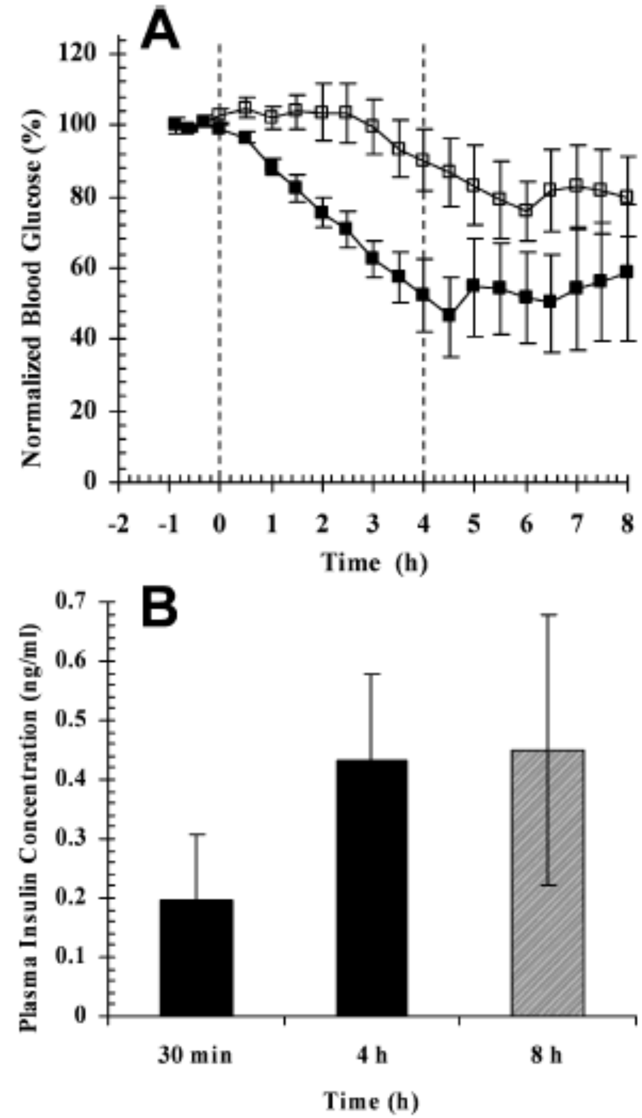
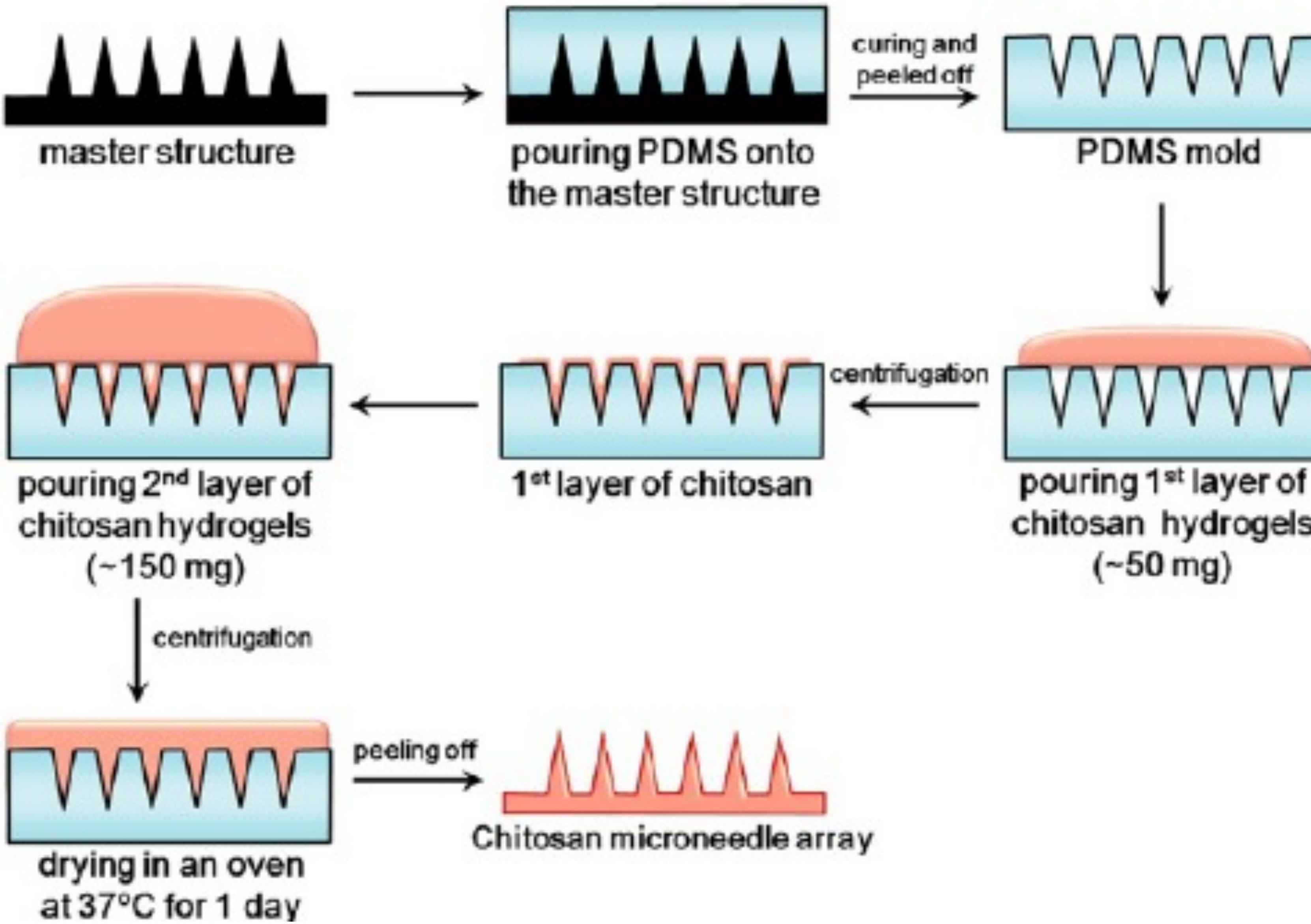


Fig. 6. Effects of transdermal insulin delivery using microneedles on blood glucose level and plasma insulin concentration in diabetic, hairless rats. (a) Blood glucose level before, during, and after transdermal insulin delivery either using microneedles inserted into the skin (■) ($n = 5$) or across intact skin (□) ($n = 4$). Glucose levels are normalized relative to the average blood glucose level during the 1.5-h period before treatment, 465 mg/dl. The dashed lines indicate the beginning and end of the 4-h insulin delivery period. (b) Concentration of human insulin in rat plasma during (0.5 and 4 h, black bars) and after (8 h, gray bar) transdermal delivery using microneedles. Average $\pm$ SEM is shown for $n \geq 3$ replicates.

Biodegradable microneedles



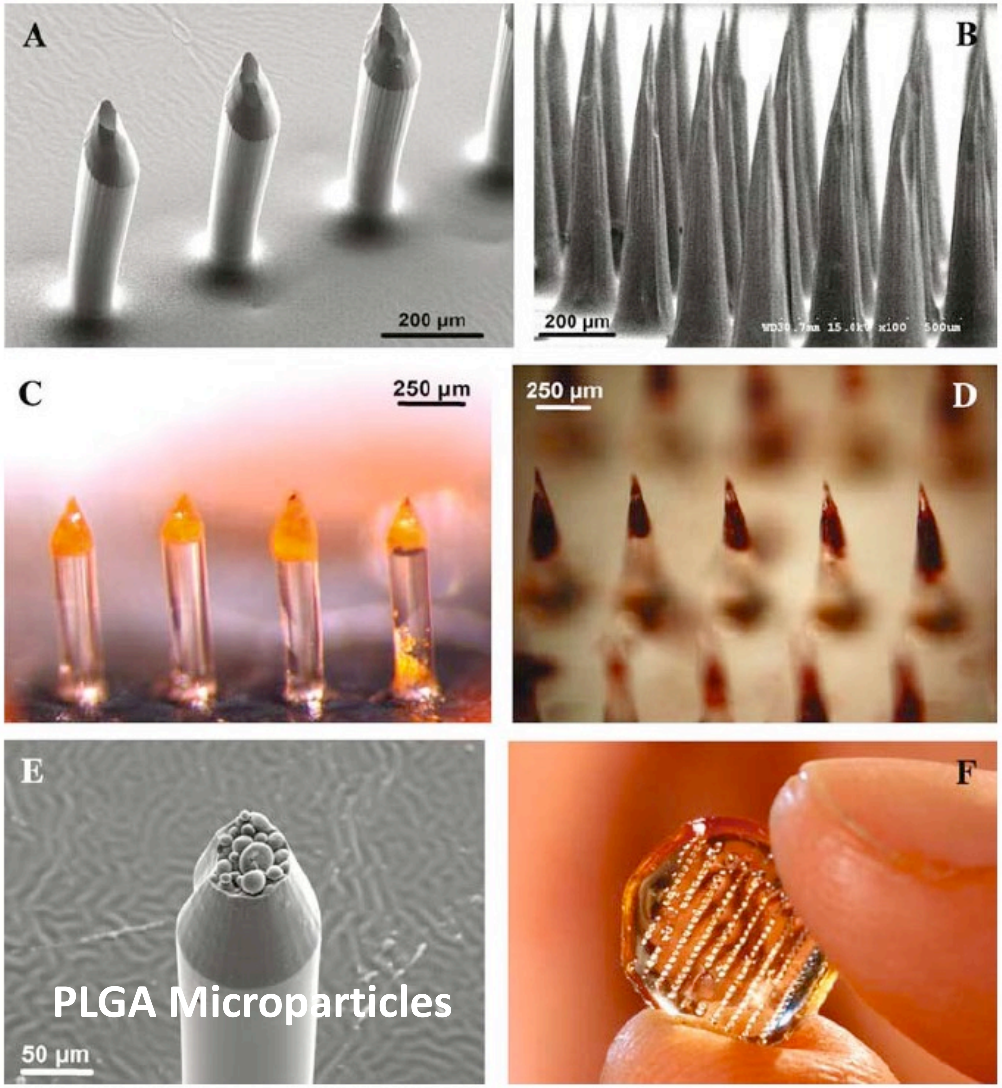


Fig. 3. Microscopy images of microneedles. A section of an array of (A) bevel-tip microneedles and (B) tapered-cone microneedles used as master structures (imaged by SEM). Making a mold and using it to prepare polymer microneedles as described in Fig. 2 yielded (C) bevel-tip and (D) tapered-cone microneedles made of PLGA and encapsulating calcein within their tips (imaged by fluorescence and bright-field microscopy, respectively). Using a double-encapsulation method produced microneedles that encapsulate microparticles that, in turn, encapsulate calcein. (E) Cutting off the tip of a PLGA microneedle reveals the PLA microparticles within (imaged by SEM). (F) A complete 20 × 10 array of PLGA microneedles is shown (imaged by flash photography).

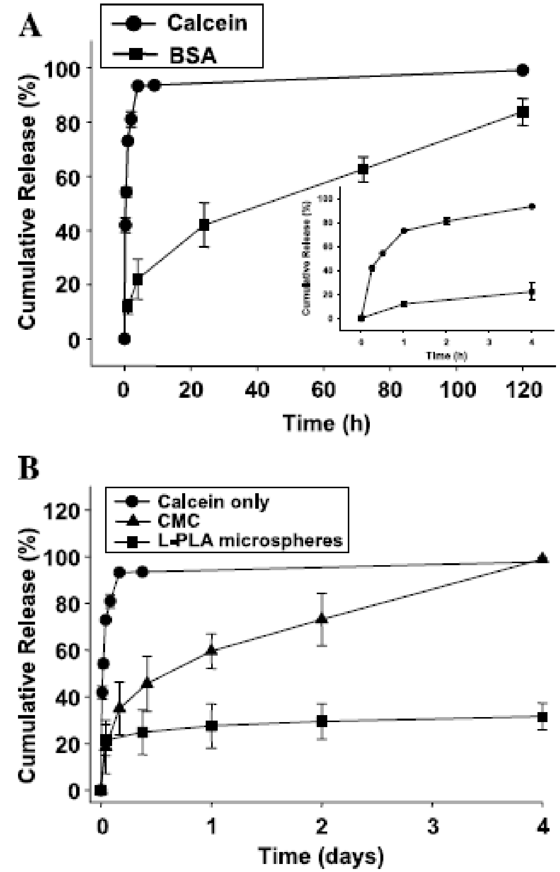


Fig. 4. Cumulative release of model compounds from PLGA microneedles prepared using different formulations to control release kinetics *in vitro*. (A) Release of calcein (●) and BSA (■) encapsulated within microneedles. The inset has an expanded time axis to better show release over the initial 4 h. (B) Release of calcein from microneedles that encapsulated (●) calcein only, (▲) CMC microparticles containing calcein, or (■) PLA microparticles containing calcein. Data are presented as average ± SEM ($n = 3–5$).

Pharmaceutical Research, Vol. 23, No. 5, May 2006 (© 2006)
DOI: 10.1007/s11095-006-0028-9

Inhalers

- Advantages
  - By-pass FPM
  - Good for delivery to lungs
  - Patient compliant

- Disadvantages
  - Systemic absorption only occurs for molecules that reach deep lungs
  - Solids and liquids can be absorbed in systemic circulation only if size is below 100 nm

Inhalation based drug delivery

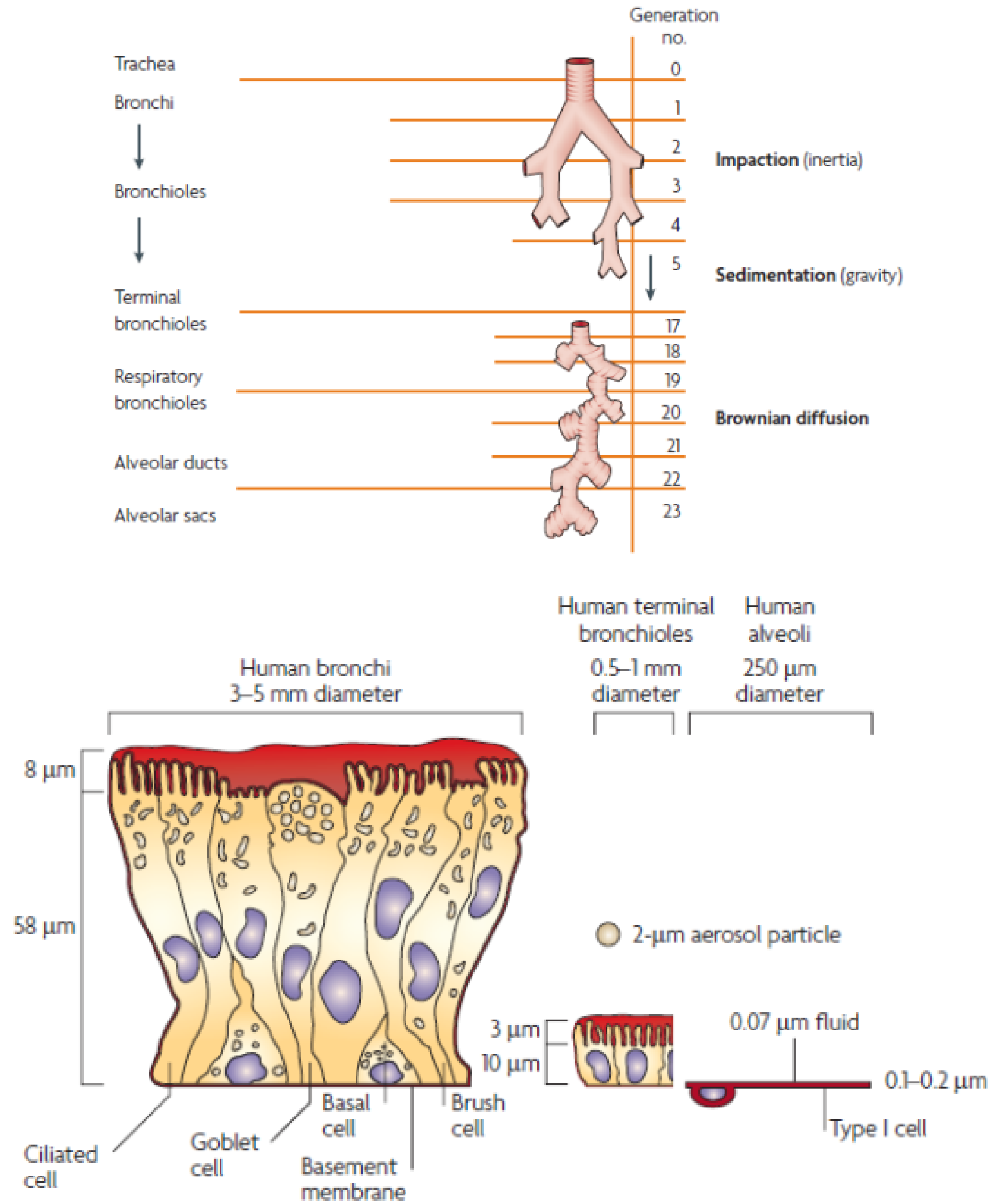
Table 1 | **Examples of inhaled medicines**

Year	Formulation/device	Molecule(s)	Disease
1500 BC	Egyptians used 'vapors'	?	?
1662 AD	Bennet's inhalation treatment	?	Tuberculosis
1802	Potter's cigarettes	?	Asthma
1860	Sales-Giron's portable nebulizer	?	?
1925	Aqueous/nebulizer	Insulin	Diabetes
1945	Aqueous/nebulizer	Penicillin	Lung infections
1951	Aqueous/nebulizer	Isoprenaline	Asthma
1955	Aqueous/nebulizer	Hydrocortisone	Asthma
1956	First metered-dose inhaler (MDI) (freon)	Albuterol	Asthma
1960	First dry powder inhaler (DPI)	Norsadrenaline	Asthma
1988	First multidose DPI	Terbutaline	Asthma
1996	First protein aqueous/nebulizer	DNAse	Cystic fibrosis
1998	First antibiotic aqueous/nebulizer	Tobramycin	Cystic fibrosis
1997	First hydrofluoralkane MDI	Albuterol	Asthma
2006	First protein DPI	Insulin	Diabetes

NATURE REVIEWS | DRUG DISCOVERY VOLUME 6 | JANUARY 2007 | 67

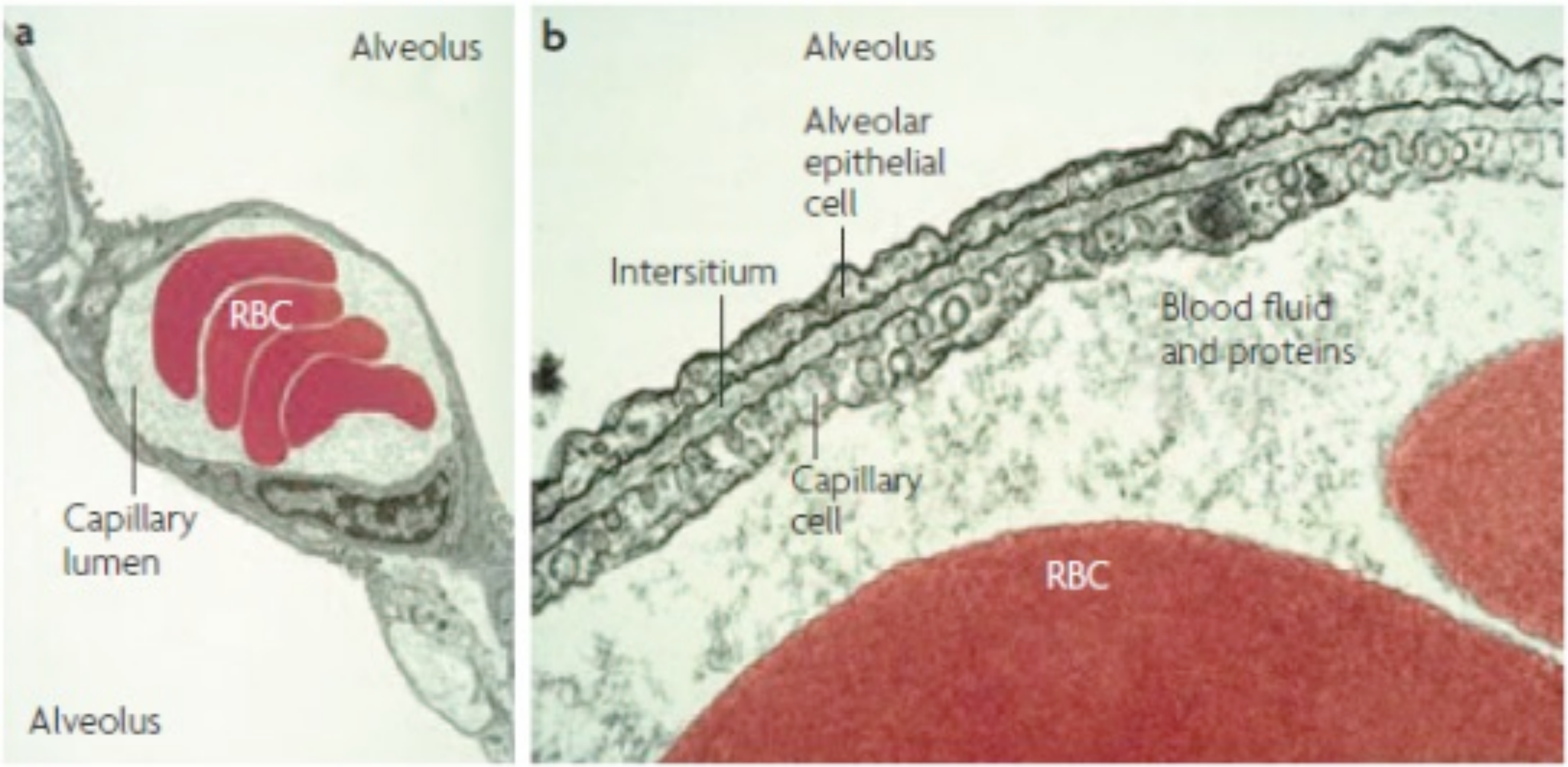
Inhalation

- The airways branch roughly 16–17 times before alveolar sacs are encountered
- About 500 million alveoli in human lungs
- Each alveoli is patrolled by 10-15 alveolar macrophages

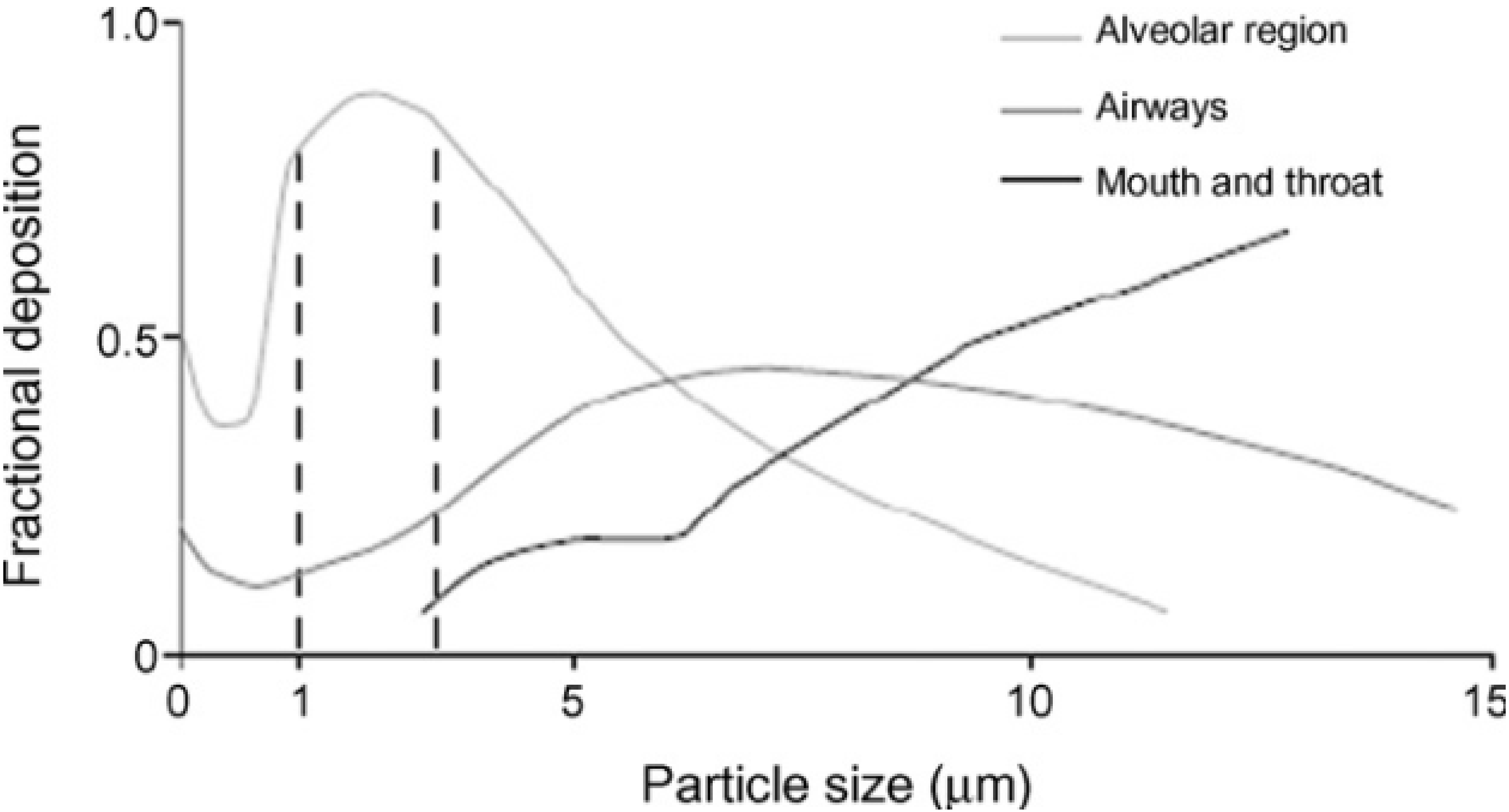


NATURE REVIEWS | DRUG DISCOVERY VOLUME 6 | JANUARY 2007 | 67

Lung-capillary permeability



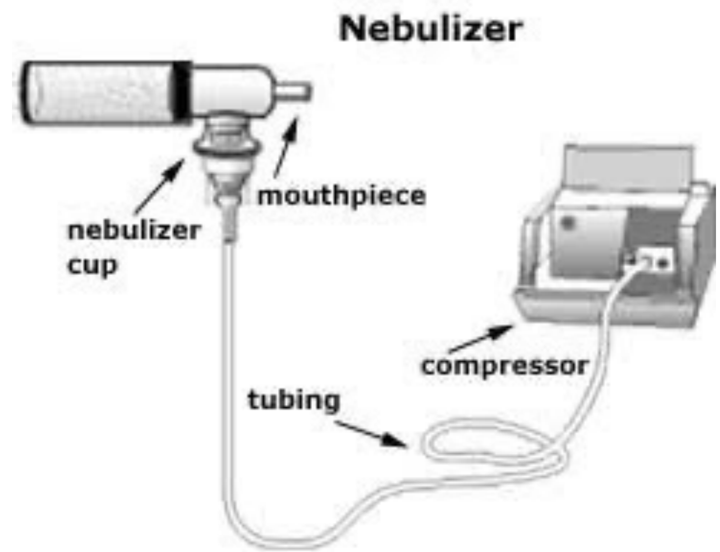
Inhalation



W. Yang et al. / International Journal of Pharmaceutics 356 (2008) 239–247

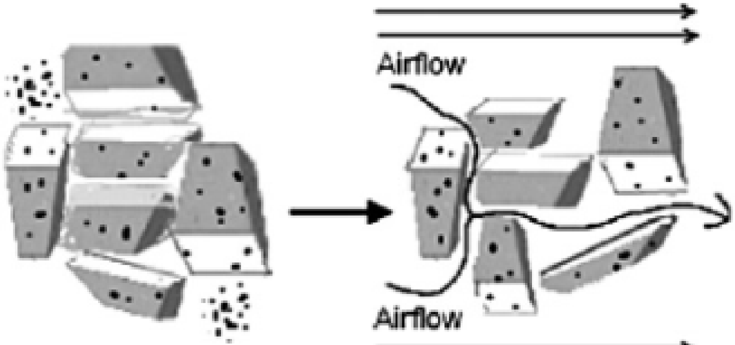
Nebulizers

- Generates mist that are inhaled in the lungs
- Use compressed air and ultrasonic power to break solutions into aerosol droplets
- Works well in a hospital setting but not very easy and convenient to use otherwise



Dry powder inhalation

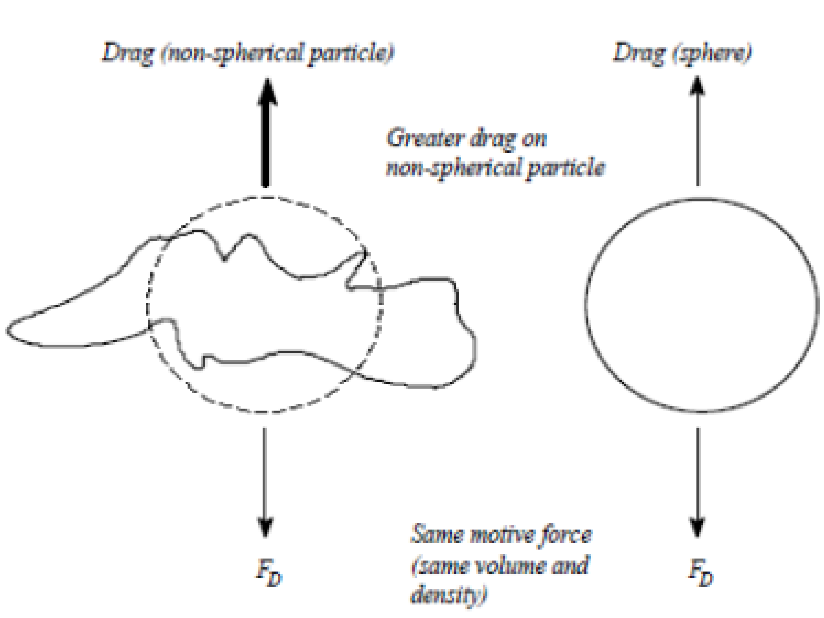
- Patient compliant/ease of use
- Drug stability is increased
- Powder particles should be around 1-5 μm for deep lung delivery
- Excipients are used to carry drug adsorbed on the surface to deep lungs
- Sugars such as lactose are commonly used
- Doesn’t result in slow release as all the drug is administered immediately
- Particles of can also be used for this application and will result in depot at lung alveolar space

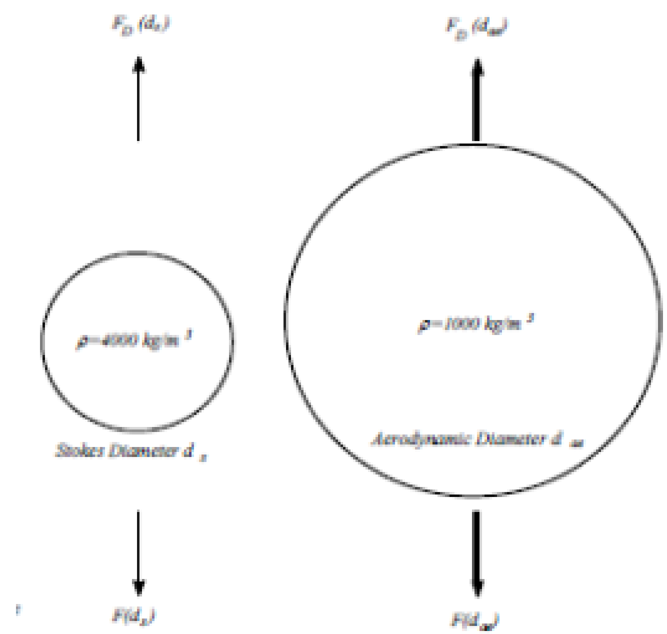


Inhalable Carriers: Aerodynamic Diameter

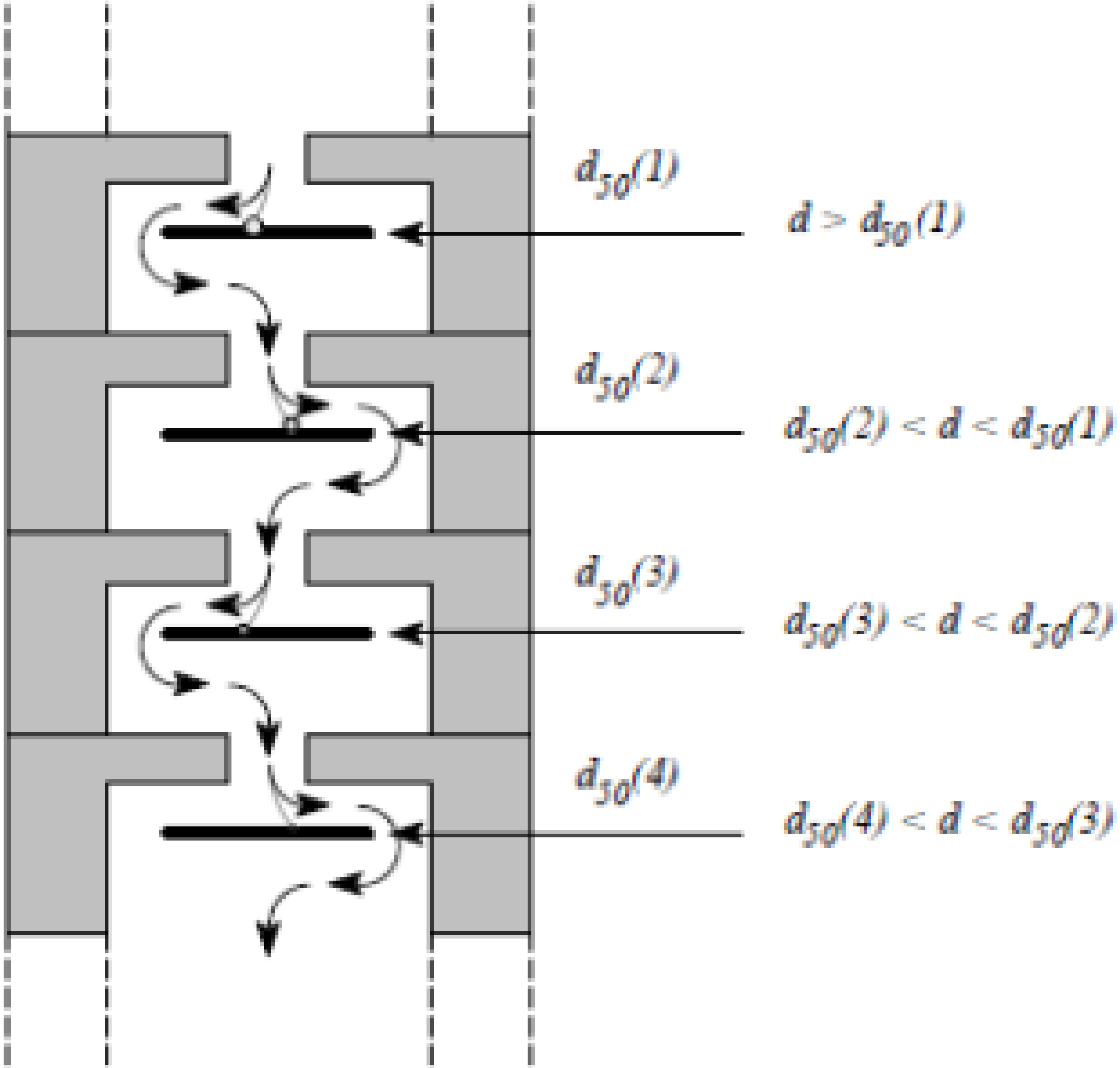
- Particle size for deep lung delivery via pulmonary route: 1-5 µm in aerodynamic diameter
- Aerodynamic Diameter: The diameter of sphere of density 1000 kg/m^3 with the same settling velocity as particle of interest
- Standardizes for shape (sphere) and density (water droplet)
- Measured by cascade impactor

Drag force for different shape of particle





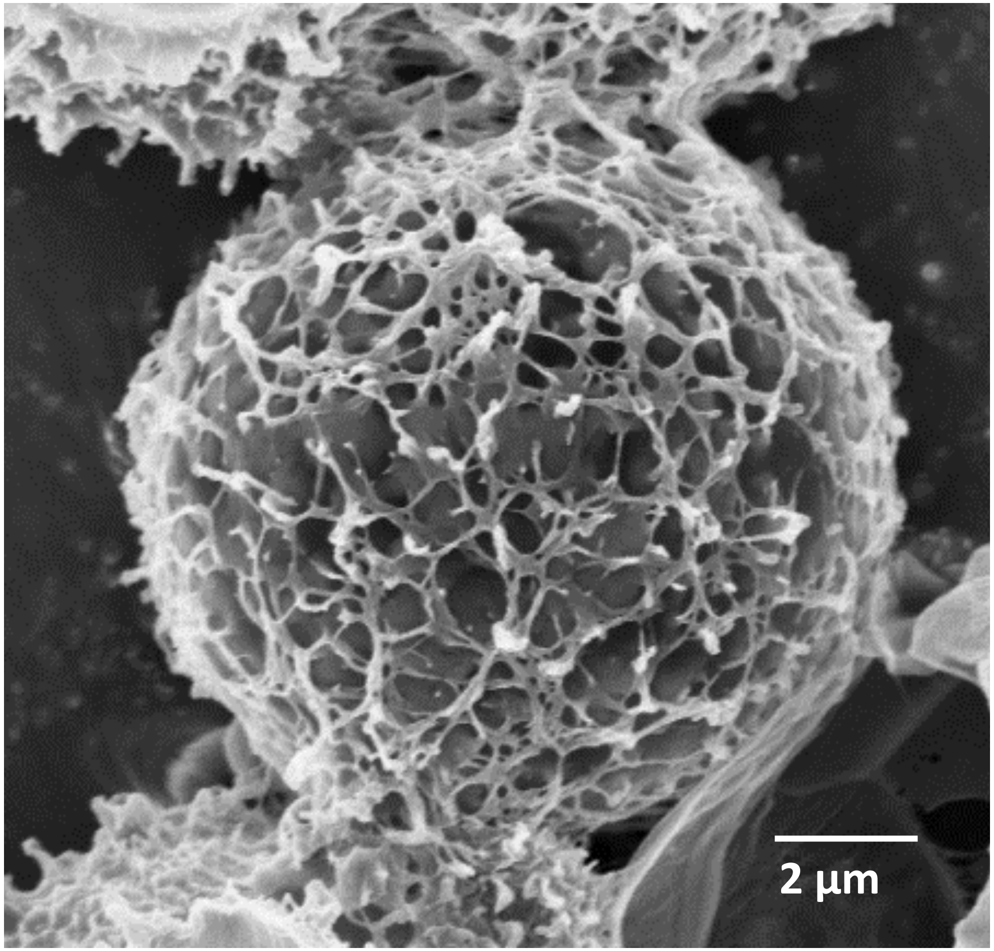
Cascade impactors



Inhalable Carriers: Porous Particles

- Particle size for macrophage uptake: 1-5 μm
- Porosity achieved by including porogens or effervescent molecules during synthesis
- Porous & larger particles
  - 8-10 μm in physical diameter
  - Porosity reduces particle density to match 2-5 μm d_{aer}
  - Cannot be phagocytosed by macrophages
  - Larger size: less aggregation and better aerosolization

$$D_{aero} \propto D\rho^{1/2}$$



Agarwal et al. Nature Biomedical Engineering 2, 841 (2018).

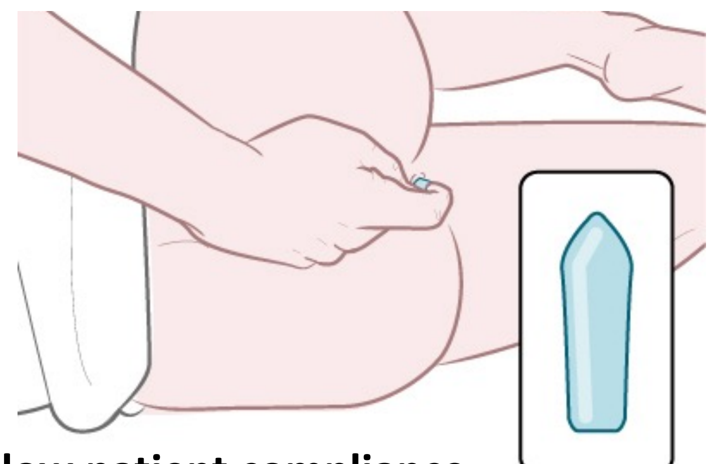
Relenza: Anti-flu medication

- Blocks release of virus from cell surface
- Not much in use these days as replaced by other drugs

Rectal

- Advantages
  - By-pass first pass metabolism
  - Useful for children as might be difficult to find blood vessels
- Disadvantages
  - Absorption depends on disease state
  - Degradation by bacterial flora
  - Uncomfortable

- Traditional delivery system/devices
  - Suppository
    - a small, round or cone-shaped object that dissolves rapidly
  - Enema



Not heavy research is being done here due to low patient compliance

Example from Industry: Valeant Pharmaceuticals

- Diastat AcuDial: diazepam rectal gel for prolonged seizures
- High need for children and an administration route that parents can easily administer
  - Even in hospital setting, its hard to find the vein for babies and toddlers
- High lipid solubility
  - Rapidly diffuses into all high lipid regions
- Oral administration is slow to take effect and muscular administration causes necrosis in the region

Rectal administration

- Fecal microbiota transplant
- Clinically used for treatment of *Clostridium difficile* infection
  - Diarrhea
  - Fever
  - Abdominal pain
  - Vancomycin unable to clear it in many cases
- Stool from close family member is administered using enema
- Invasive, so not often used for other treatments

Intra-articular administration

- Direct injection to the joints
- Specific for joint disease

Zilretta

- 35-55 µm PLGA particles with around 500nm channels
- Active ingredient is triamcinolone acetonide, which is a common short acting corticosteroid
- Longer lasting pain relief for more than 12 weeks

$500 per injection
$111 million sales in 2023

https://www.pharmaceutical-technology.com/comment/commentwith-fda-approval-secured-flexions-zilretta-poised-to-shake-up-knee-osteoarthritis-marketplace-5947301/

http://www.limu-chemicals.com/en/News/2018/0418/39.html

Zilretta

- 32 mg of triamcinolone acetonide per injection in 5mL.
- Drug loading is 25%w/w so around 96mg of PLGA per injection
- Can only be given again atleast after 12 weeks of first administration

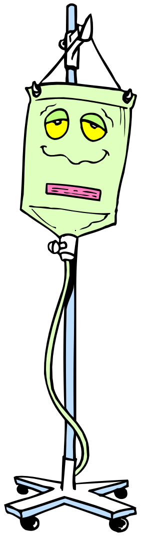
https://www.accessdata.fda.gov/drugsatfda_docs/label/2017/208845s000lbl.pdf

Intravenous (IV)

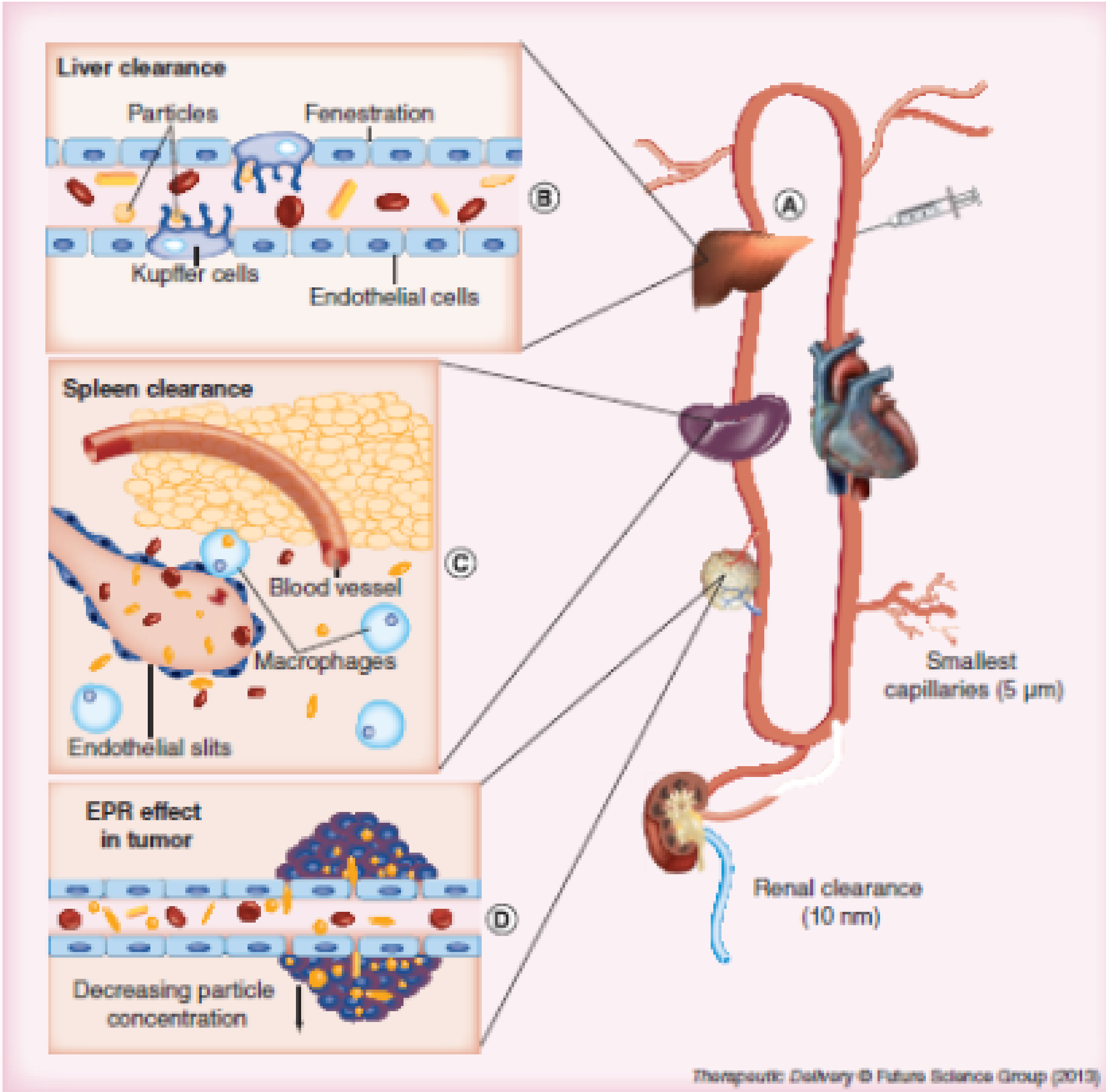
- **Advantages**
  - **Drug 100% bioavailable**
  - **Rapid response**
  - **Total control of blood concentration**
  - **By-pass FPM**

- **Disadvantages**
  - **Invasive**
  - **Trained personnel**
  - **Possible toxicity due to incorrect dosing**
  - **Sterility**

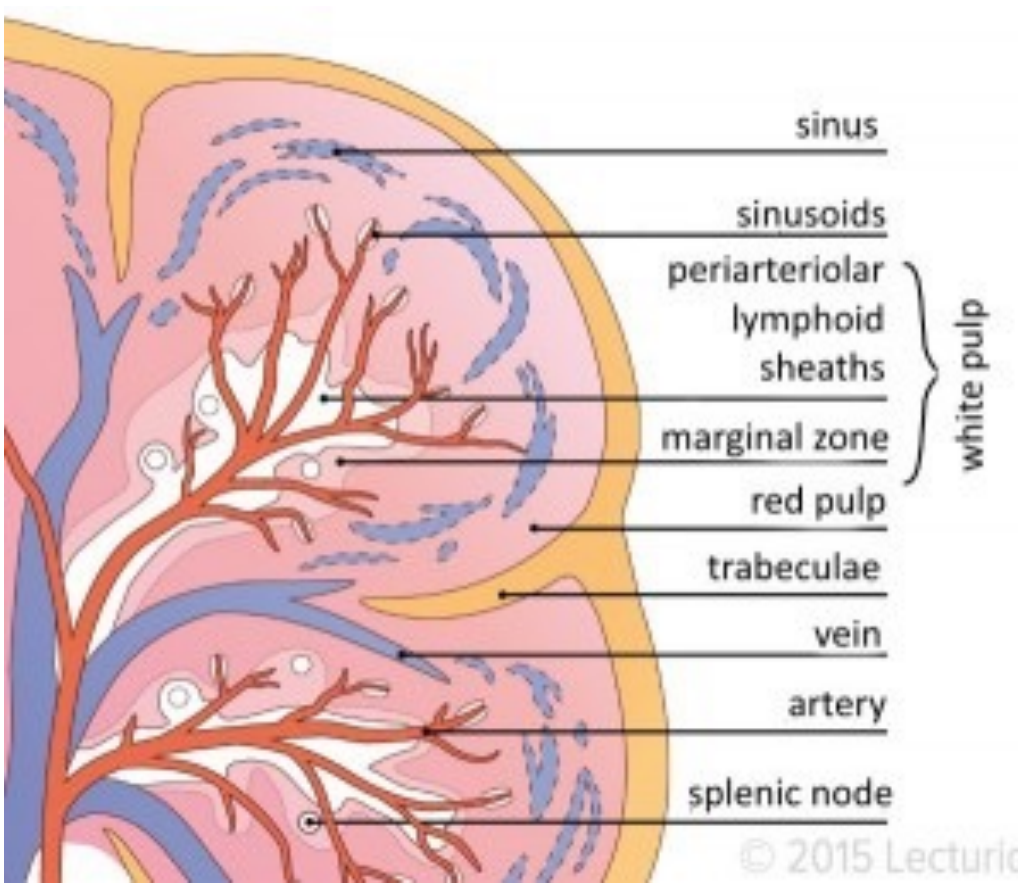
- Traditional delivery system/devices
  - Injection-bolus
  - IV bag - infusion

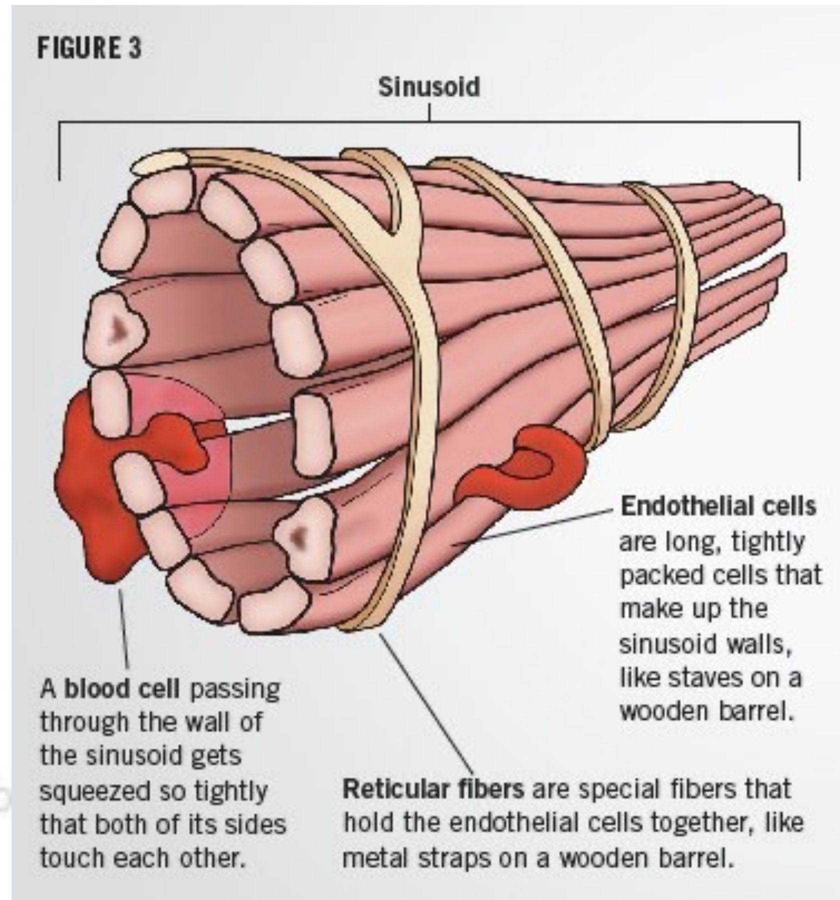


Size requirements



Spleen blood filtration



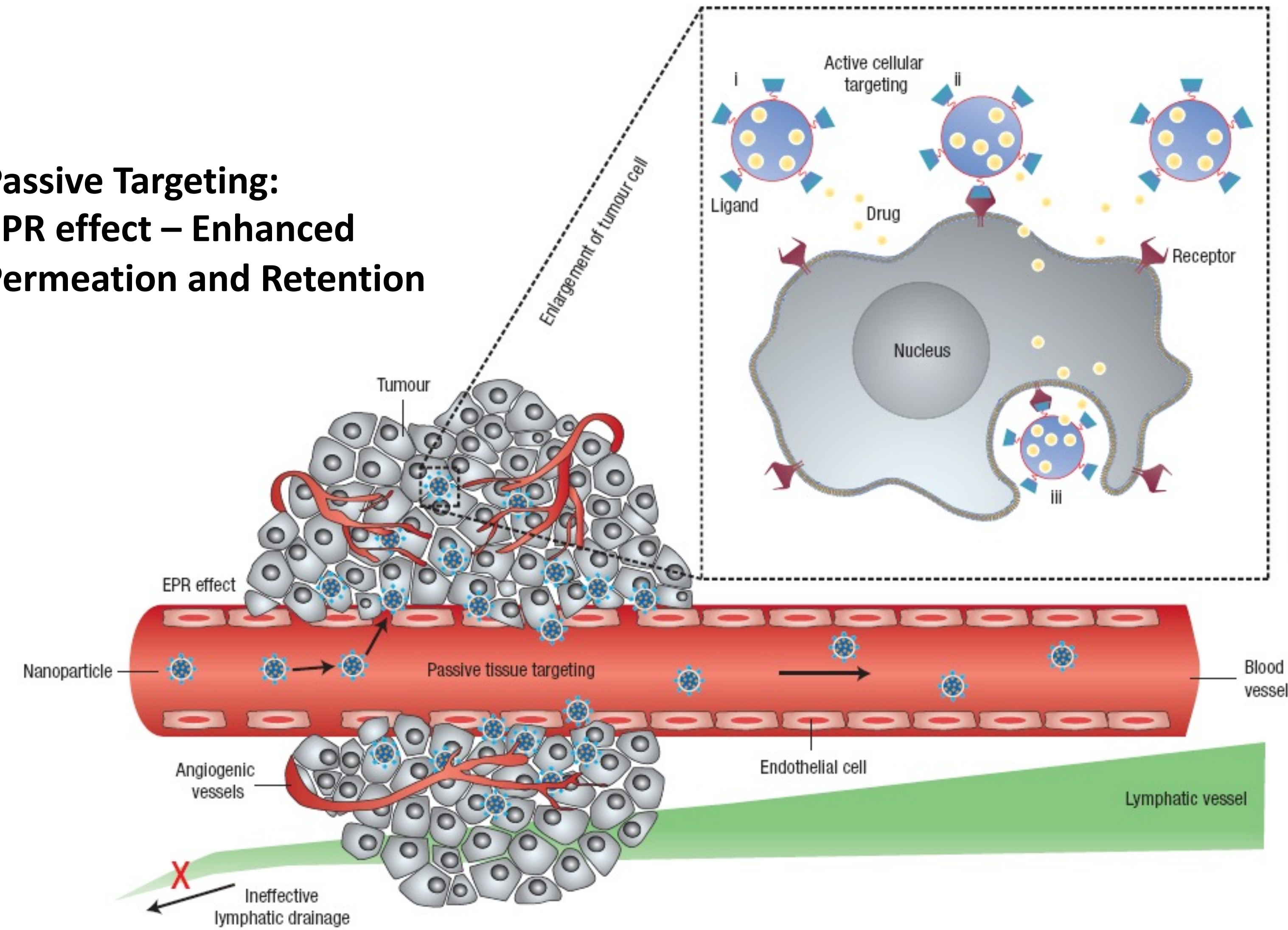


https://www.lecturio.com/magazine/spleen/

https://answersingenesis.org/human-body/the-mysterious-spleen/

Tumors grow blood vessels that are leaky

Passive Targeting:
EPR effect – Enhanced Permeation and Retention

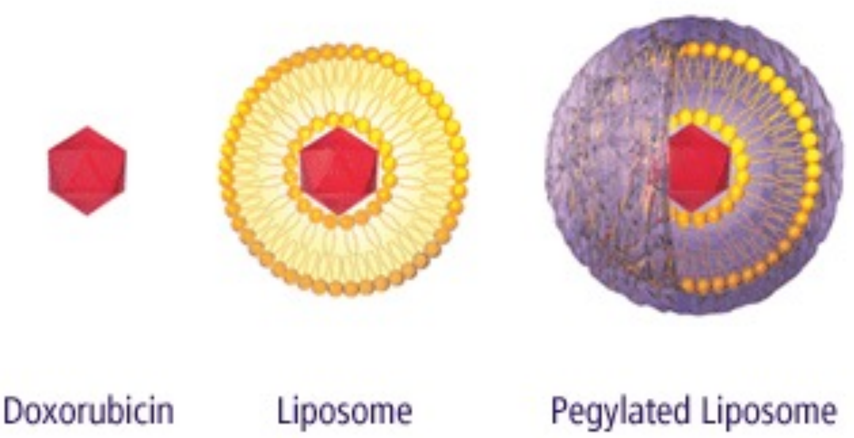


Peer, D, et al. *Nature Nanotechnology* **2007**, 2, 751-760

Approved nanocarriers for cancer: Doxil

Doxorubicin is effective anti-cancer drug but its treatment has high associated cardiotoxicity due to systemic circulation

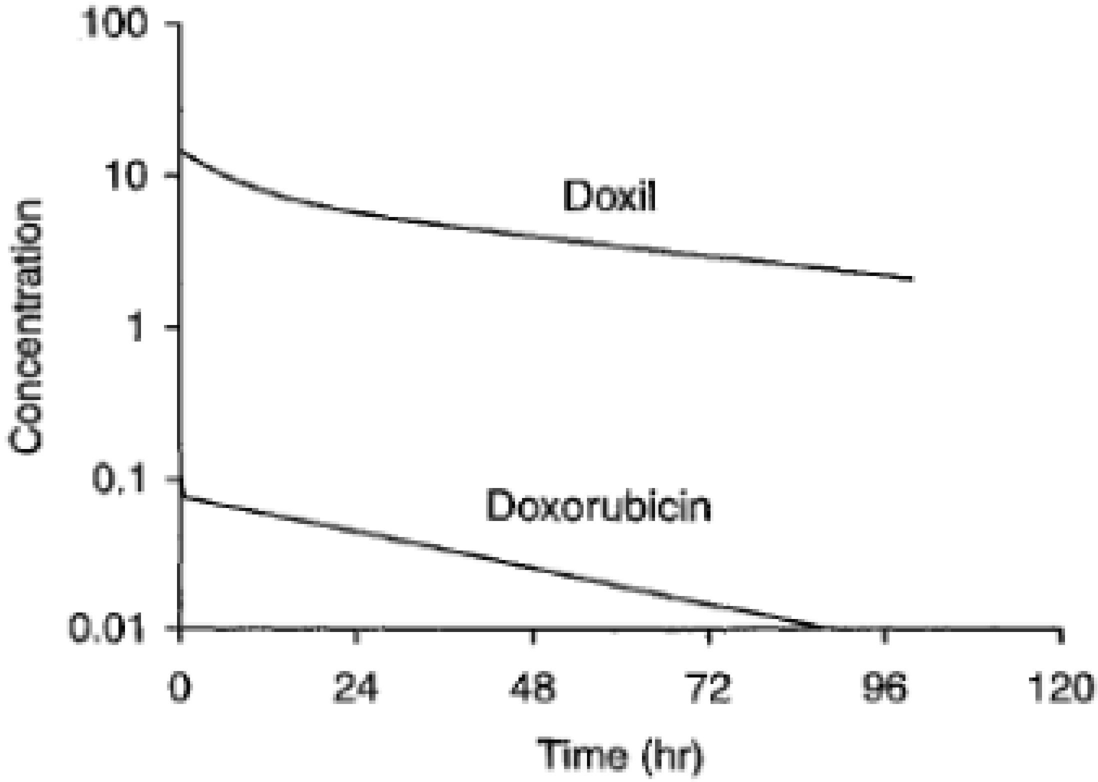
Limits the dose of Dox that can be administered



Drug: doxorubicin
Chemotherapy agent

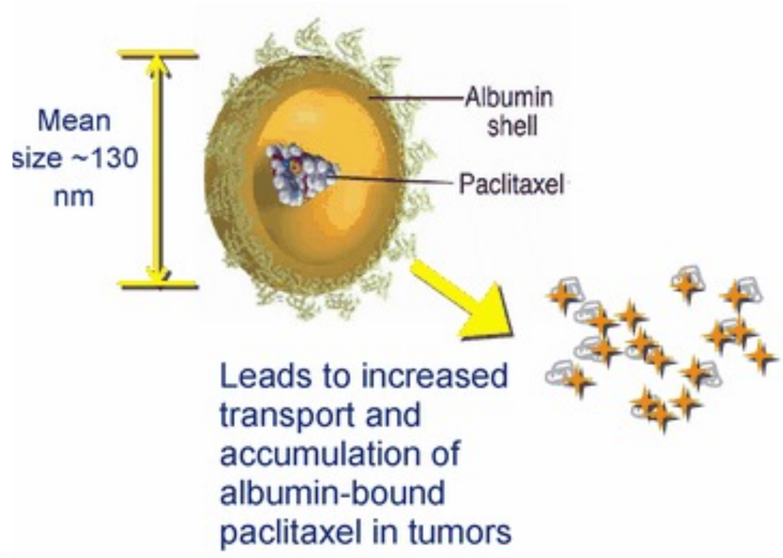
Name of product: Doxil
Reduced cardiotoxicity

Http://www.doxil.com



Drug Delivery: Engineering Principles for drug therapy by Mark Saltzman, 2001

Nano Carriers Example: Abraxane



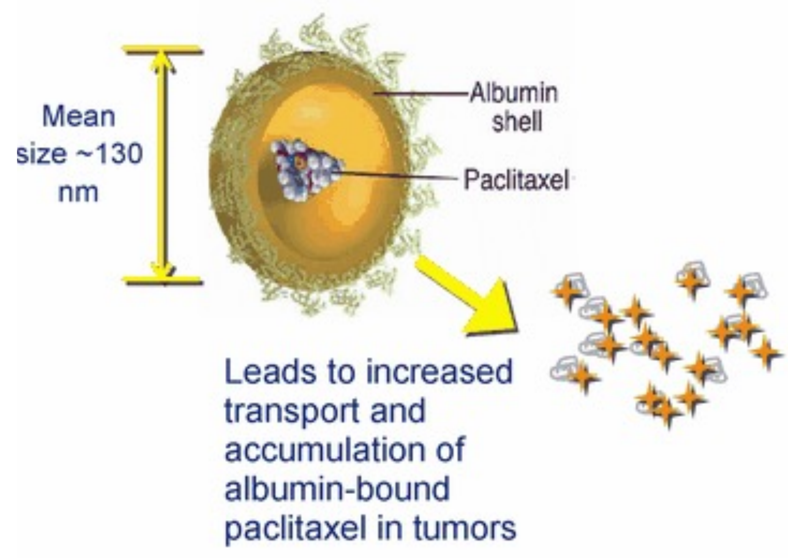
Drug: Paclitaxel

Chemotherapy for breast cancer: Very low solubility in water

Chemotherapeutic bound to protein nano-particle: Albumin bound paclitaxel

http://www.abraxisbio.com

Nano Carriers Example: Abraxane



Paclitaxel is dissolved in an organic solvent to form an oil phase. This is then mixed with an aqueous phase – which is the albumin solution. Coarse emulsion which is then processed using high shear to create a nanoemulsion of 130nm.

Name of product: Abraxane

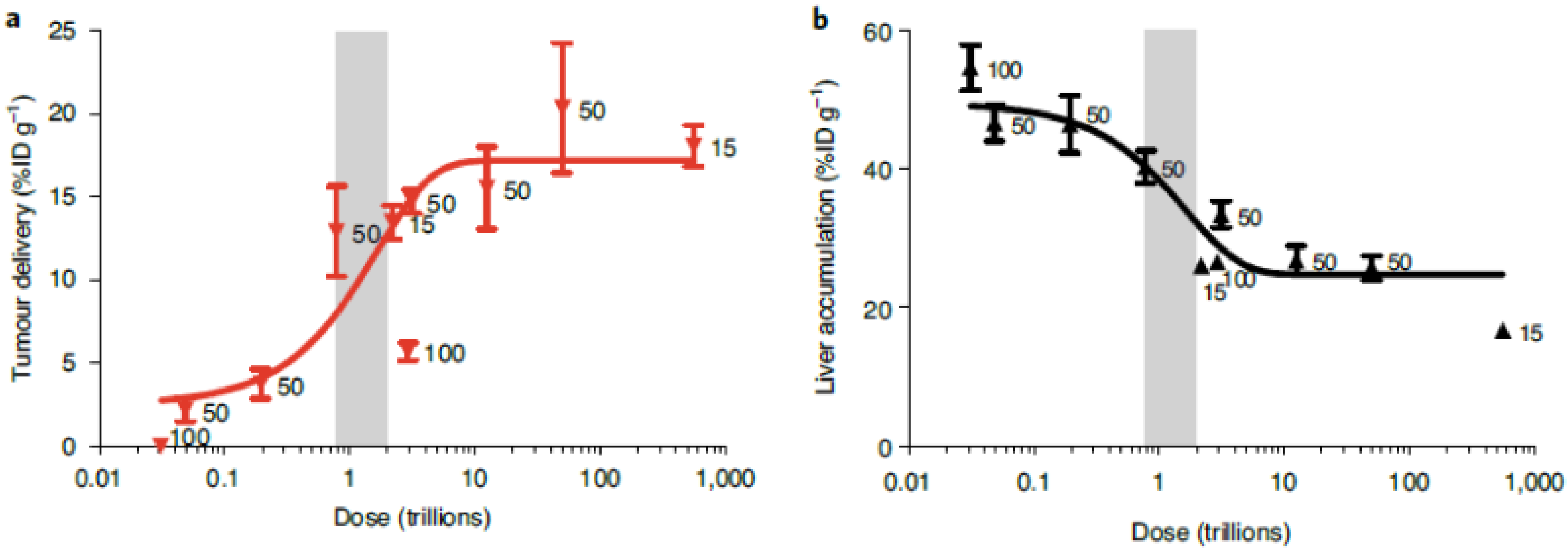
Approved in 2005 ($134 million in sales that year)

http://www.abraxisbio.com

https://www.microfluidics-mpt.com/applications/nanoparticle-albumin-bound-nab-drug-delivery

EPR effect: challenges

- Only about 0.7% of nanoparticles reach tumor
- Liver is a major sink of IV injected nanoparticles
- Uptake sites in liver can saturated at high doses that can increase tumor delivery



Numbers indicate particle diameter

Nature Materials: https://doi.org/10.1038/s41563-020-0755-z

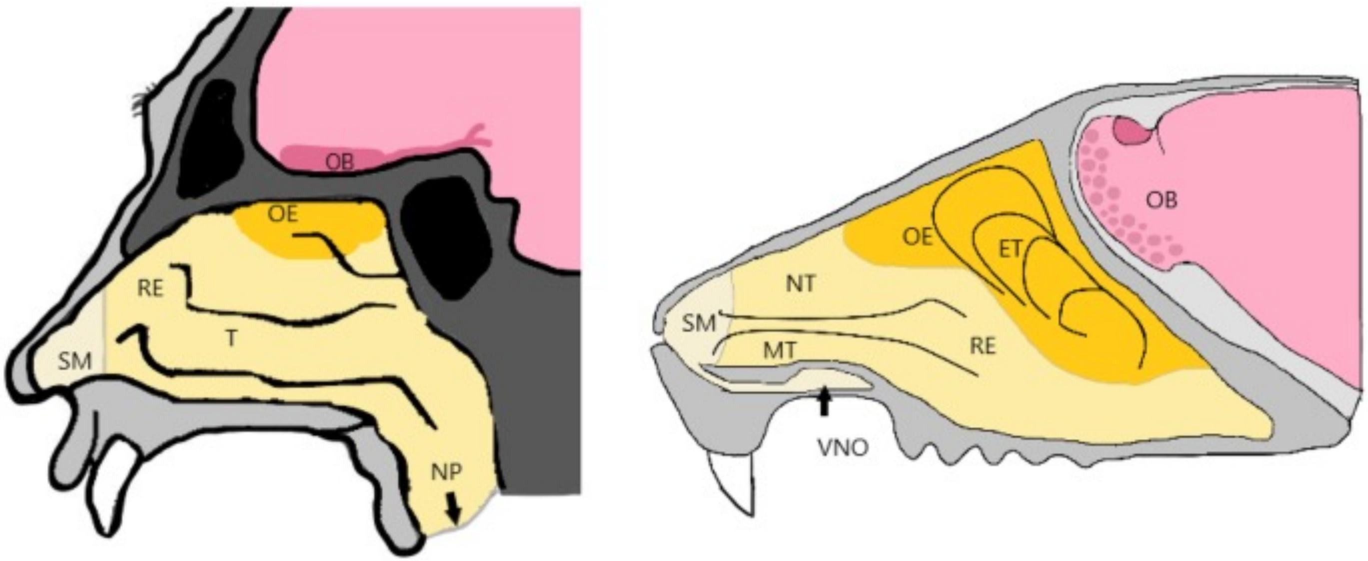
EPR effect: challenges

- Injecting particles through any other route than IV and relying on them to reach systemic circulation is difficult
- Even if they do reach at a low efficiency, you would then expect to only deliver <1% of systemically circulating particles to tumor site.
- Please remember this and not design your group projects like this!

Brain delivery

Two possible ways:

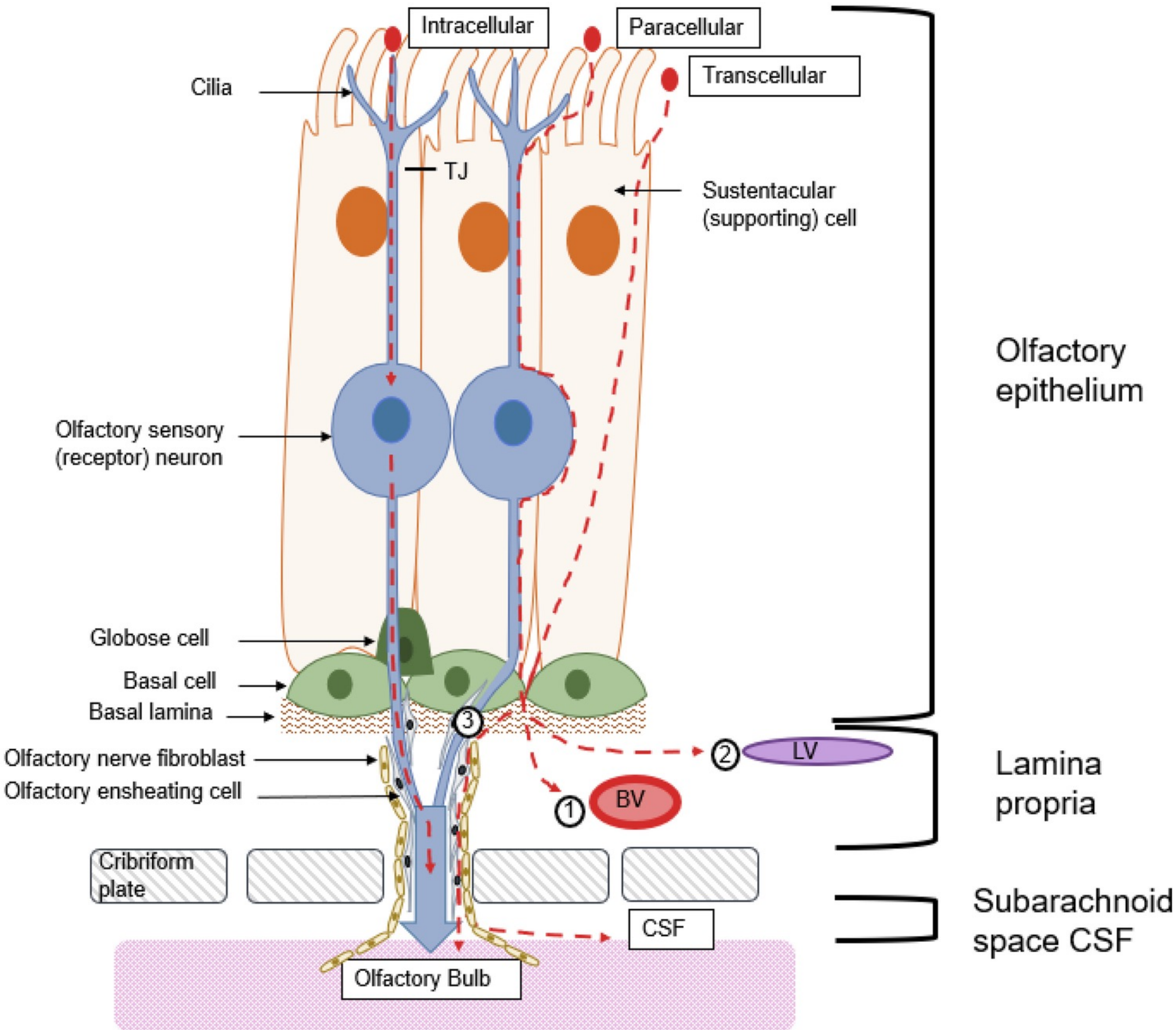
1. IV delivery
2. Intra-nasal delivery



Nasal cavity of humans (left) and rodents (right), showing the different regions within the nose. Starting with the squamous mucosa (SM) right at the nostril openings. The respiratory epithelium (RE) covers the main part of the nasal cavity in humans. This is then connected with olfactory epithelium (OE). The olfactory bulb (OB) is in close proximity to the cribriform plate and connected to the OE via the axons of the sensory neurons projecting towards the brain.

Drug Deliv. and Transl. Res. (2021). https://doi.org/10.1007/s13346-020-00891-5

Brain delivery through Intra-nasal



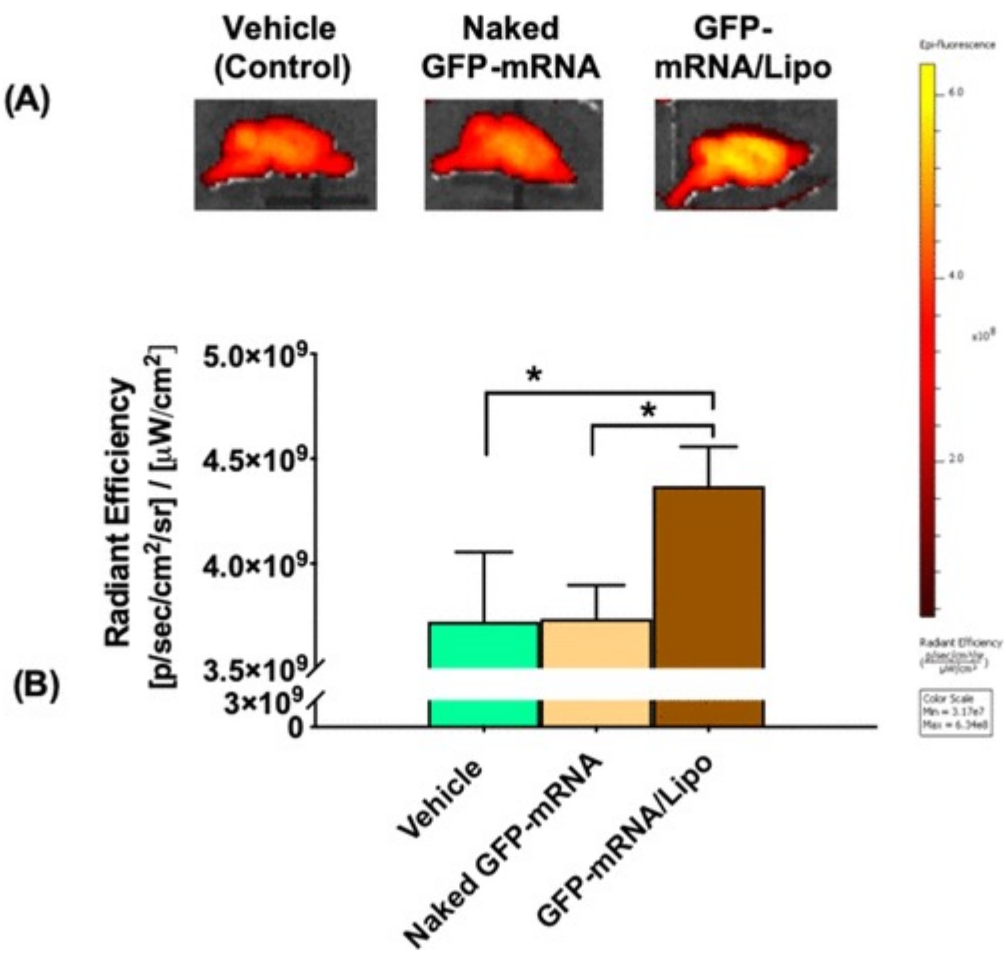
Three different pathways

1. The substance can bind to a receptor and be internalised by for example receptor-mediated endocytosis. Then, it travels through the olfactory sensory neuron (OSN) towards the olfactory bulb (OB).
2. Substance uses leaky passages within the OE and travels paracellular into the lamina propria.
3. Substance can also be transported transcellular through the sustentacular cells (SUS) to the lamina propria.

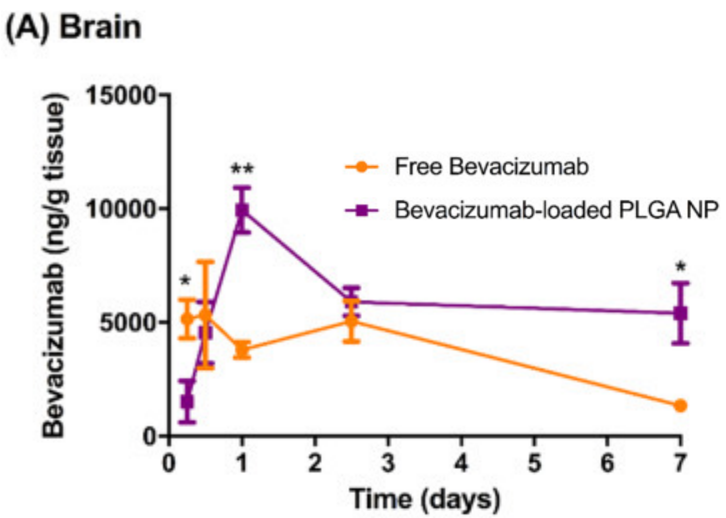
After passing the cribriform plate, the substance can theoretically also reach the cerebrospinal fluid and distribute through the different brain regions

Drug Deliv. and Transl. Res. (2021). https://doi.org/10.1007/s13346-020-00891-5

Brain delivery through particles



200nm liposomes



200nm PLGA particles

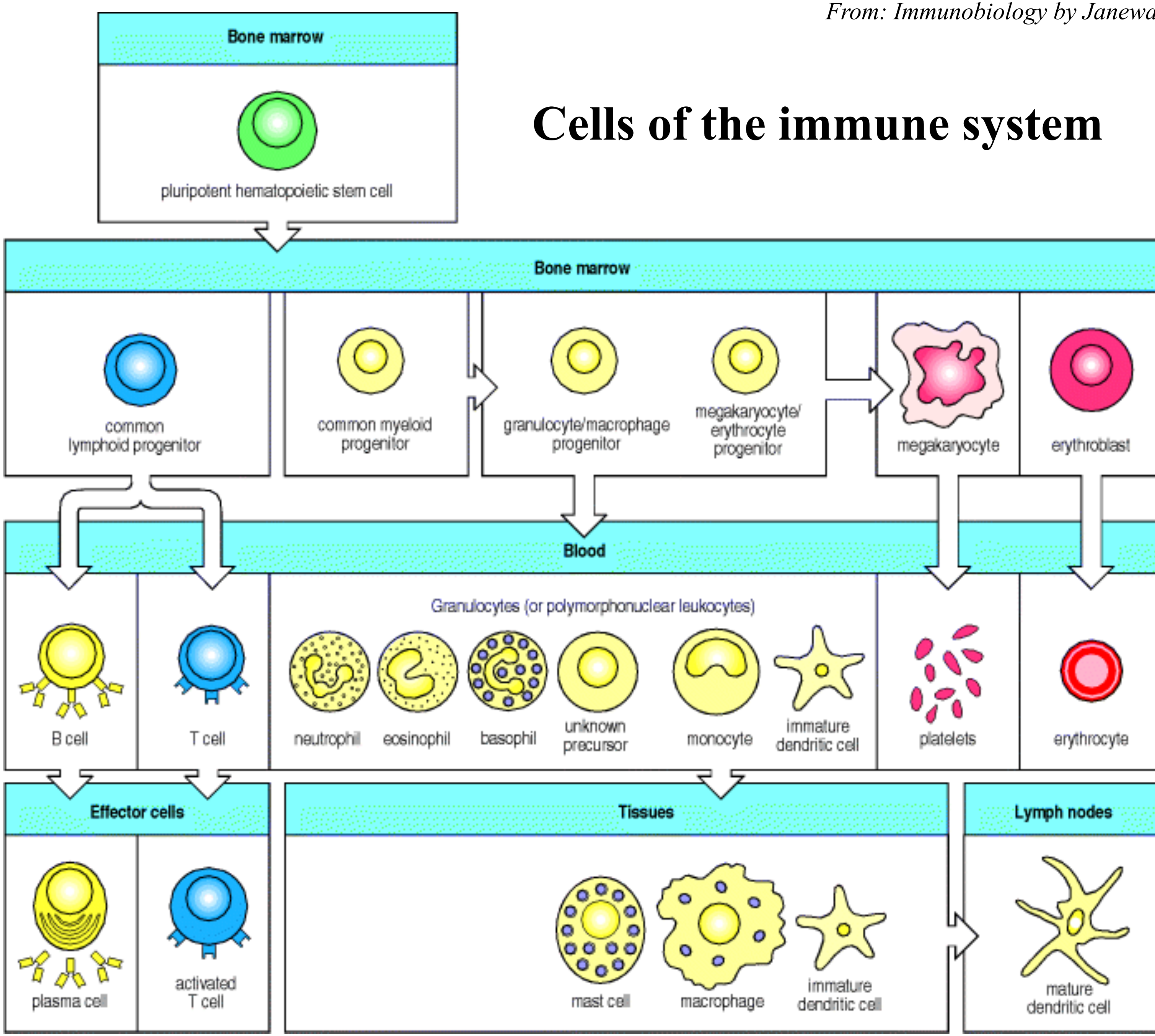
https://doi.org/10.1016/j.jconrel.2019.07.033

https://doi.org/10.1021/acs.molpharmaceut.0c00170

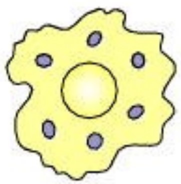
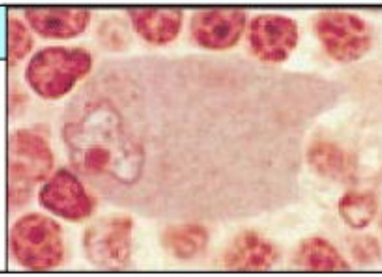
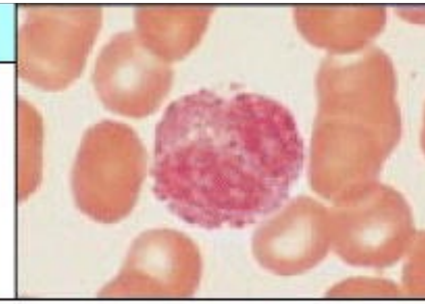
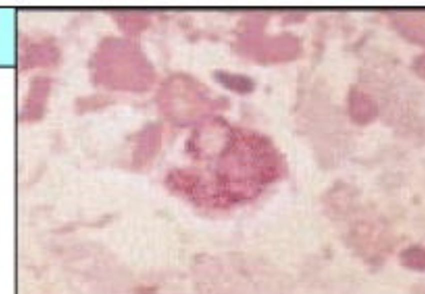
Immune response to materials

From: Immunobiology by Janeway and Travers

Cells of the immune system

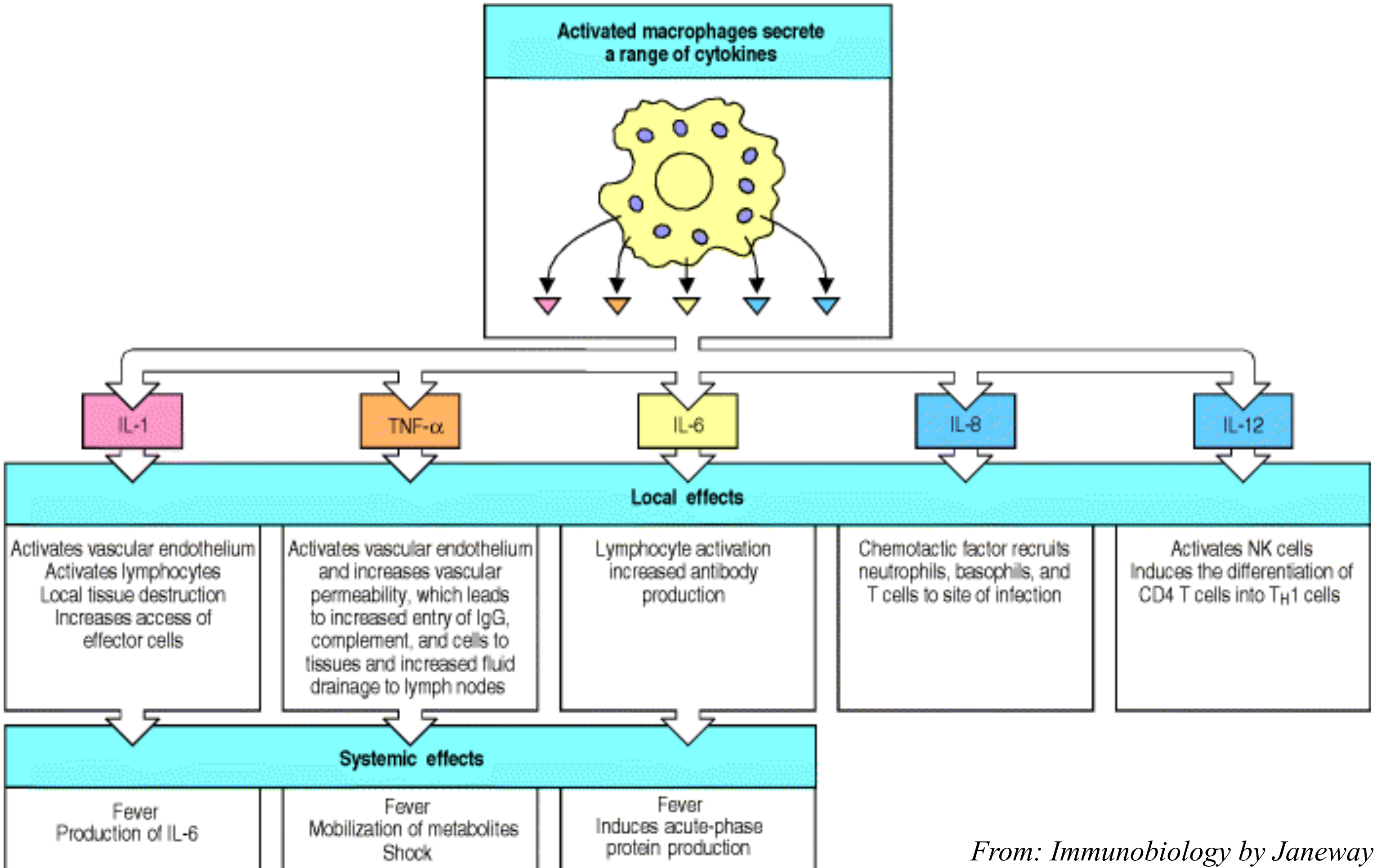


From: Immunobiology by Janeway and Travers

Cell			Activated function
Macrophage			Phagocytosis and activation of bactericidal mechanisms Antigen presentation
Dendritic cell			Antigen uptake in peripheral sites Antigen presentation in lymph nodes
Neutrophil			Phagocytosis and activation of bactericidal mechanisms
Eosinophil			Killing of antibody-coated parasites
Basophil			Inflammatory reactions and allergies
Mast cell			Release of granules containing histamine and other active agents

The Onset of Inflammation

- There are always some phagocytic cells in the tissue at all times
- Upon tissue injury or infection, these cells become active
- Continue digesting foreign Ag and necrotic cells; secrete CYTOKINES



From: Immunobiology by Janeway and Travers

Cytokines and chemokines

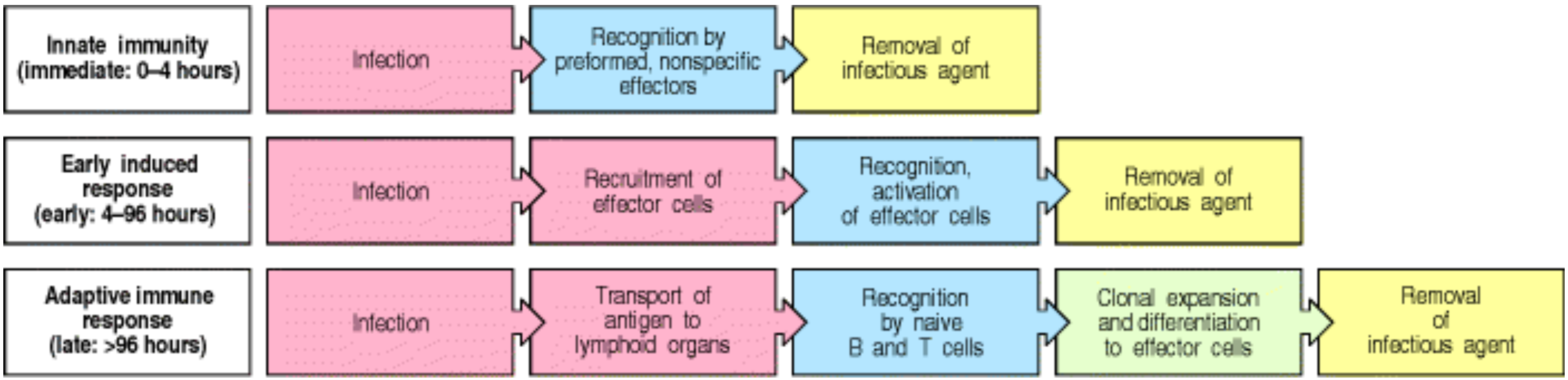
- **Cytokines** are small proteins (~25 kDa) that are released by various cells in the body, usually in response to an activating stimulus, and induce responses through binding to specific receptors.
  - They can act in an **autocrine** manner, affecting the behavior of the cell that releases the cytokine,
  - or in a **paracrine** manner, affecting the behavior of adjacent cells.
  - Some cytokines can act in an endocrine manner, affecting the behavior of distant cells, although this depends on their ability to enter the circulation and on their half-life.
- **Chemokines** are a class of cytokines that have chemoattractant properties, inducing cells with the appropriate receptors to migrate toward the source of the chemokine.

Inflammation: Physiological Events

- Tissue injury + foreign entities → activation of tissue macrophages and neutrophils (phagocytic cells) → phagocytosis → cytokine secretion
- Local cytokine release triggers a cascade of events
  - Vasodilation (increase in diameter of the capillaries, redness)
  - Increase in capillary permeability → influx of protein-rich fluid and cells (exudate) → edema (swelling, pain)
  - Influx of leukocytes and phagocytic cells into the tissue
  - Healing, tissue regeneration → formation of new capillaries and scar tissue

From: Immunobiology by Janeway and Travers

Response to an infection



Materials and the immune system

- Main job of the immune system is to protect us from pathogenic invasions
- **Materials in the body → equivalent to foreign invasions**
- The body has employed a range of interactions to counter this invasion
- **Most materials, following implantation or administration, comes in contact with blood**
  - Even if it is not delivered into the blood, the tissue injury caused by implantation causes blood contact
- Degree or intensity of the tissue reaction → Measure of material biocompatibility
- These host reactions are tissue dependent, organ dependent and species dependent

Blood response to materials

- **Complement System**
  - COMPOSED OF MORE THAN 20 DISTINCT PLASMA PROTEINS THAT WORK IN AN ORCHESTRATED CASCADE
  - PURPOSE: NON-SPECIFIC RECOGNITION AND ELIMINATION OF FOREIGN ELEMENTS FROM THE BODY
- **Blood coagulation on material surfaces**
  - PHYSIOLOGICAL FUNCTION: TO STOP BLEEDING FROM INJURED VESSELS, THE TISSUE SURFACE PLAYS AN IMPORTANT ROLE
  - BIOMATERIAL SURFACES CAN "TRIGGER" THE COAGULATION CASCADE

Not covering in this course but covered in FoB2

Implications in drug delivery

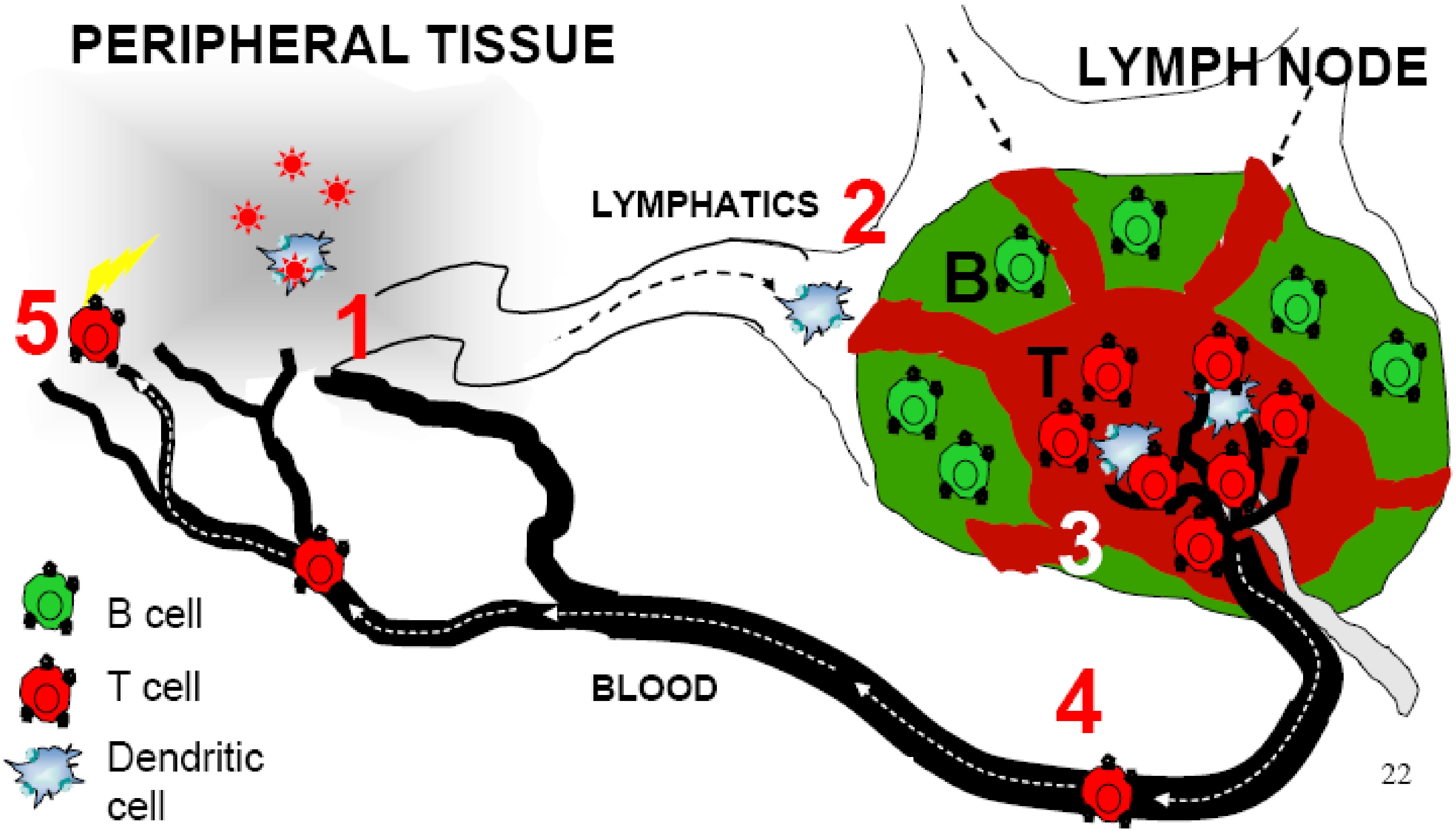
- Effects on drug delivery?
  - Fibrous capsule formation
  - Diffusion limitations: released drug cannot diffuse to the body
  - Limited exchange of fluid
  - Slower rate of hydrolysis and degradation
  - Leads to slower rate drug release
  - Danger to patient

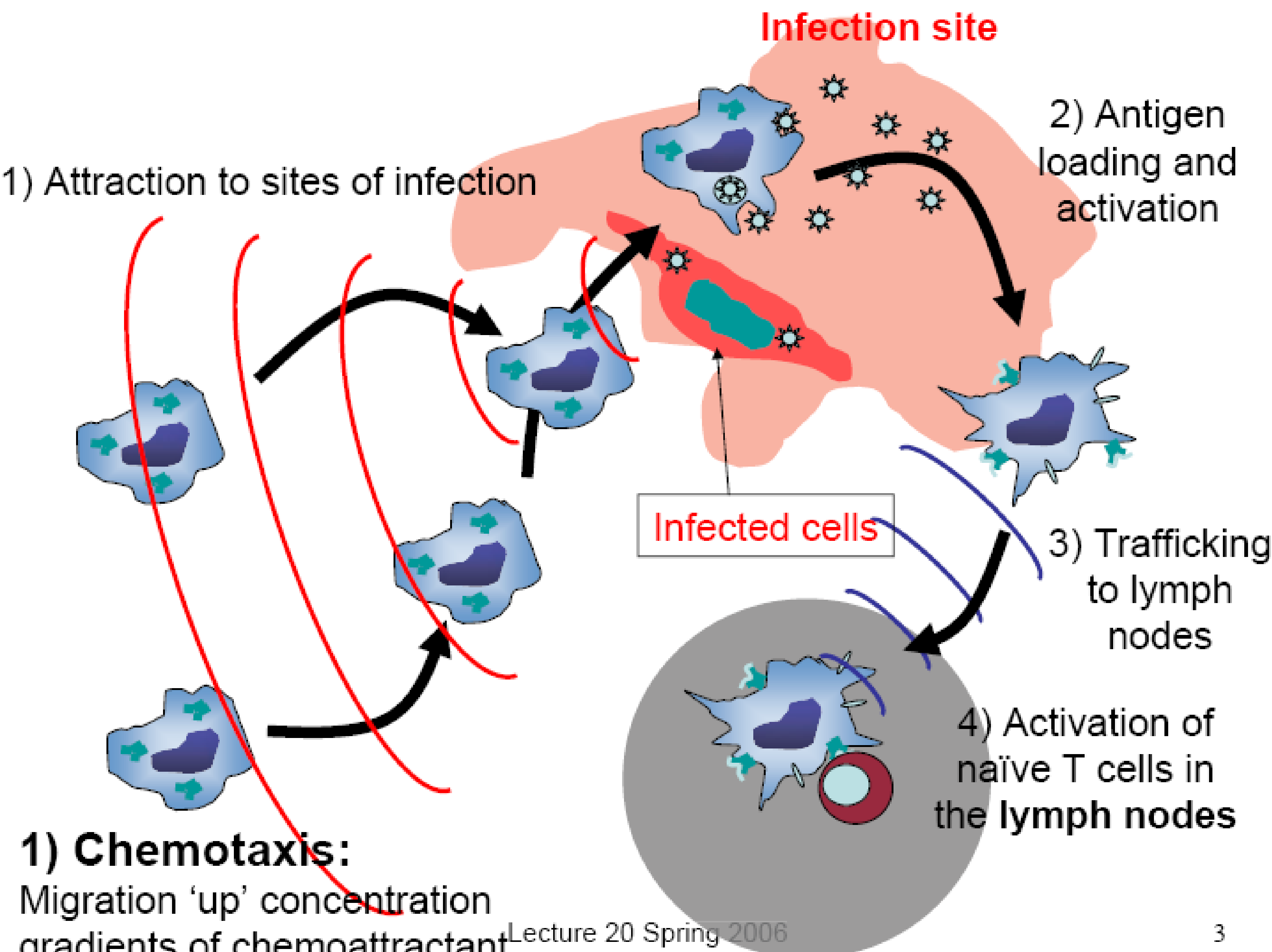
Adaptive Immune Response

Cells of the Adaptive Immune System

- Dendritic cells and macrophages
  - Antigen Presenting Cells
  - Present in the tissue and survey the surroundings
- T cells
  - Helper T cells: Boost and help stronger immune response
  - Cytotoxic T cells: Directly kill the mutant cell
  - Regulatory T cells: Involved in tolerance
- B cells
  - Antigen presentation
  - Antibody generation

Physiology of the primary immune response

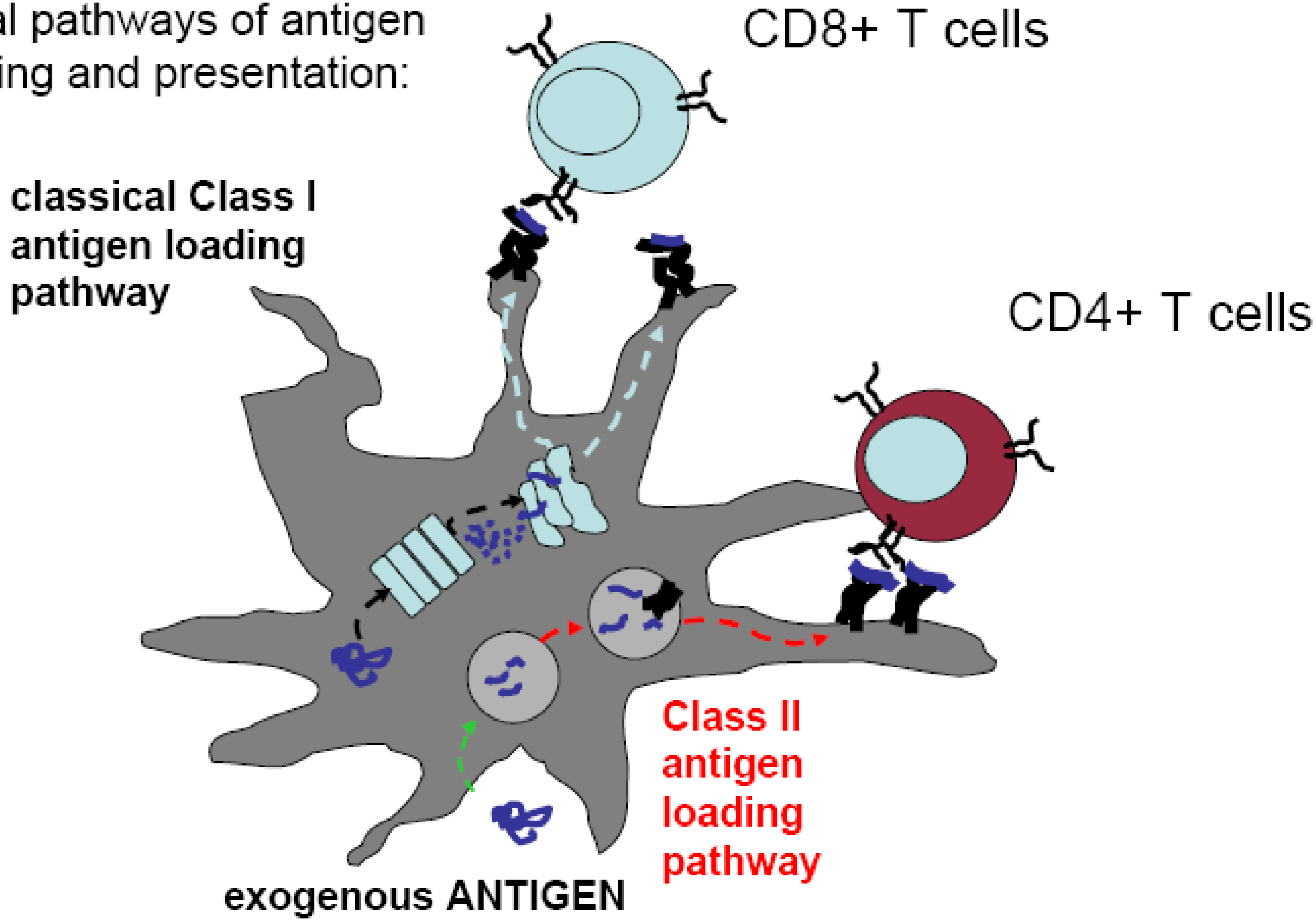




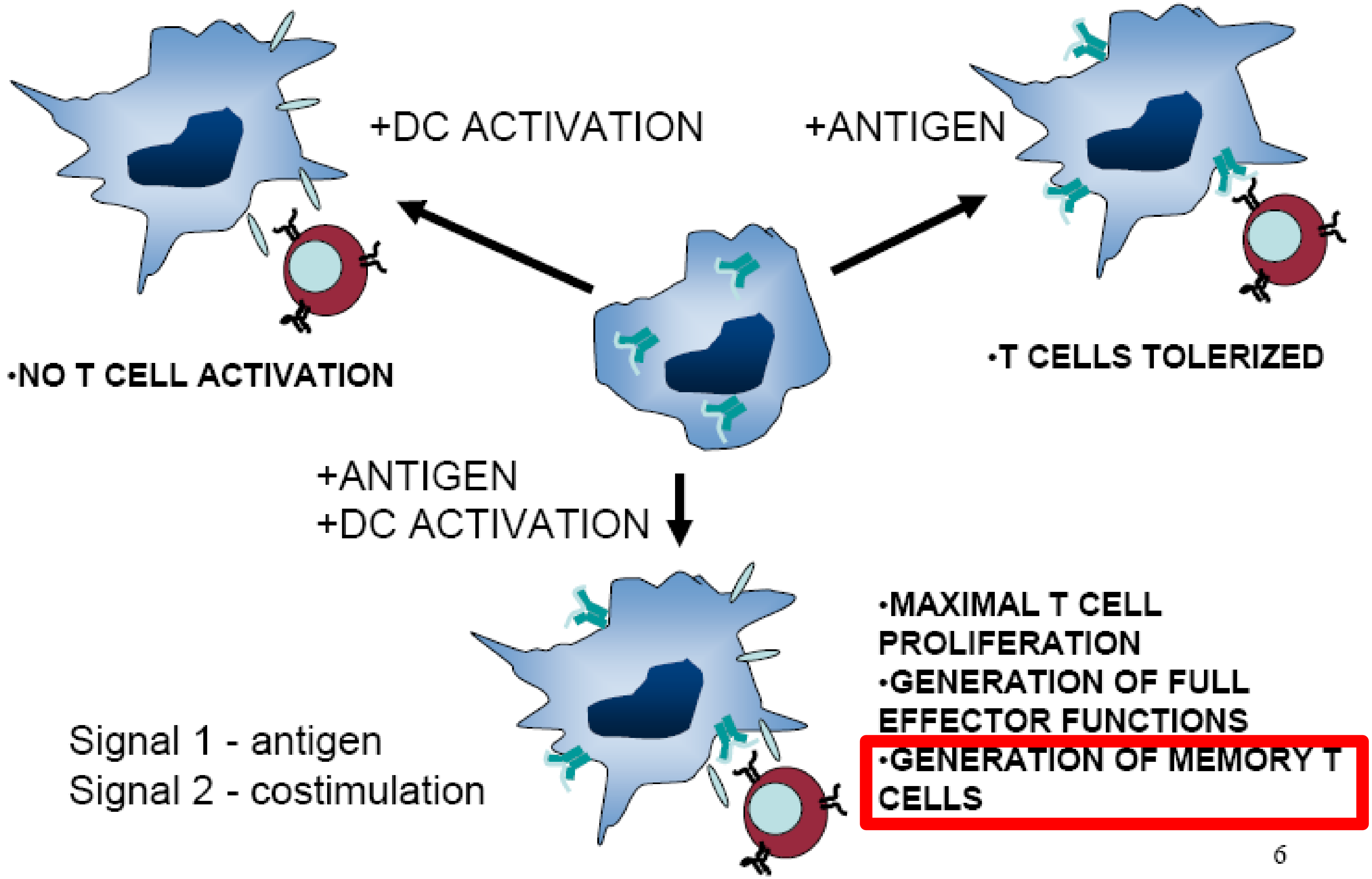
1) Chemotaxis:
Migration 'up' concentration gradients of chemoattractant

Biology of dendritic cells in T cell activation

Classical pathways of antigen processing and presentation:



Antigen is *one* of (at least) *two* signals that must be delivered by a vaccine



Adaptive immune response against biomaterials?

- More relevant for natural polymers (i.e. organism derived proteins, peptides, polysaccharides etc. that the body considers to be "foreign")
- However, some synthetic polymers can also be recognized as foreign
  - These "polymers" can be processed as "antigens" by dendritic cells and macrophages, presented to T cells which can lead to production of antibodies and cytotoxic T cells.
  - Important criteria while choosing peptides, proteins, polysaccharides etc. as biopolymers for drug and gene delivery.

Immuno-isolated Cell Therapy

- Transplanting cells that can secrete a therapeutic protein over a long period and in response to physiological stimuli
  - Pancreatic Islet Transplantation
  - Long-term gene therapy (using transformed cells)
- The cells transplanted are allogeneic or xenogeneic and can generate immune reaction (rejection)
- A protective polymer encapsulation, that allows nutrients and small molecules (oxygen, electrolytes, nutrients, i.e. < 50 Kd) to reach the cells, but prevents antibody molecules and T cells,→ immuno-isolation
  - Replaces constant immuno-suppression
  - Allows xenogeneic transplantation

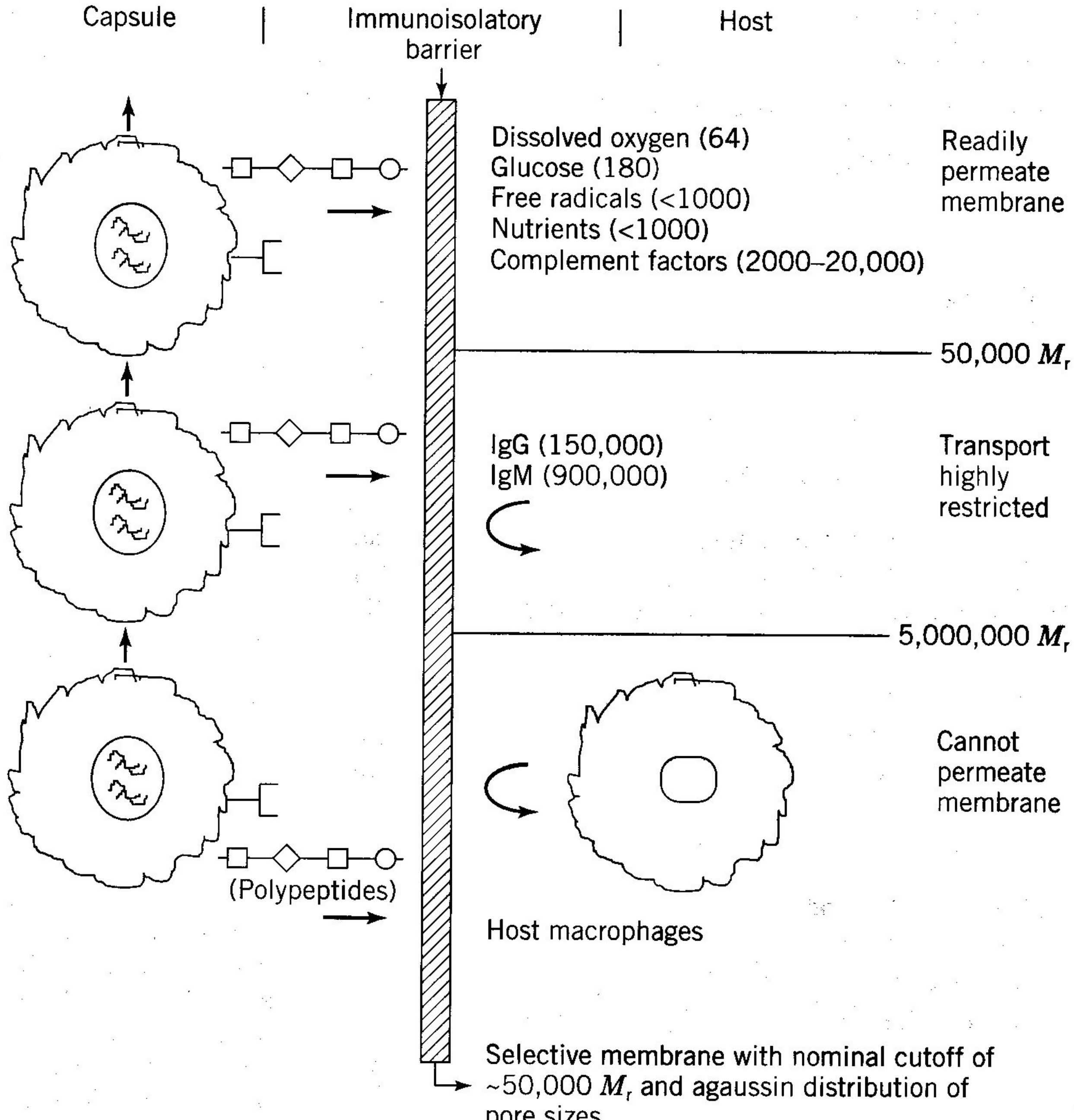


Figure 1. Immunoisolated cell therapy relies upon a membrane to separate foreign secretory cells from a host. Molecules secreted by the cells, usually proteins, can pass through the membrane if they are smaller than 50,000 M_r. In similar fashion, oxygen glucose and molecules such as insulin can diffuse from the host to the cells. However, the membrane blocks the passage of host cells and restricts transport of host immunoglobulins. Membranes allow passage of complement factors and free radicals that appear to be generated primarily when an implant containing xenogeneic cells is directly implanted into host tissue.

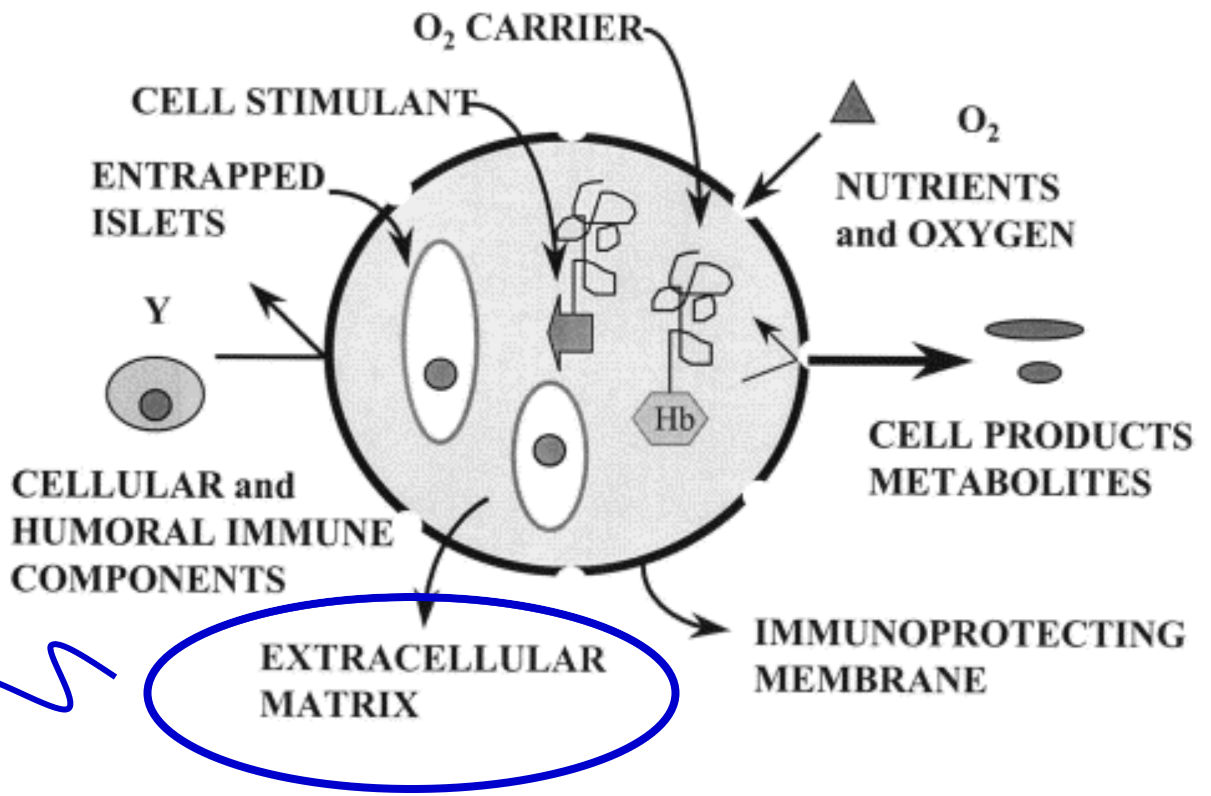
Components of Immuno-isolated Implants

- Polymeric components:
  - Must be non-erodible (remains intact over entire transplantation duration)
  - Must be mechanically stable (cannot break)
  - Must be biocompatible (fibrous encapsulation, adhesions etc. Would severely impair diffusion)
  - Membranes
    - Hemodialysis membranes: Polysulfone, PAN-VC with cutoff ~ 50,000 daltons → can also be reinforced (cast on a support)
  - Matrices
    - Hydrogels: Natural polymers such as alginate or collagen
    - Scaffolds: Polyurethanes coated with hydrophilic polymer

Components of Immunoisolatory Implants

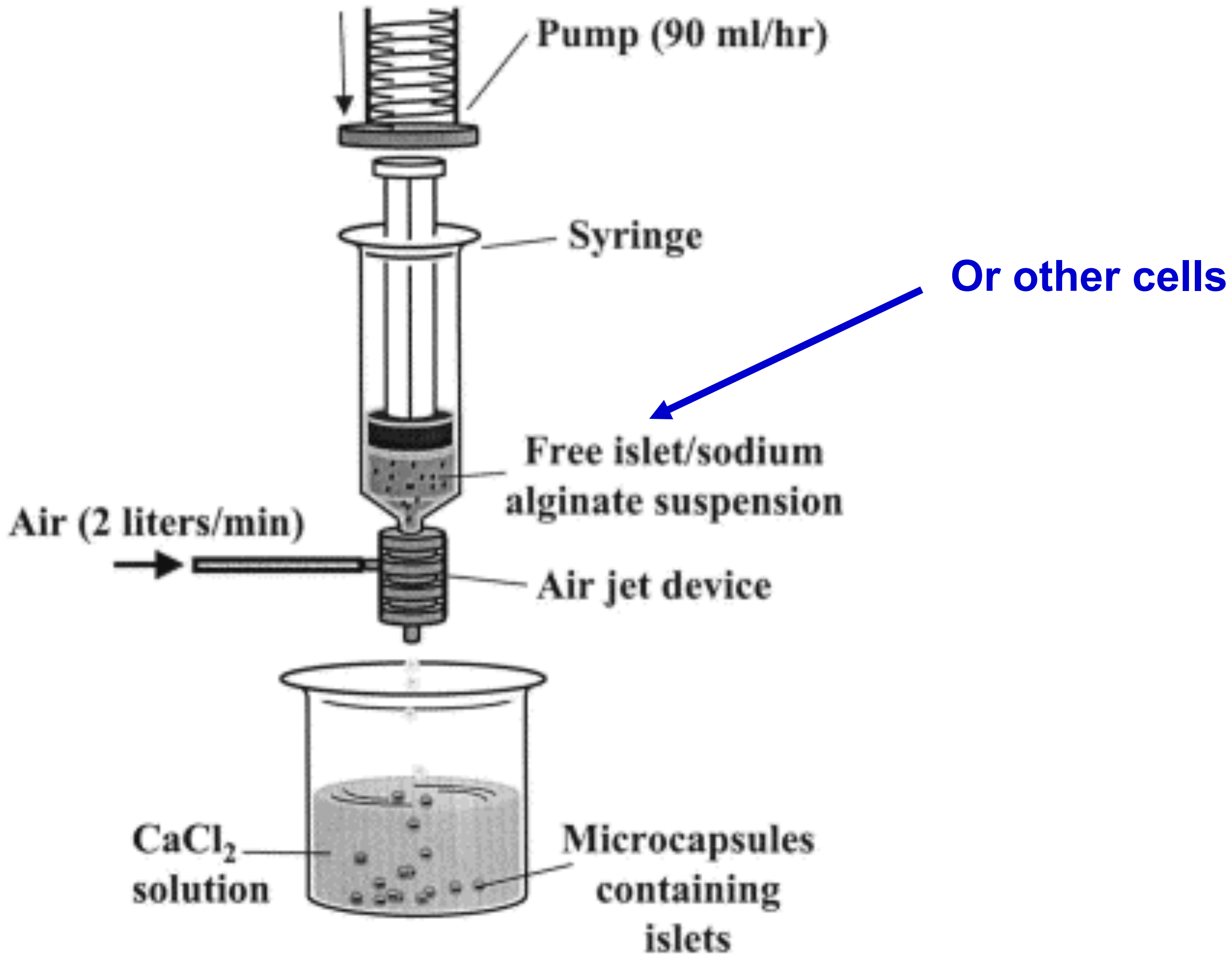
- Cells used for encapsulation
  - Primary
    - Islets (Insulin therapy for diabetes)
    - Hepatocytes (Liver replacement)
  - Cell lines
    - E.g. PC-12 cells secreting dopamine (Parkinson's)
  - Engineered
    - Cell lines transformed with genes encoding for therapeutic proteins like Factor VIII (Hemophilia) or EPO (Anemia)
    - Engineered cells that also contains factors preventing complement activation, apoptosis etc.
    - Chemical "switch" that can be turned on or off by addition of drugs

Model configuration



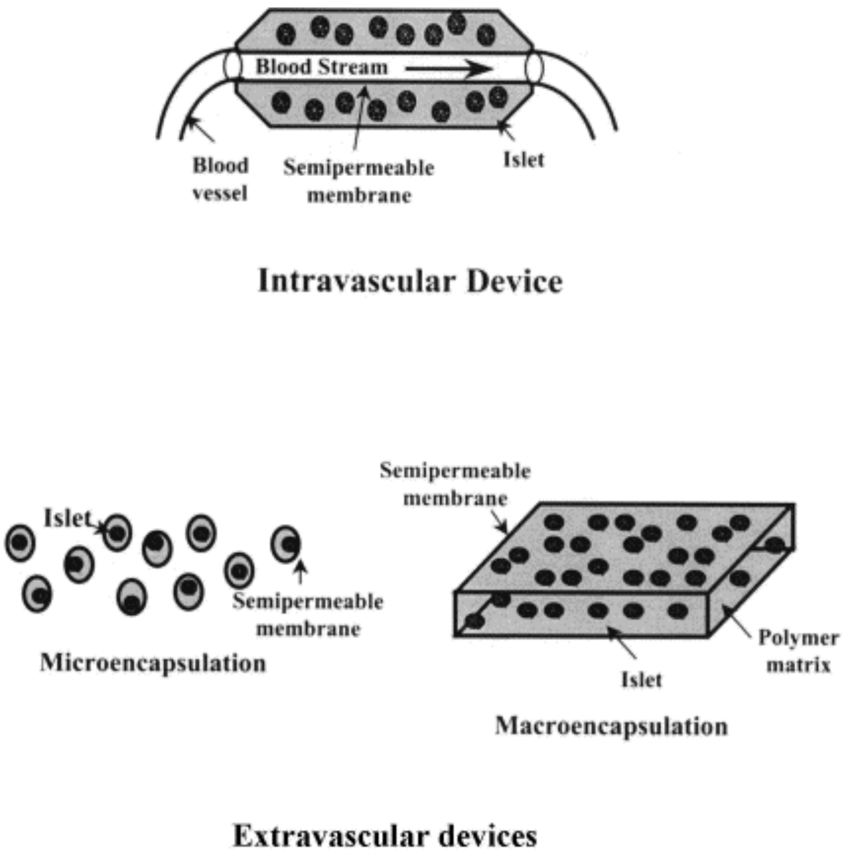
1. Immobilization of the cells to prevent fusion, aggregation and necrosis
2. Improved oxygen transport, such as is provided with perfluorocarbons
3. Presentation of an acceptable physical environment for attachment-dependent cells
4. Provision of an acceptable chemical environment (e.g., collagen gels have been shown to induce epithelial cells to grow into duct-like structures and to promote neonatal endocrine pancreatic cells to organize into islet-like structures)
5. Physical limitation of the cell volume within the chamber
6. Secondary immunoprotection

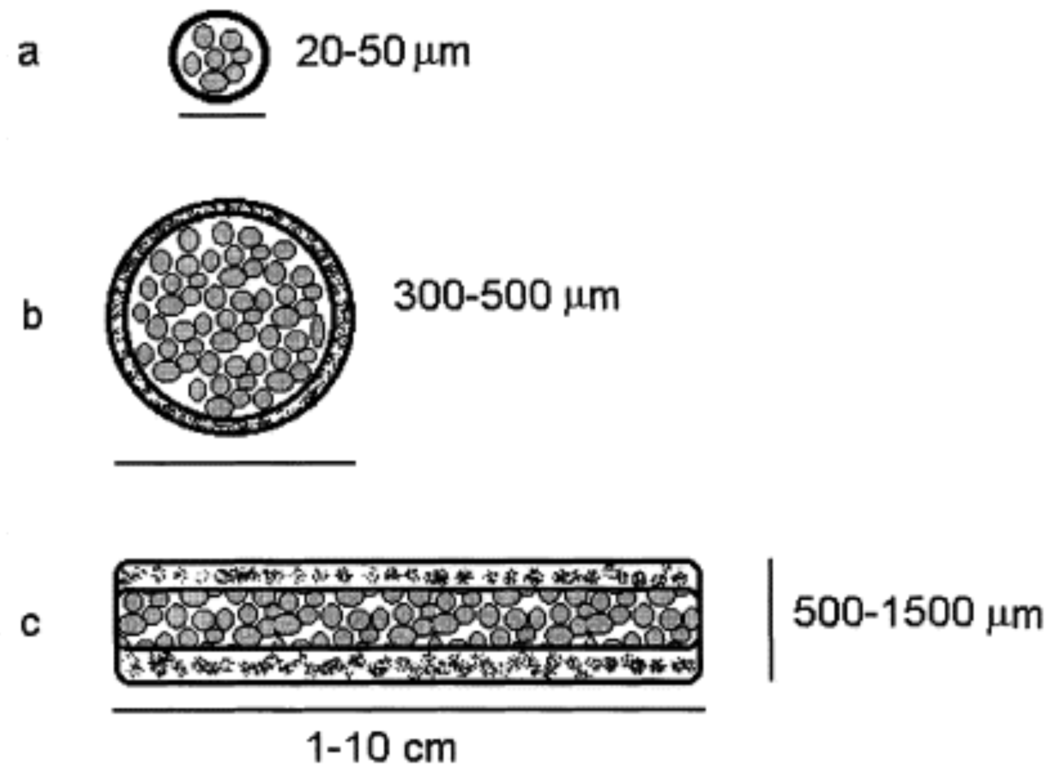
Schematic of micro-encapsulation process



Implant design ?

- Grafted cells must be less than 200 to 500 µm away from host environment → membrane thickness or matrix properties should be such that oxygen and nutrients can effectively diffuse through





Microfabricated biocapsules for insulin delivery

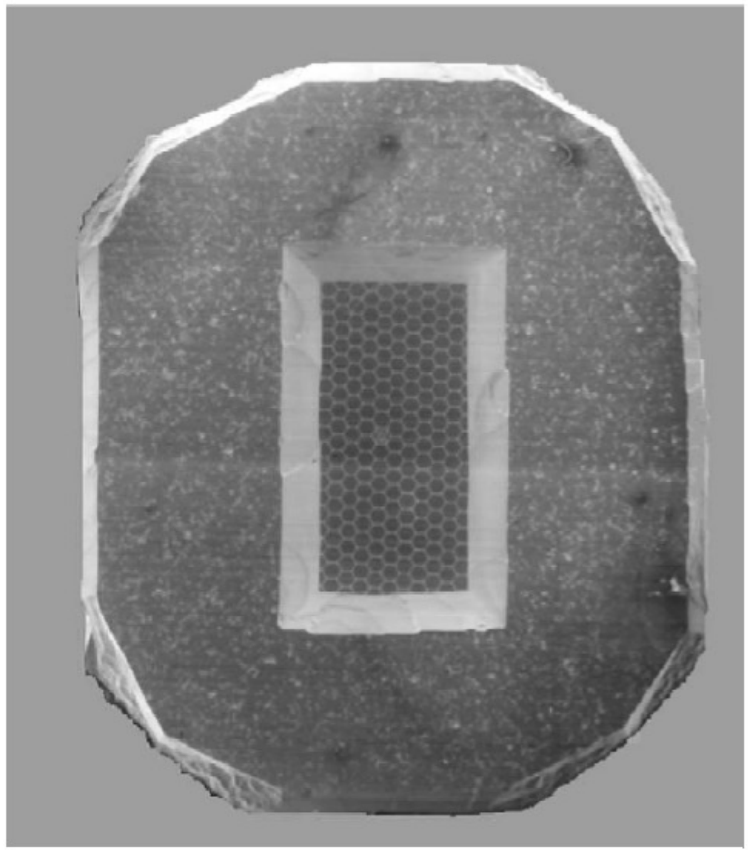


Fig. 7. Micrograph of a biocapsule membrane with 24.5-nm pores.

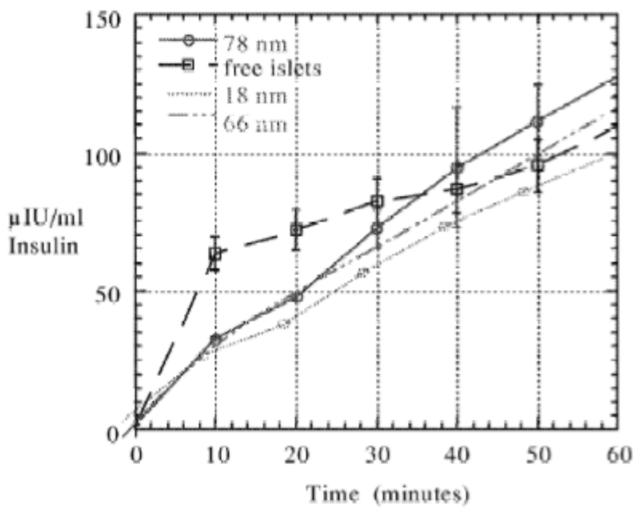
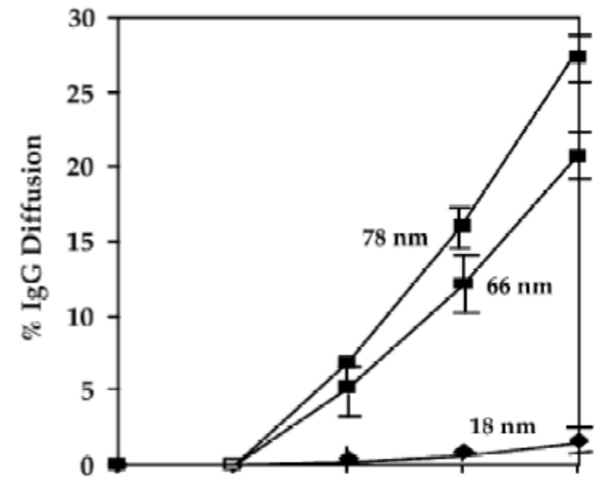


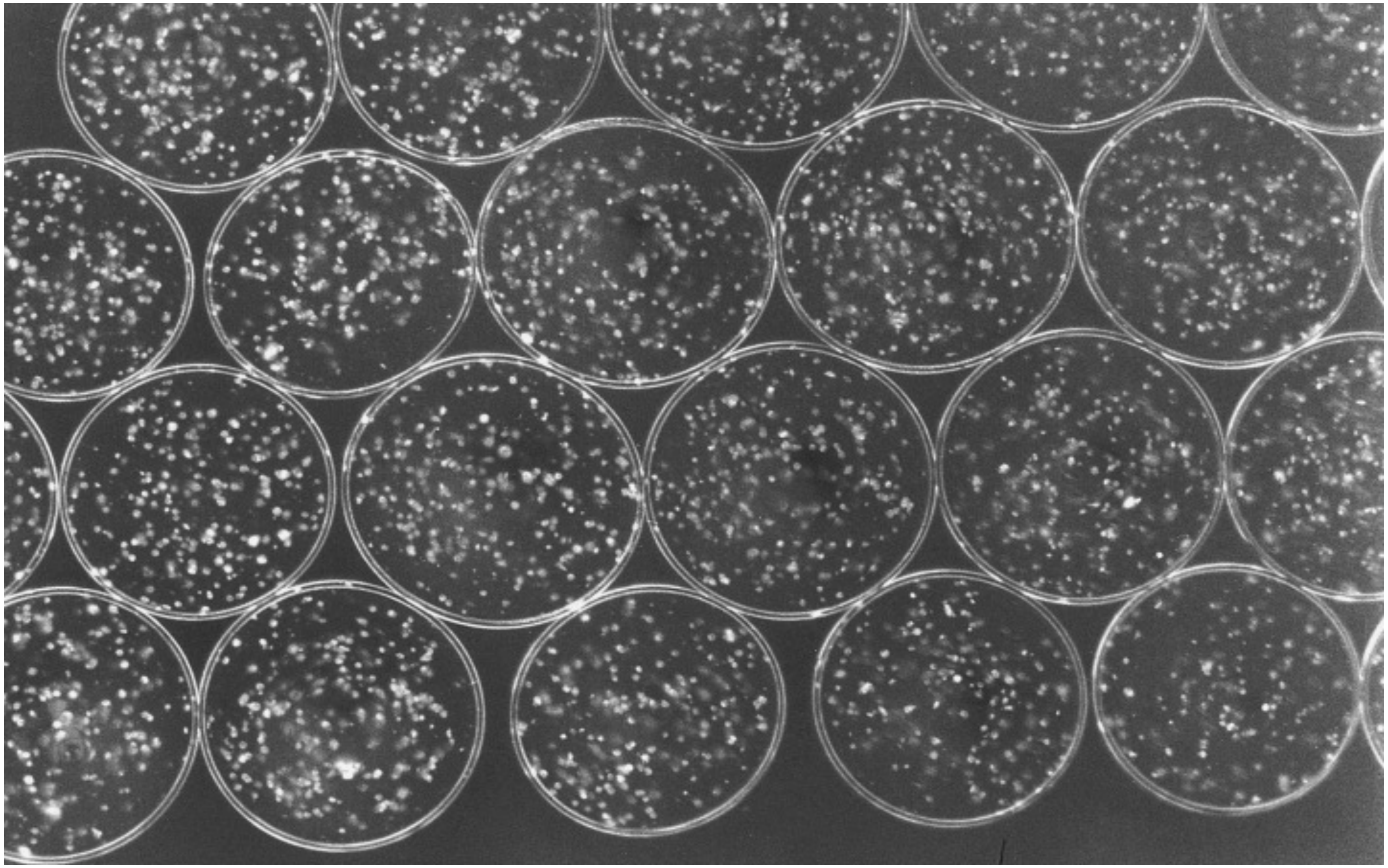
Fig. 8. Insulin secretory profile through differing pore sizes.



. 9. IgG diffusion through microfabricated biocapsules of three
erent pore sizes.

Gene Therapy

ENCAPSULATED MYOBLASTS



Implantation in immuno-deficient mouse

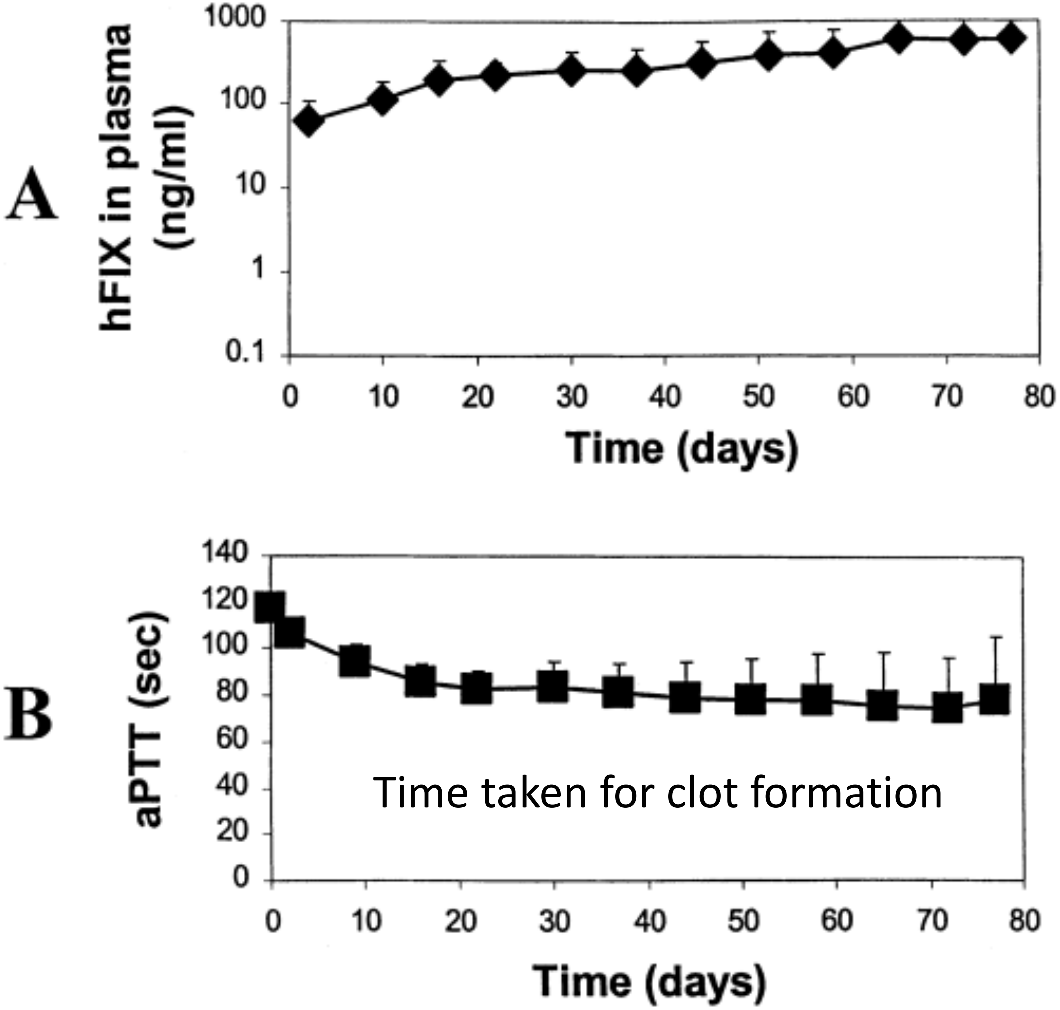
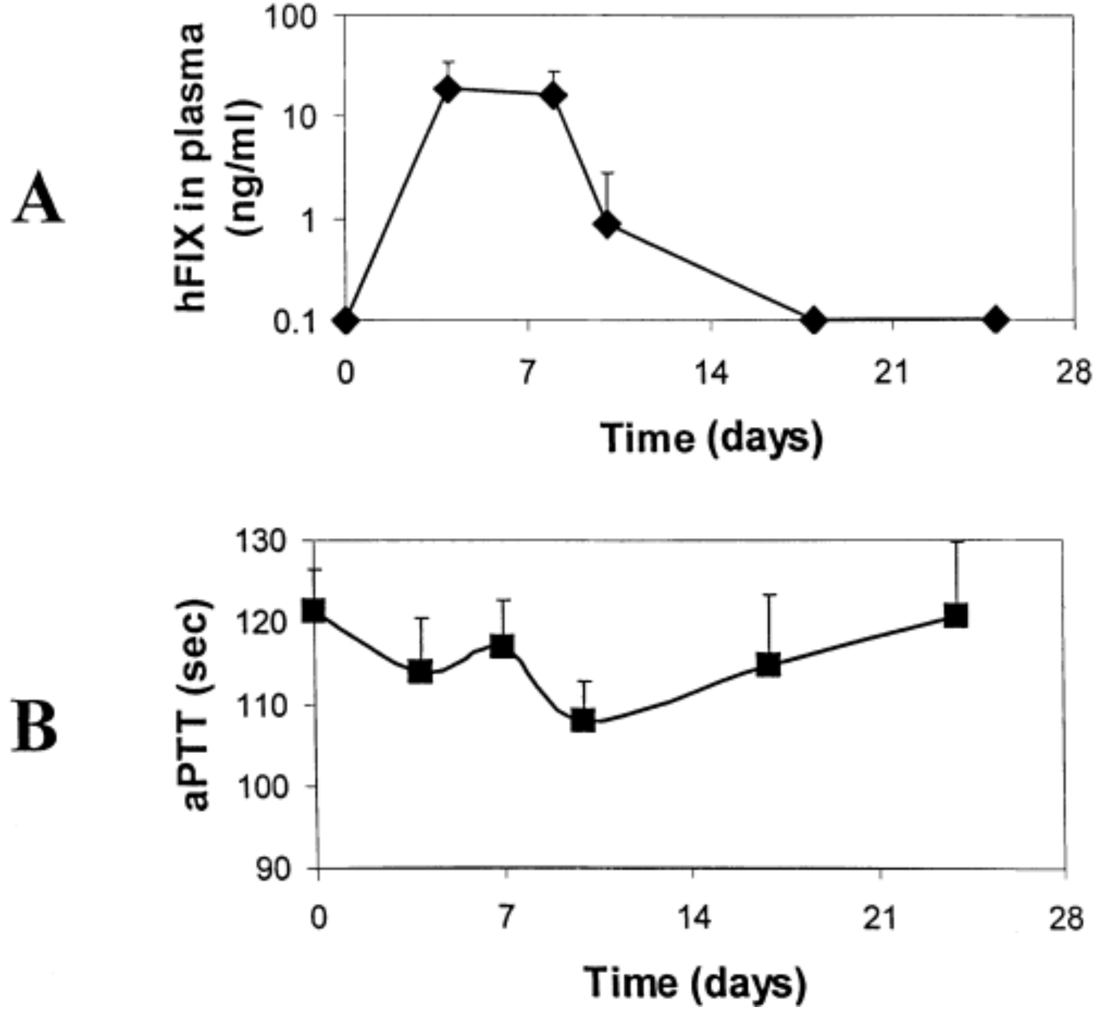


Fig . FIX and APTT in nude haemophilic mice. Determination of hFIX antigen in the circulation of nude haemophilic mice. Haemophilic immunodeficient mice (*n*=7) were implanted with microencapsulated mMyohFIX-18 myoblasts. At various times postimplantation, mice blood samples were collected through retroorbital bleeding. (**A**) hFIX antigen levels in plasma were assayed by ELISA. (**B**) Clotting activity was determined by APTT. Error bars are the standard deviations. Antigen hFIX levels were sustained in treated mice for at least 77 days (length of experiment).

Implantation in immuno-competent mouse



Factor IX and APTT in immuno-competent haemophilic mice. Delivery of hFIX in immuno-competent haemophilia mice implanted with encapsulated myoblasts transfected with pLNMIXIL. Haemophilic mice (n=7) were implanted with microencapsulated myoblasts. At various times postimplantation, blood samples were collected from each mouse through retroorbital bleeding. (A) hFIX antigen levels in plasma were assayed by ELISA. (B) The functional activity of the hFIX delivered was determined by the APTT of the plasmas obtained from the implanted mice.

Human protein in mouse: Antibodies

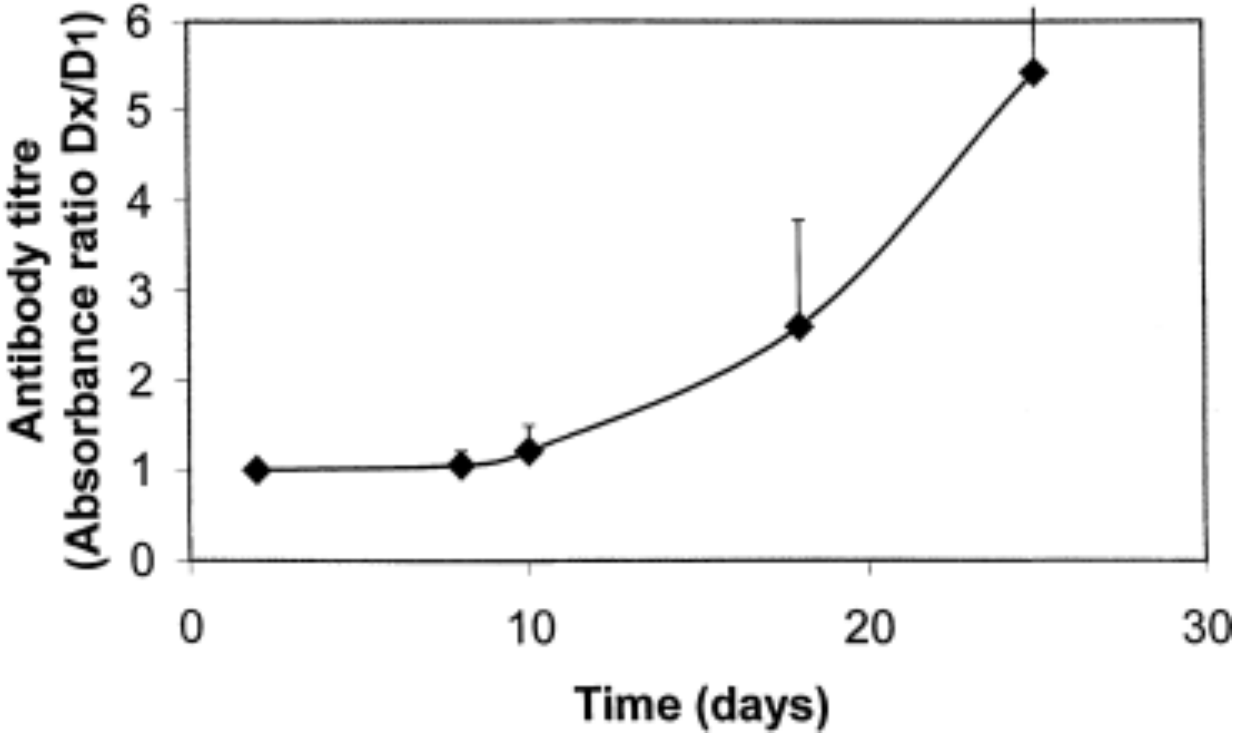


Fig. Antibodies to hFIX in immunocompetent mice implanted with microcapsules. AntihFIX antibodies in the plasmas of immunocompetent mice implanted with encapsulated myoblasts. The presence of antibodies to hFIX in mouse plasma was detected by ELISA, using mouse plasma diluted 1 : 100. The ratio of absorbance (OD at 405 nm) obtained for a given blood sample (Dx) over that of the blood sample from the same mouse at day 0 (D0), was plotted to measure the antibody titre. Each value shows the average antibody titre in the plasmas of each group of mice. Error bars are the standard deviation.

What is Gene Therapy (no cell delivery)?

Gene Therapy means treating a disease by introducing some genetic material (nucleic acid) into our cells. These genes can:

a. Replace a missing or defective gene

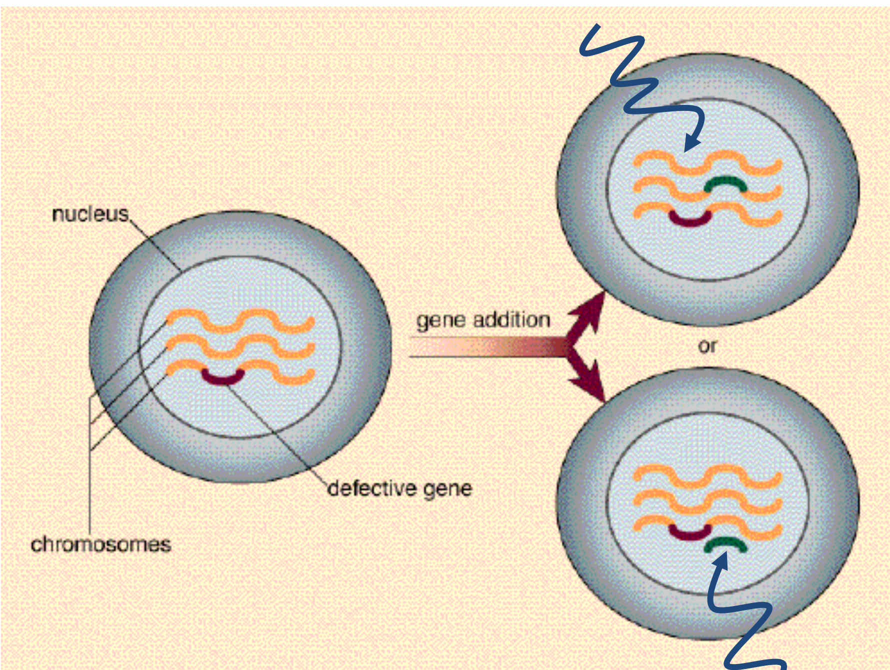
b. Repair a non-functional or mal-functional gene

c. Produce a therapeutic protein that helps to treat the disease

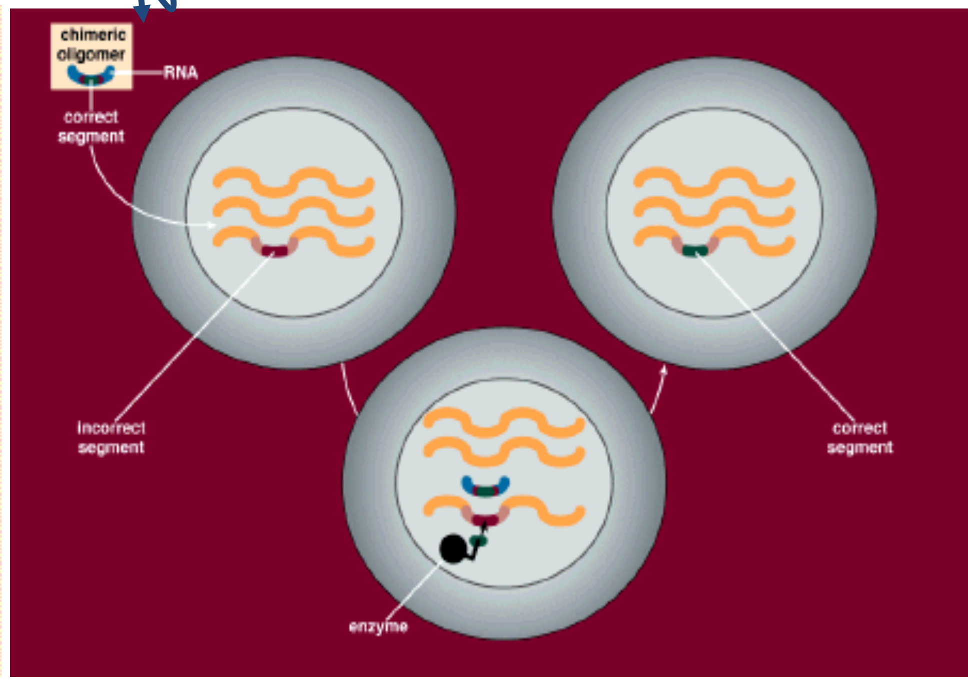
d. Can vaccinate the patient against an infection, etc.

What is Gene Therapy?

Chromosomal insertion



Correct template



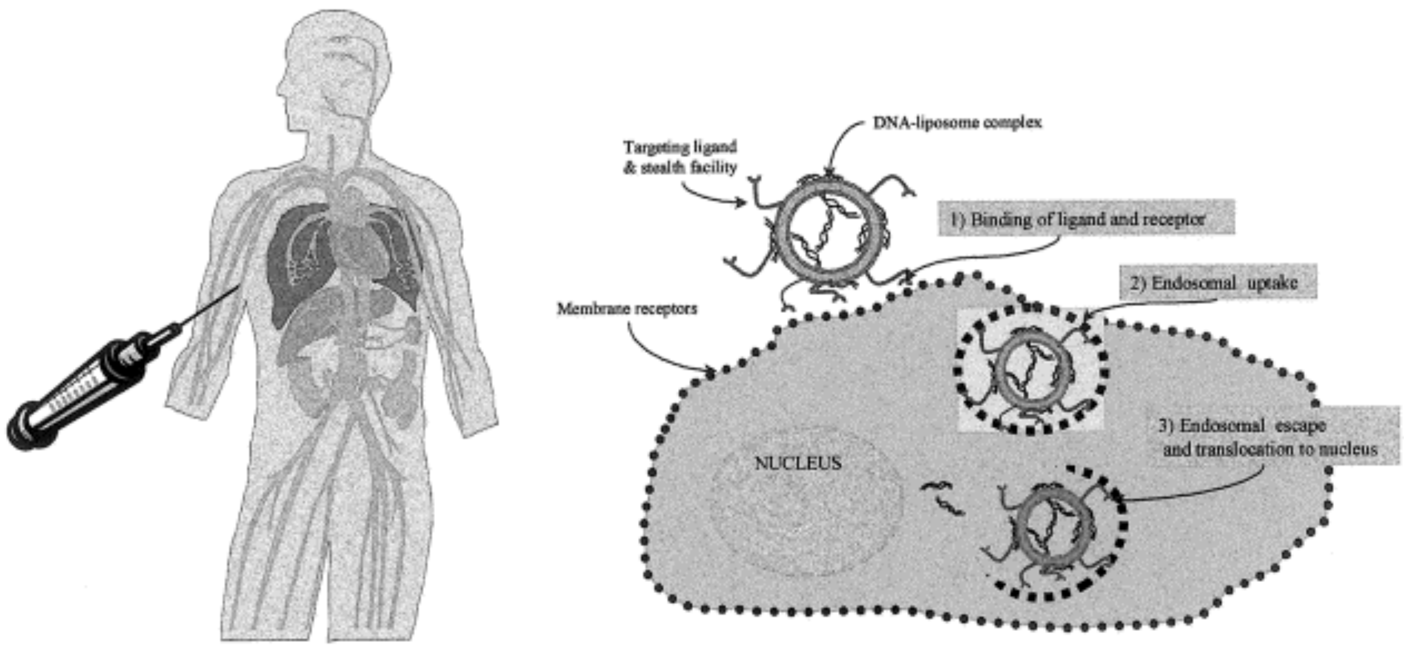
Gene Addition or Replacement

Gene Repair

Extra-chromosomal

From: Eric B. Kmiec, American Scientist, May-June 1999

Barriers in Gene Delivery



EXTRACELLULAR BARRIERS

Degradation of DNA in plasma
Uptake of DNA by reticuloendothelial system
Inability to target DNA to specific organs
Largely ineffective via the oral route – except for immunisation
Transfection inhibited by mucus

INTRACELLULAR BARRIERS

Endosomal escape of DNA
Lysosomal degradation of DNA
Cytoplasmic stability of DNA
Translocation of DNA to the nucleus

Understanding the barriers: Intracellular studies

$$x_{rms} = \sqrt{2Dt}$$

Efficient active transport of gene nanocarriers to the cell nucleus

3878–3882 | PNAS | April 1, 2003 | vol. 100 | no. 7

Junghae Suh*, Denis Wirtz†‡§, and Justin Hanes*†§

Departments of *Biomedical Engineering, †Chemical and Biomolecular Engineering, and ‡Materials Science and Engineering, Molecular Biophysics Program, The Johns Hopkins University, 3400 North Charles Street, Baltimore, MD 21218

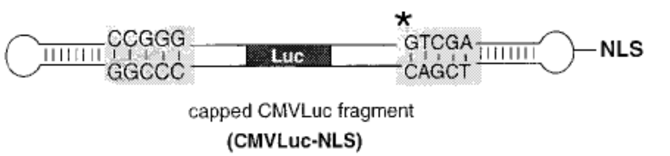


FIG. 1. Strategy for the preparation of a double-stranded DNA fragment coupled to an NLS peptide. A functional luciferase gene of 3,380 bp was cut out of pCMVLuc with *Xma*I and *Sal*I. Further digestion with *Xmn*I and *Bsp*HI cut the unwanted restriction fragment into small fragments (970, 875, 768, and 240 bp) that were removed by sucrose gradient centrifugation. The capped CMVLuc-NLS DNA was obtained by ligation of the ^{32}P-labeled (*) oligonucleotide-peptide and oligonucleotide-cap hairpins to the restriction fragment.

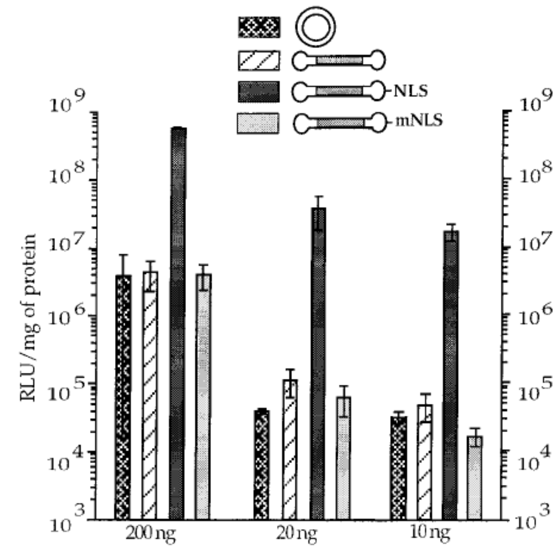


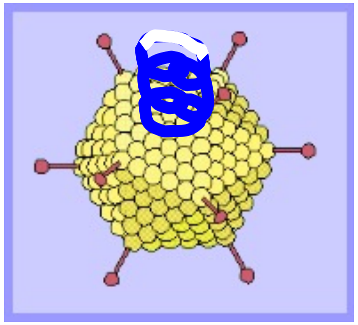
FIG. 5. Sustained luciferase expression levels are due to the nuclear localization peptide. HeLa cells were transfected with various amounts of DNA complexed to ExGen500 PEI (N/P = 5 in a 5% glucose solution) in the presence of 10% serum.

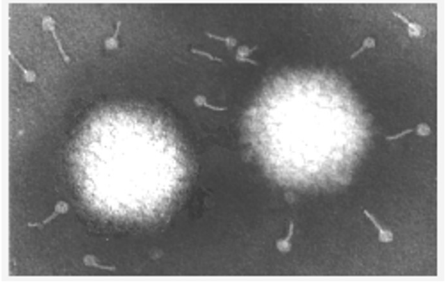
Carriers for putting genes inside cells?

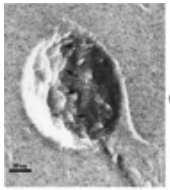
We need a carrier

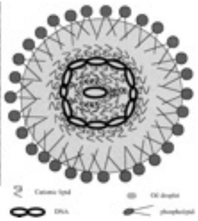
Virus

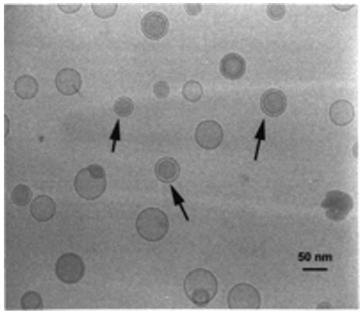
Polymers and Lipids (Non-viral)











The State of the Science: Viral & Non-viral

	Viral Vectors	Non-viral Vectors
Pros	1. High Transfection Efficiency 2. Natural Tropism (ability to infect different cells) 3. Evolved mechanisms for endosomal escape 4. Natural transportation mechanism of DNA into nucleus	1. Low immunogenicity 2. Can be made to be non-toxic 3. Easy to synthesize: quality control for mass production 4. Potentially targetable 5. No limit on plasmid size 6. No integration: Can be administered as drugs
Cons	1. Strong immune reactions against viral proteins prohibit multiple administrations 2. Possibility of chromosomal insertion and proto-oncogene activation 3. Complicated synthesis process 4. Limitation on gene size 5. Toxicity, contamination of live virus	1. Low transfection efficacy 2. Currently most vehicles are toxic at high doses 3. No natural tropism, endosomal escape or nuclear transport mechanisms

Betibeglogene autotemcel

- Beta thalassemia is caused by mutations to or deletions of the HBB gene leading to reduced or absent synthesis of the beta chains of hemoglobin that result in variable outcomes ranging from severe anemia to clinically asymptomatic individuals.
- LentiGlobin BB305 is a lentiviral vector which inserts a functioning version of the HBB gene into a recipient's blood-producing hematopoietic stem cells (HSC) **ex vivo**.
- The resulting engineered HSCs are then reintroduced to the recipient.
- Approved for medical use in the European Union in May 2019

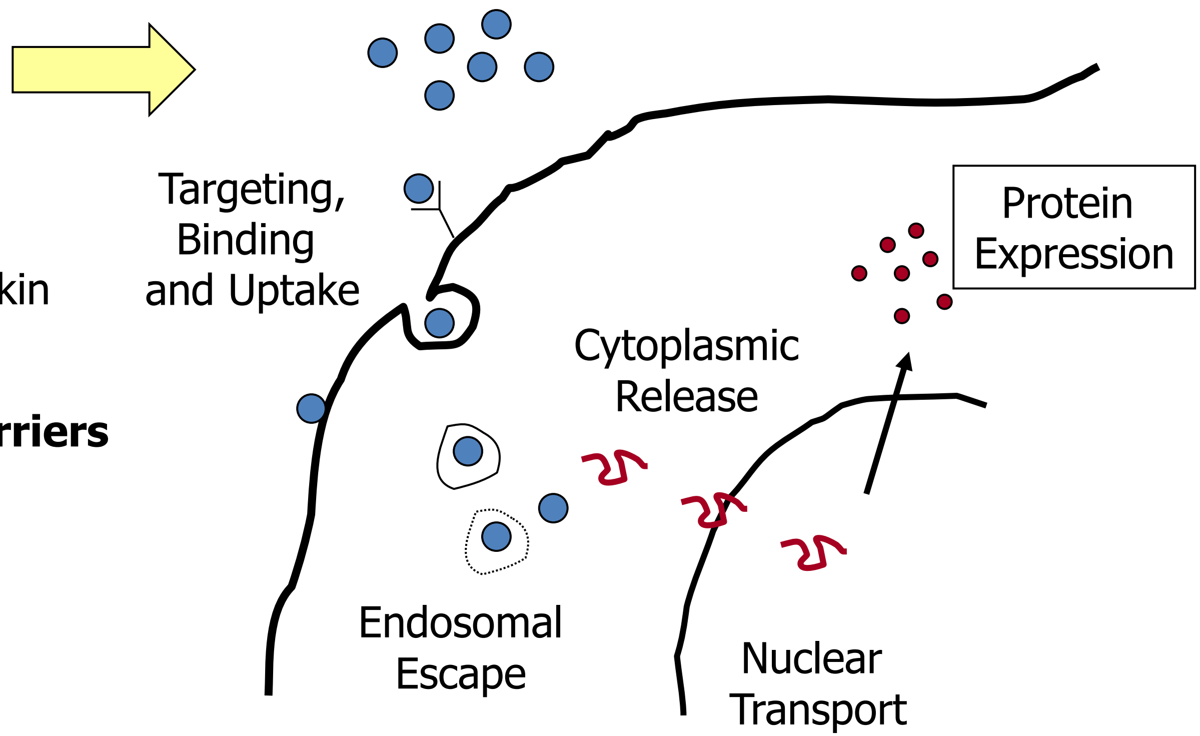
Gene Delivery: Where and How?

Macroscopic Delivery

Injections, Nasal spray, Oral, Skin

Engineering novel carriers

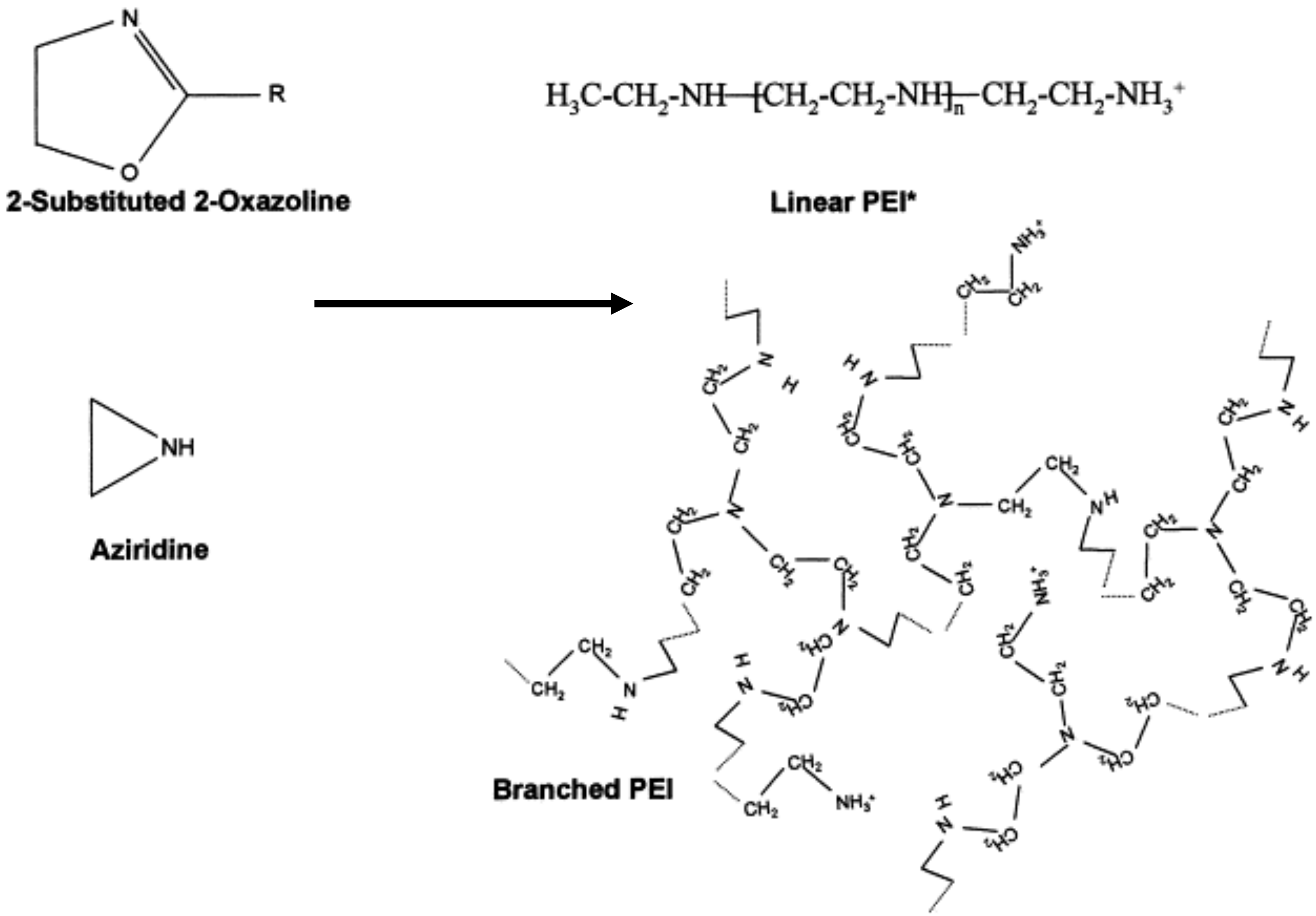
Microscopic Delivery



Polymers used in Gene therapy

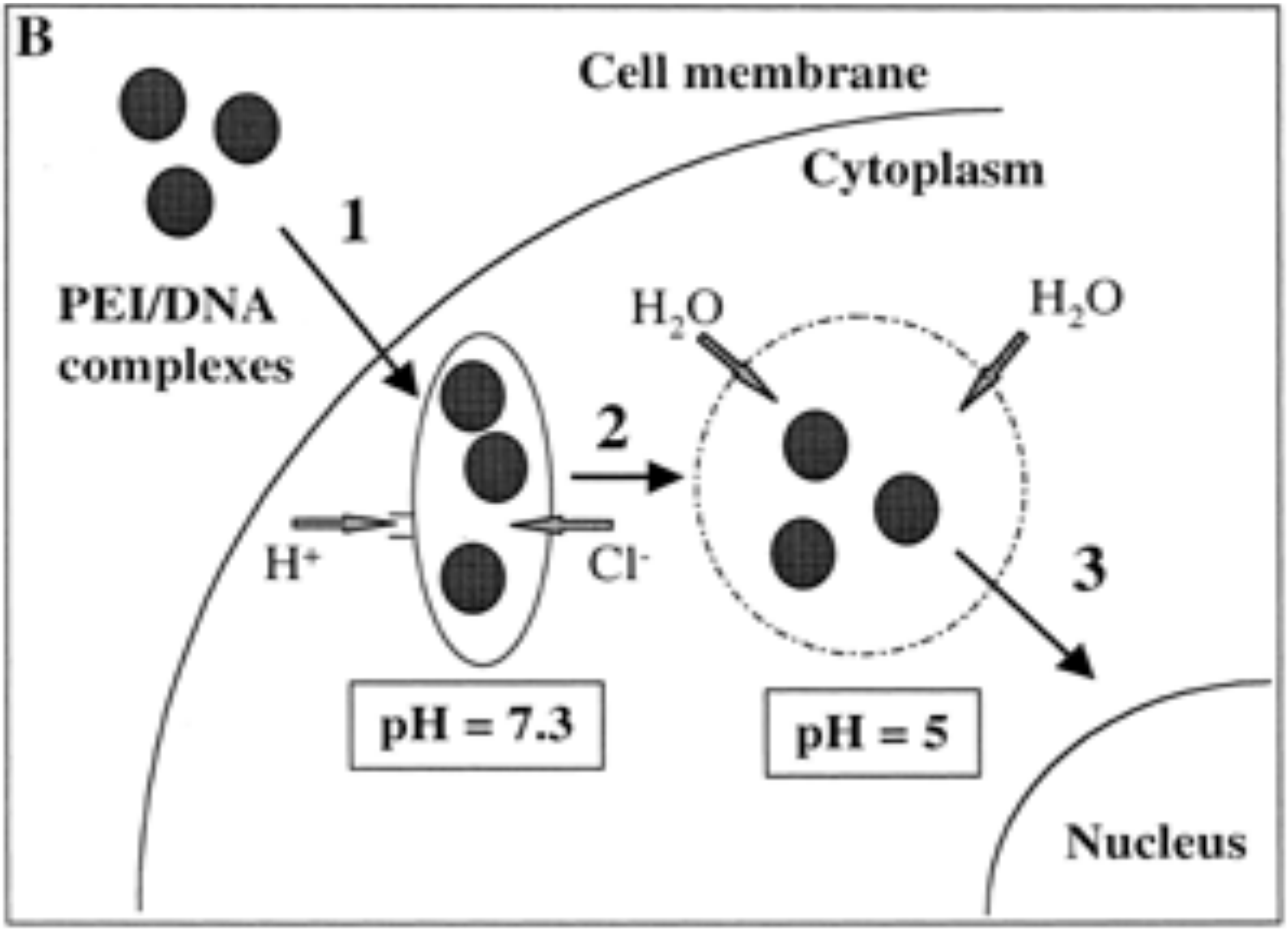
- Cationic polymers
  - Why Cationic? DNA itself is a highly negatively charged polymer
  - Can form complexes with DNA and create “packets” or “particles”
  - One can modify them to attach targeting ligands
  - Can be “eaten” (endocytosis) by cells → transfection
  - E.g: Poly(lysine), Poly(ethyleneimine), Chitosan, Polyamidoamine dendrimers
- Other polymers
  - Can slowly release genes (DNA) from polymer devices inside the body
  - Can be in the form of small particles or matrices or gels
  - E.g: Poly(lactide-co-glycolide) or PLGA, Polyvinylpyrrolidone (PVP) etc.

Polyethyleneimine as gene carrier



Cationic chain-growth polymerization process

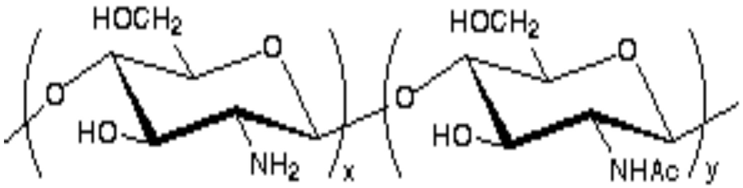
Polyethyleneimine as gene carrier



PEI can be toxic and hence care must be taken before use

Chitosan as a gene carrier

- Chitosan
  - Cationic bio-polymer that can complex with anionic DNA
  - Natural polysaccharide, **non-toxic**, non-immunogenic, biocompatible
  - Suitable for mucosal delivery (mucoadhesive)
- Complex coacervation
- Available amine groups on surface: Ligand attachment and potential for targeted delivery



General Structure of Chitosan

Vaccine

- Antigen: Molecule against which the immune response needs to be generated
  - Mostly protein based
- Adjuvant: Molecules that stimulate immune response to the target antigen
  - Amplifies the response
  - Can be used to direct the response towards a certain type: Cellular vs Humoral
  - Several different types and function

Table 1
Description of Toll-like receptor (TLR) ligands

TLR-L number	Ligand	Description (Source)
1	Pam_3Cys	Synthetic lipoprotein
2	LTA	Lipoteichoic acid (*St. aureus*)
	PGN	Peptidoglycan (*St. aureus, B. subtilis, E. coli*)
	HKML	Heat Killed *L. monocytogenes* (Gram+)
	LPS	Lipopolysaccharide (E. coli)
3	dsRNA	Oligonucleotide
	PolyI:C	Synthetic analog of dsRNA
4	LPS	Lipopolysaccharide (*E. coli*)
5	Flagellin	*S. thyphimurium*
6	PGN	Peptidoglycan (*St. aureus, B. subtilis, E. coli*)
7	ssRNA	Oligonuleotide with repeating G and U motives
	Loxoribine	Guanosine analogue
	Imiquimod and Resiquimod	Small synthetic antiviral imidazoqionolines
8	Imiquimod and Resiquimod	Small synthetic antiviral imidazoqionolines
9	dsDNA	Oligodeoxynuleotide with repeating C and G motives (CpG)
10	Unknown	Unknown
11	Unknown	Unknown

Vaccine type by time of delivery

- Prophylactic vaccine
  - Given prior to the disease
  - Creates memory for the immune system
  - Widely used
  - Example: Polio
  - Rabies
    - Post exposure prophylaxis
- Therapeutic vaccine
  - Given after the disease
  - Purpose is to strengthen the immune system to fight back against the active disease
  - Still in nascent phase
  - Example: Cancer vaccines

Vaccine type by content

- Live-attenuated vaccines
  - Very strong immune response
  - Storage conditions should be stringent
  - Risky
  - Example: Smallpox, Measles
- Inactivated vaccines
  - Killed germ, example: β-propiolactone used in Covaxin
  - Lower response than live vaccines: may need several doses
  - Example: HepA, Flu, Polio
- Subunit, recombinant, polysaccharide, and conjugate vaccines
  - Can be given to anyone
  - Needs booster shots
  - Example: HPV, HepB
- Toxoid vaccines
  - Use toxin
  - Create immunity against the toxin and not the germ
  - Example: Diptheria, Tetanus

Delivering Vaccine Antigens to APCs: Particle-based Approaches

- Microparticles: Why?
  - DCs and macrophages take up particles in large quantities
- Co-delivery of adjuvants
  - Possible to co-deliver the antigen and adjuvant for better efficacy
- Protein, peptide and DNA antigens
  - Can carry all types of antigens
- Formulation issues
  - Do suffer from effect of synthesis and loading process on antigen and adjuvant structures and activity

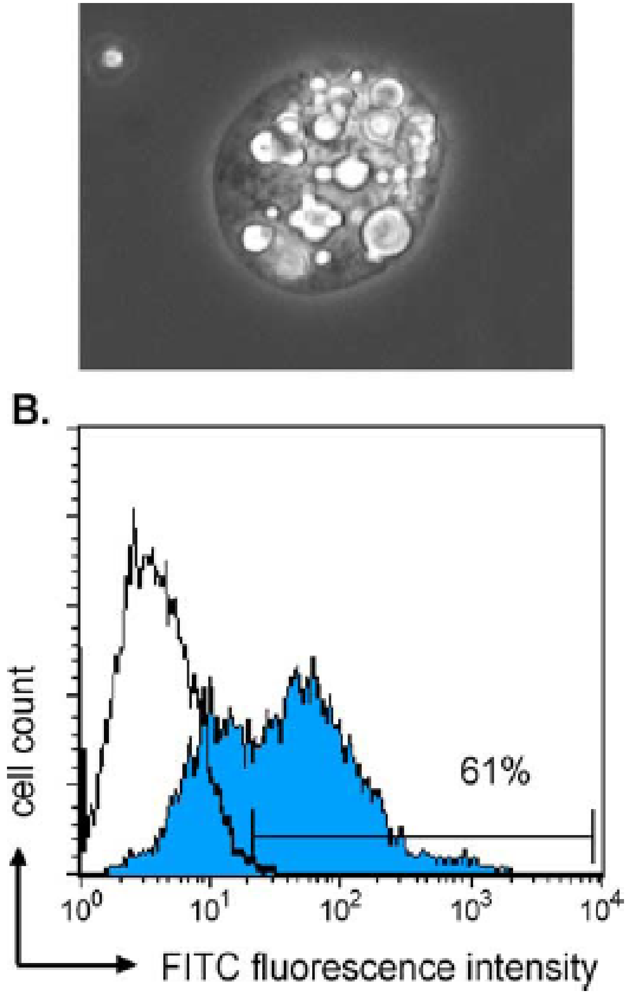
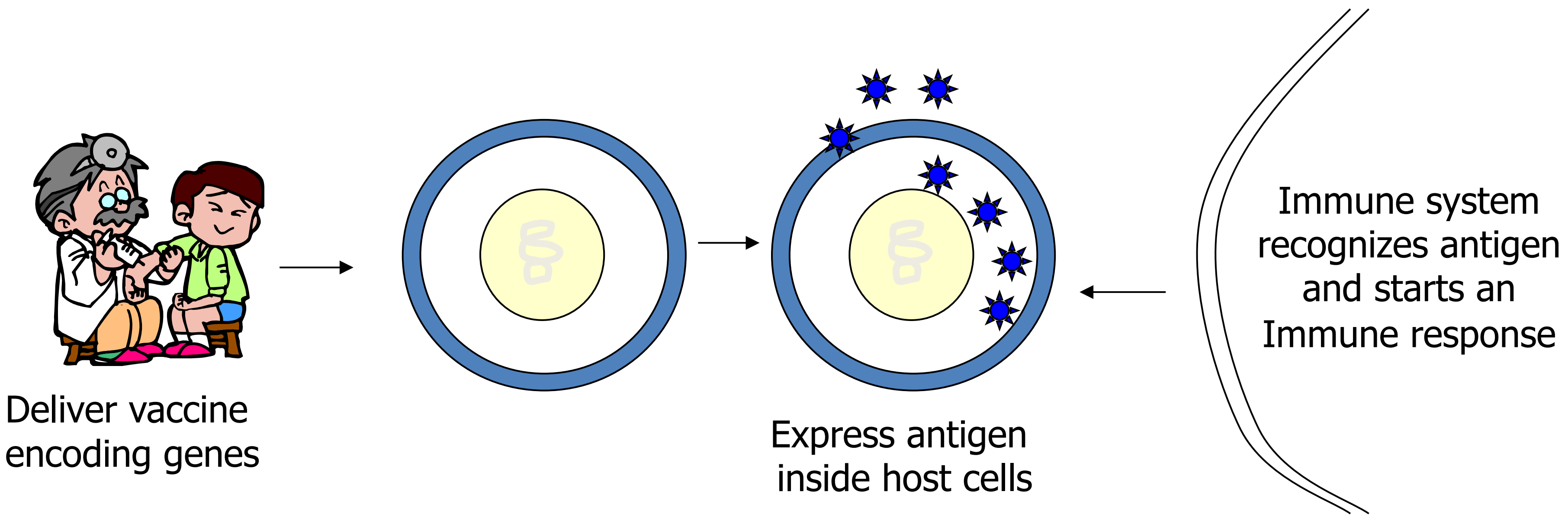


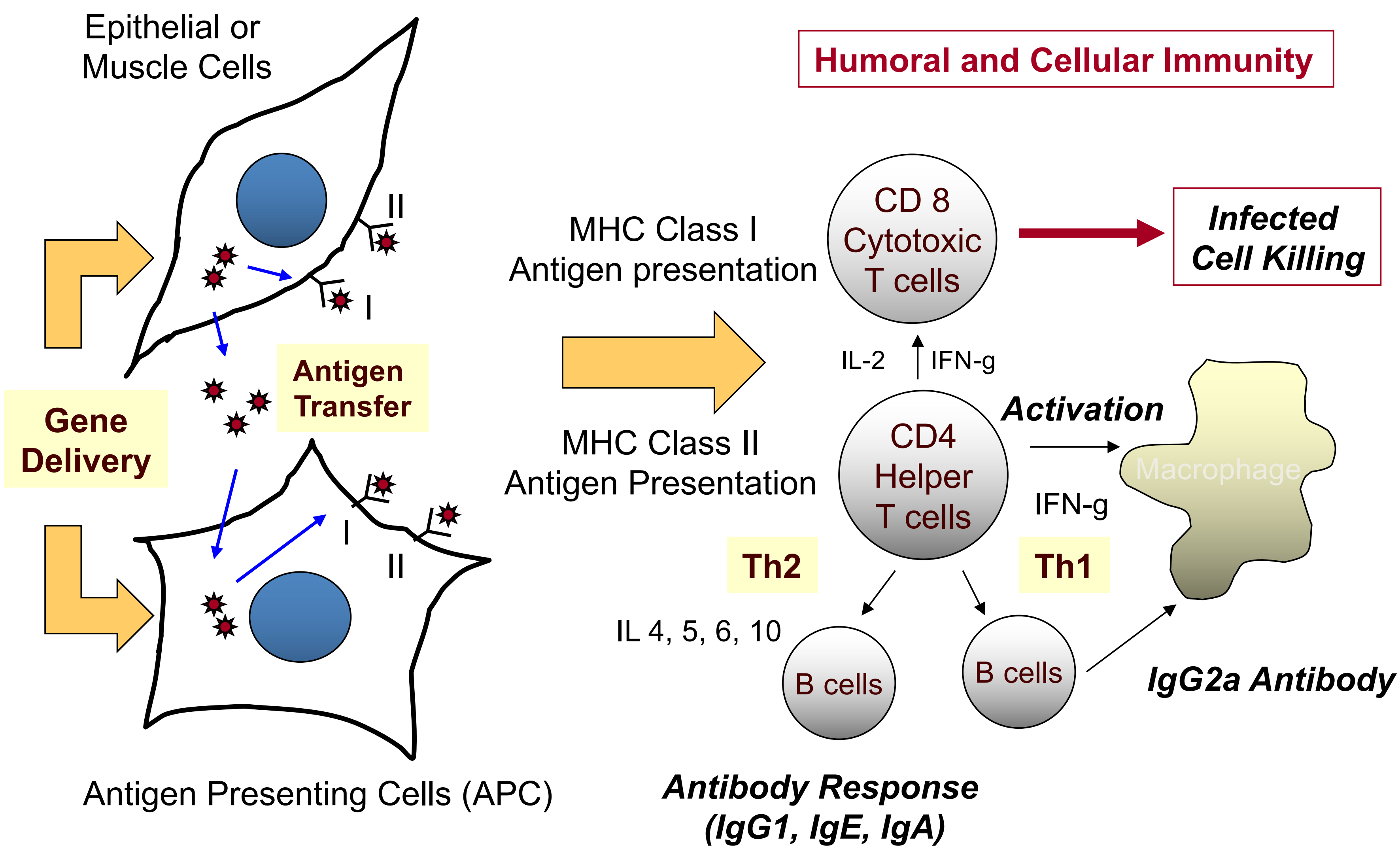
Fig. 1. Phagocytosis of fluorescent microspheres (MS) by human MoDC. Immature hMoDC were incubated with FITC containing MS for 18 h at 37 °C. Cells were then collected, washed and examined by phase contrast microscopy. (A) One immature hMoDC containing many MS is shown at magnification 40×. (B) Quantitative uptake of fluorescent MS by immature hMoDC is examined by flow cytometry. The number within the histogram plot represents the percentage of MS-FITC fluorescent hMoDC based on the whole pregated hMoDC population.

Genes as Vaccines: How?

- Conventional vaccines: Replication deficient bacteria, recombinant proteins (antigens) etc.
  - The antigen (immunogenic component of the pathogen) is processed by the hosts immune cells which generates a protective response
- DNA vaccines: Genes encoding for specific antigenic proteins of interest
  - Instead of delivering the pathogen or recombinant protein, the host cells themselves produce the antigen



DNA Vaccination: The Players



Cationic microparticles for DNA vaccines

- PLGA microparticles prepared using w/o/w or o/w emulsion techniques in the presence of a cationic surfactant
- Produces cationic surface on which plasmid DNA can be efficiently adsorbed
- Could have better "bioavailability" of the DNA compared to encapsulated plasmids
- Increases the efficacy of DNA to target Antigen presenting cells and enhances immune response

Table 1. Cationic microparticles with adsorbed DNA: Particle size, net surface charge, loading efficiency, and DNA loading levels based on hydrolysis of PLG-DNA formulation

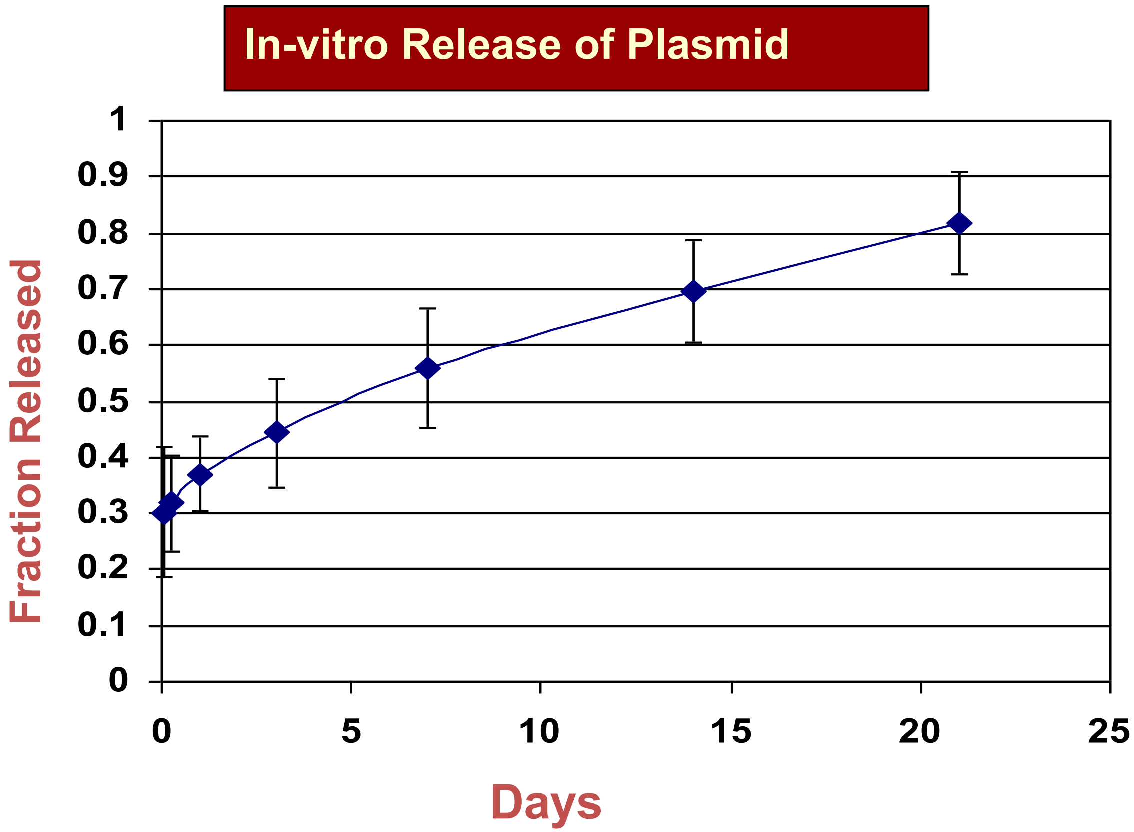
Formulation	Mean size, μm	Zeta potential, mV	Loading level, % wt/wt	Mean loading efficiency, %
PLG/CTAB-p55 DNA	1.54 ± 0.20	+36 ± 6	0.92 ± 0.12	92.0
PLG/DDA-p55 DNA	2.20 ± 0.12	+30 ± 10	0.68 ± 0.08	68.0
PLG/DOTAP-p55 DNA	0.98 ± 0.11	+24 ± 8	0.62 ± 0.14	62.0

PLG miroparticles prepared with polyvinyl alcohol as a particle stabilizer, but without cationic surfactants have a zeta potential of -13 ± 3 mV. Values are mean ± SE of triplicate measurements.

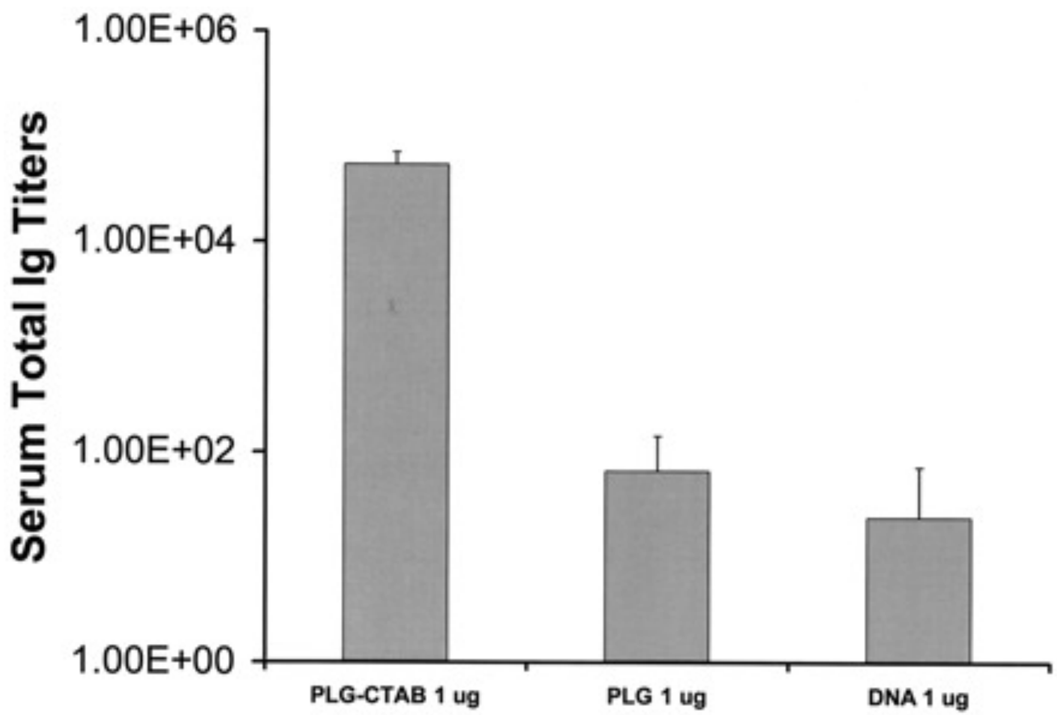
Developed by: Chiron Corporation (NOVARTIS), Emeryville, CA

Characterization: DNA Release Profile

In-vitro release of plasmid DNA from PLGA microspheres:
Spheres were suspended in PBS/EDTA buffer and incubated at 37^{o}C (end over end tumbling) for various time intervals. DNA concentrations were measured in the release buffer following centrifugation of the microspheres



Immune response data

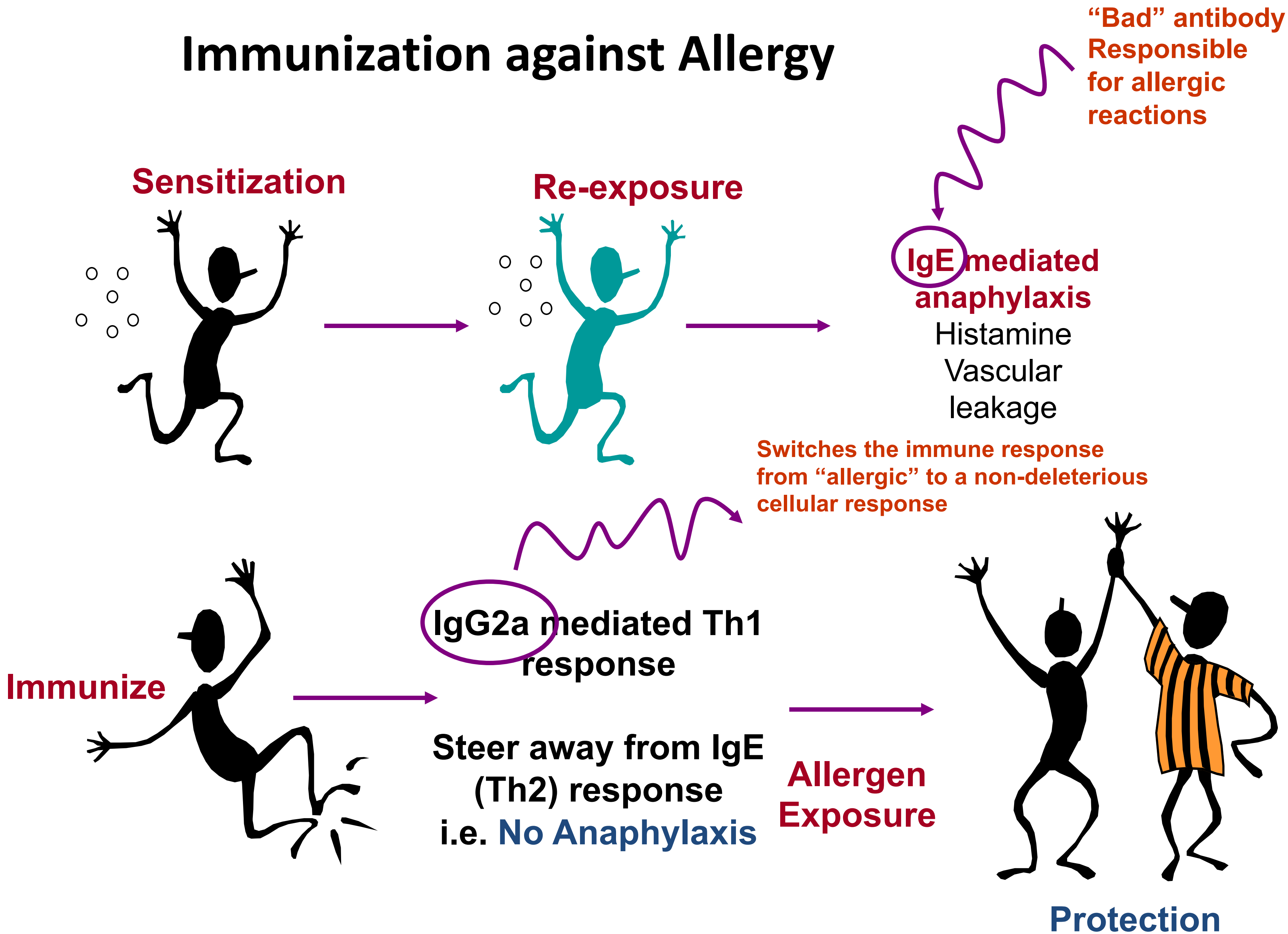


Serum IgG titers in groups (n = 10) of BALB/c mice immunized with either PLG/CTAB-p55 gag DNA, blank PLG microparticles + DNA, or DNA alone at 1 µg dose. Antibody titers are geometric mean titers ± SE at 2 weeks post-second immunization (week 6) time point. The response from the PLG/CTAB-p55 DNA was significantly higher ($P < 0.05$) than other groups.

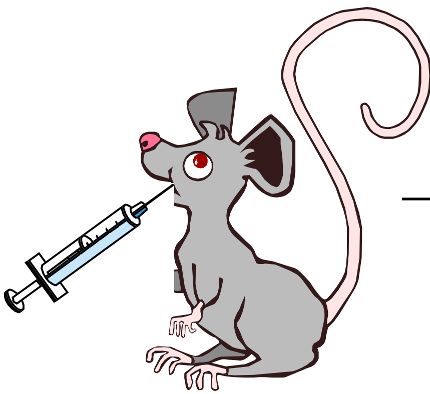
Immunization against food allergy

- Peanut Allergy:
  - Anaphylactic reactions mediated by Th2 cytokines (IgE and IgG1)
  - Potentially fatal or near-fatal especially in children
- Allergen immunization:
  - Potential preventive therapy if proven safe and effective
    - Protein administration can cause deleterious anaphylaxis
    - Need for novel immunization method that does not induce immuno-pathology
- Oral immunization:
  - High patient compliancy, ease of administration for mass vaccination
  - Generation of mucosal immunity against infectious agent

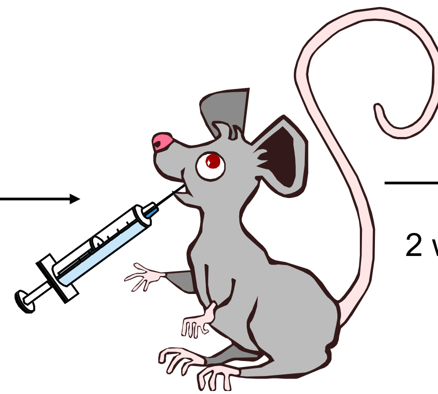
Immunization against Allergy



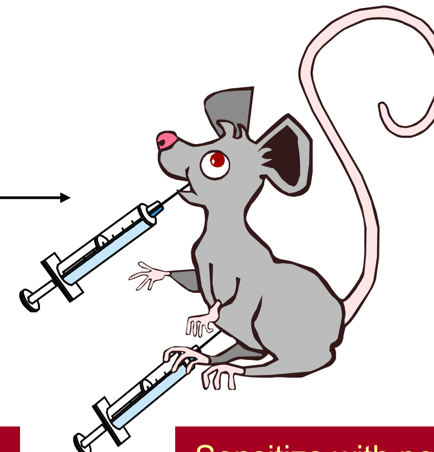
Oral DNA Delivery with Chitosan Nanoparticles (Roy et al., Nature Medicine, April 1999)



2 weeks



2 weeks



Immunize with Arah2 encoding plasmid

Oral feeding of 50 μg plasmid
in nanospheres or as naked DNA

Boost with Arah2 encoding plasmid

Booster groups fed
with same dose two
weeks following first
immunization

Sensitize with peanut extract

Oral (1mg) and i.p. (500 μg) sensitization
with crude peanut extract four weeks after
first immunization followed by three more
weekly sensitization.

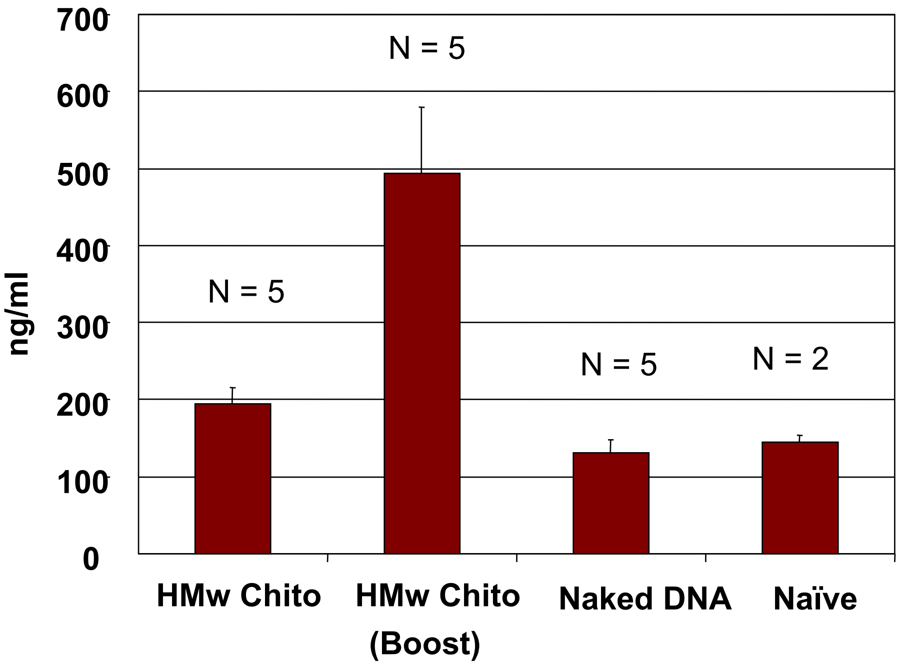
Susceptible strains of mice
that show anaphylactic
reactions to peanut

Challenge

Challenge with Arah2 protein

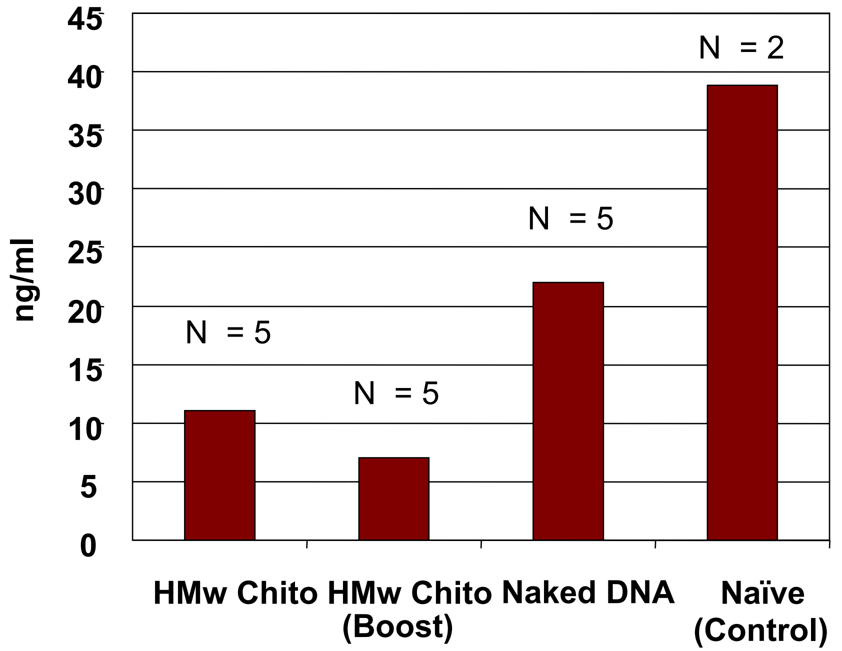
Assays: Protection against antigen challenge
Secreted IgA
Serum IgG2a, IgE
Plasma Histamine, Vascular Leakage

Chitosan-based oral DNA immunization switches the immune response from Th2 to Th1



Serum anti-Arah2 IgG2a response four weeks after first immunization (before sensitization)

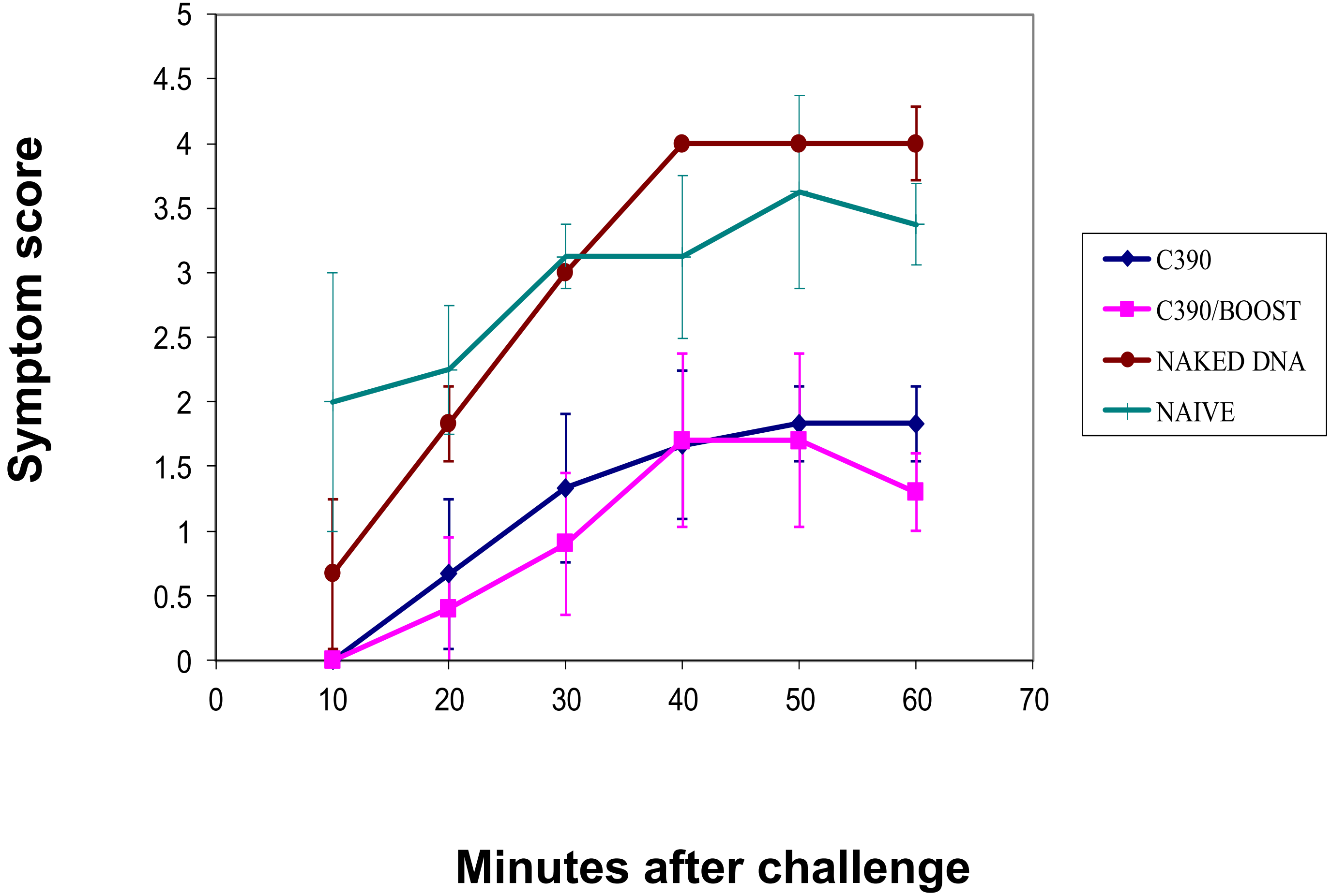
Therapeutic Th1 response



Increase in serum anti-Arah2 IgE levels after three sensitizations

Anaphylactic Th2 response

Chitosan-based oral DNA immunization protects mice from Arah2 induced anaphylaxis



0: No sign of reaction

1: scratching and rubbing around the nose and head

2: decreased activity with an increasing respiratory rate, and/or puffing around the eyes

3: labored respirations, cyanosis around the mouth and tail

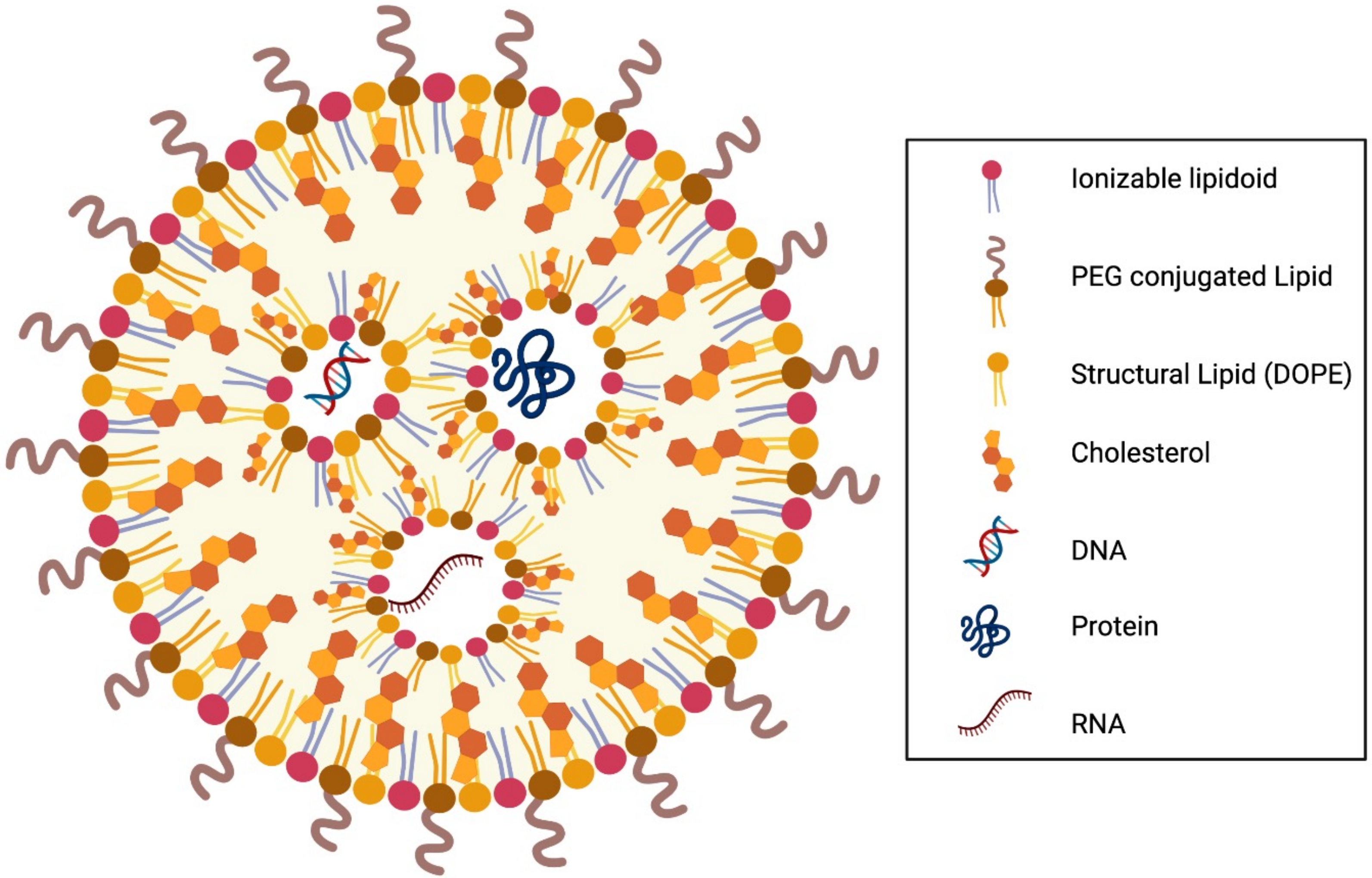
4: slight or no activity after prodding or tremors, convulsion

5: death

mRNA delivery

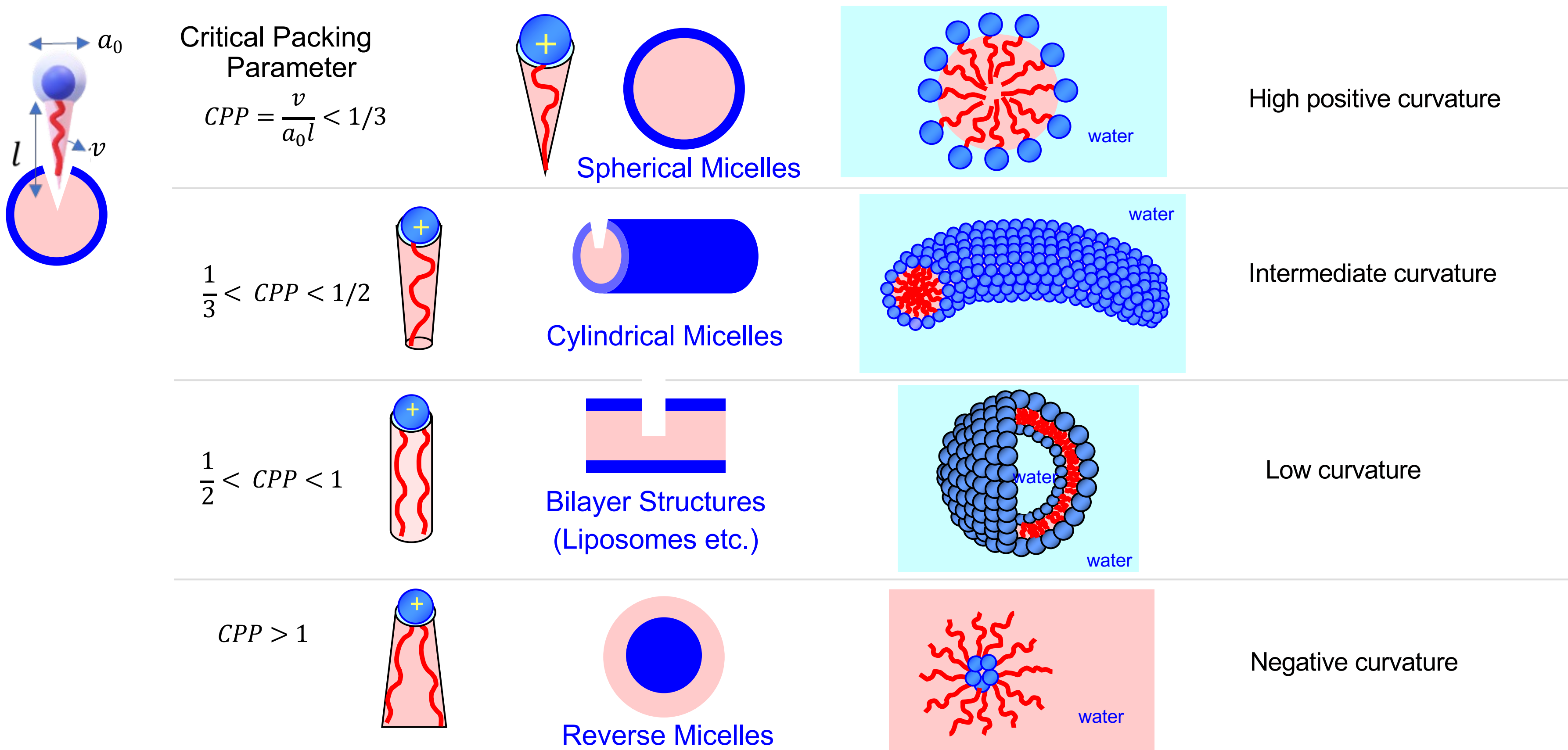
- Since gene delivery is challenging due to limitations of DNA diffusion in cell and translocation in the nucleus, mRNA delivery can overcome these barriers
- mRNA is a lot less stable than DNA
- Several discoveries (including 2023 Nobel prize in Medicine) related to mRNA modifications led to increased stability, less immunogenicity and efficient protein production!
- mRNA can be formulated with nanocarriers to increase their uptake and cytoplasmic expression and protect them from premature degradation
- A disadvantage is that the expression will be transient and would need repeated administration for sustained production
- Lipid nanoparticles have emerged as the delivery carrier of choice

Lipid nanoparticles



Iyer et al. Clinical Microbiology Reviews (2021)

Self-Assembly of Lipids and Molecular Geometry

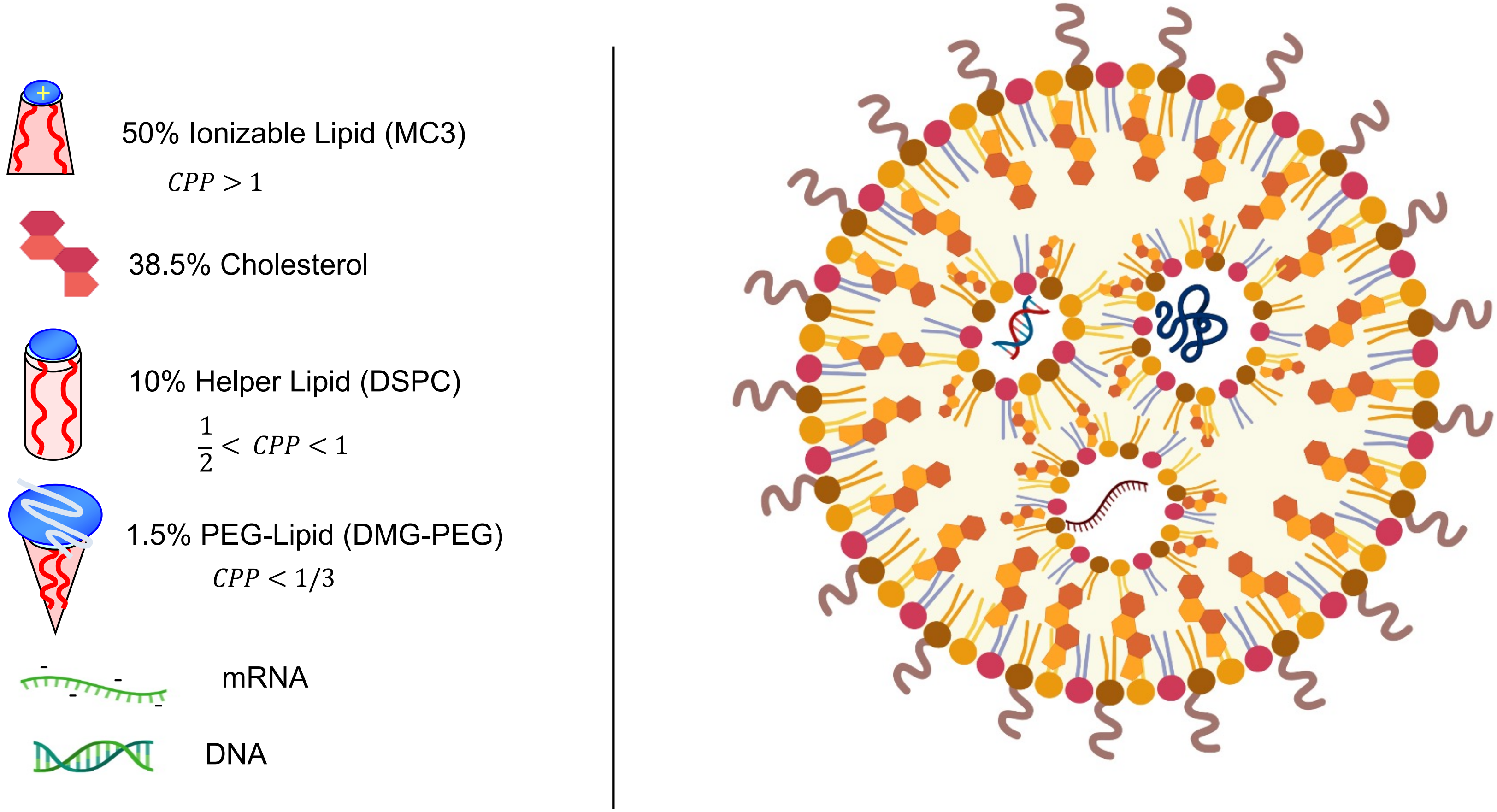


v = tail volume

a_o = head area

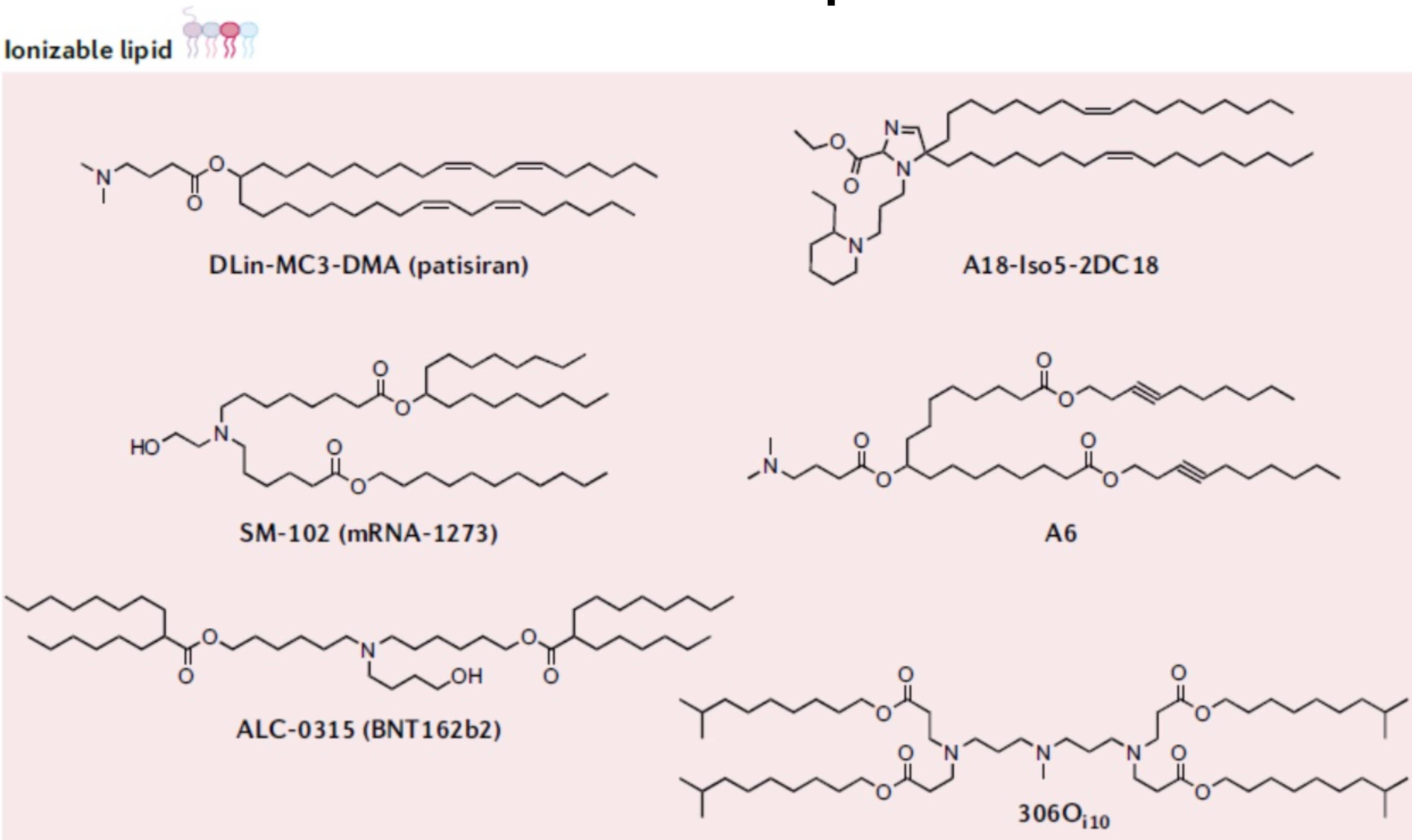
l = tail length

Structure of LNPs – Role of components



Ionizable lipids

Ionizable lipid



[*Chaudhary et al. Nature Reviews Drug Discovery*](#) (2021)

Ionizable lipids

- Trapped in endosomes, in which the pH is lower than in the extracellular environment, ionizable lipids are protonated and, therefore, become positively charged, which may promote membrane destabilization and facilitate endosomal escape of the nanoparticles
- DLin- MC3- DMA; MC3 is a key delivery component of Onpattro (siRNA for polyneuropathies induced by hereditary transthyretin amyloidosis), the first United States Food and Drug Administration (FDA)- approved siRNA drug in 2018
- Incorporation of biodegradable lipids improves the tolerability of lipid nanoparticles, by allowing fast metabolism while retaining mRNA delivery efficacy. The biodegradability of lipids can be increased by introducing ester motifs, which shows good delivery efficacy and fast elimination from the liver and plasma in vivo

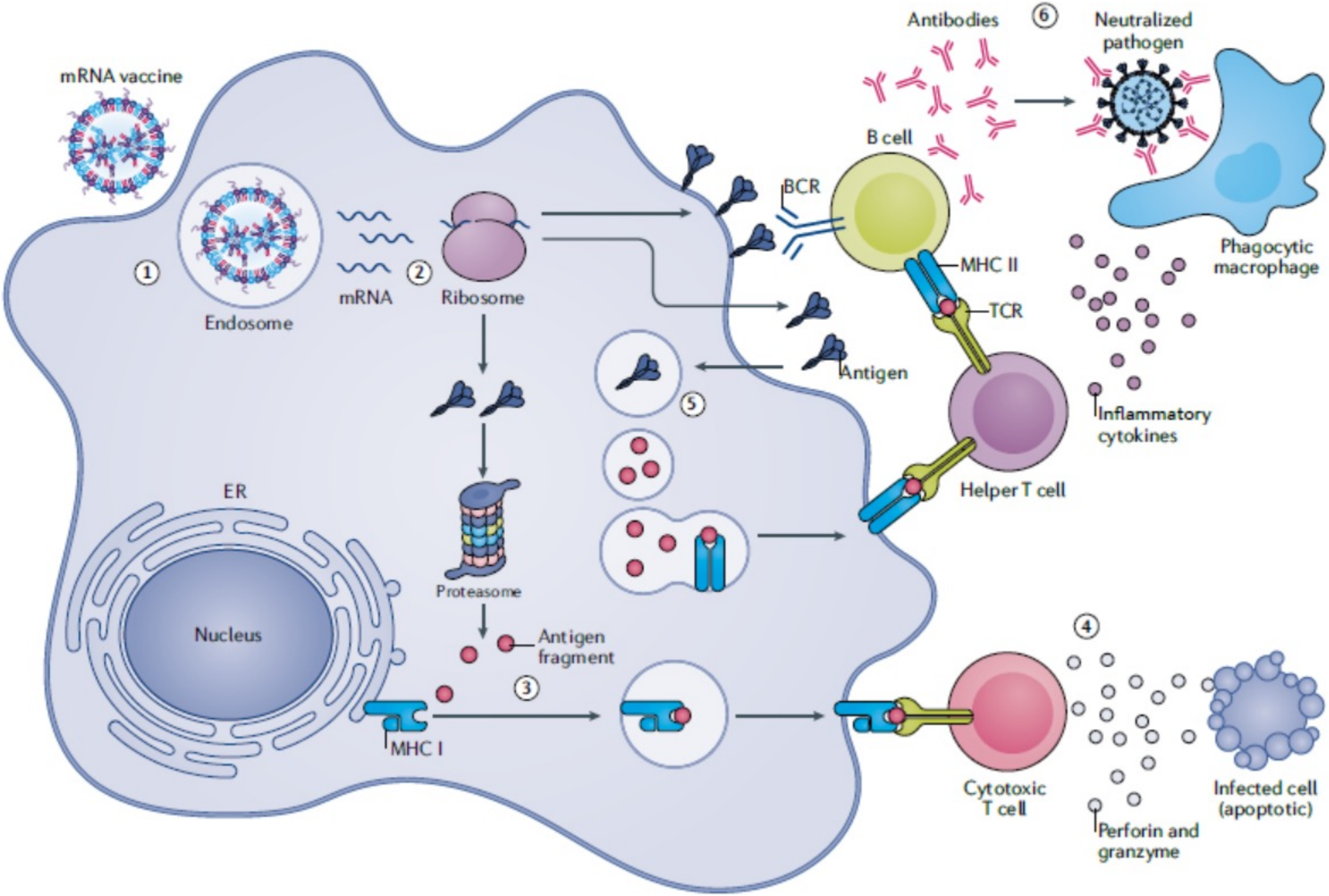
https://doi.org/10.1038/s41578-021-00358-0

Table 2 | **Representative clinical trials of lipid nanoparticle–mRNA therapeutics against infections, cancer and genetic disorders**

Name	Disease	Encoded protein	Administration route	ClinicalTrials.gov identifier	Phase
Infections					
mRNA-1944	Chikungunya virus	Antibody against chikungunya virus	i.v.	NCT03829384	I
Cancer					
mRNA 2416	Solid tumours	OX40L	Intratumour	NCT03323398	II
mRNA-2752	Solid tumours	OX40L, IL-23 and IL-36γ	Intratumour	NCT03739931	I
MEDI1191	Solid tumours	IL-12	Intratumour	NCT03946800	I
SAR441000	Solid tumours	IL-12sc, IL-15sushi, IFNα and GM-CSF	Intratumour	NCT03871348	I
Genetic disorders					
mRNA-3704	Methylmalonic acidaemia	Methylmalonyl-CoA mutase	i.v.	NCT03810690	I/II
mRNA-3927	Propionic acidaemia	Propionyl-CoA carboxylase	i.v.	NCT04159103	I/II
MRT5201	Ornithine transcarbamylase deficiency	Ornithine transcarbamylase	i.v.	NCT03767270	I/II
MRT5005	Cystic fibrosis	Cystic fibrosis transmembrane conductance regulator	Inhalation	NCT03375047	I/II
NTLA-2001	Transthyretin amyloidosis with polyneuropathy	CRISPR–Cas9 gene editing system	i.v.	NCT04601051	I

CoA, coenzyme A; CRISPR–Cas9, clustered regularly interspaced short palindromic repeats (CRISPR)–CRISPR-associated protein 9; GM-CSF, granulocyte–macrophage colony-stimulating factor; IFN, interferon; IL, interleukin; i.v., intravenous.

https://doi.org/10.1038/s41578-021-00358-0



Chaudhary et al. Nature Reviews Drug Discovery (2021)

Cancer Vaccines

Cancer

- Uncontrolled growth of cells
- Cancerous cells routinely arise in our body
- They are caused by mutations in some gene and hence express antigens that immune system can detect
- Immune system destroys them almost all the times
- Tumors are formed when immune system fails to destroy them

Cancer Prophylactic Vaccine

- Some cancer are caused by oncoviruses (e.g. cervical cancer)
- Vaccines against these viruses prevent such cancers
  - Hepatitis B vaccine to prevent liver cancer
  - Human papillomavirus vaccine that is responsible for ~70% of cervical cancers

Cancer Therapeutic Vaccine

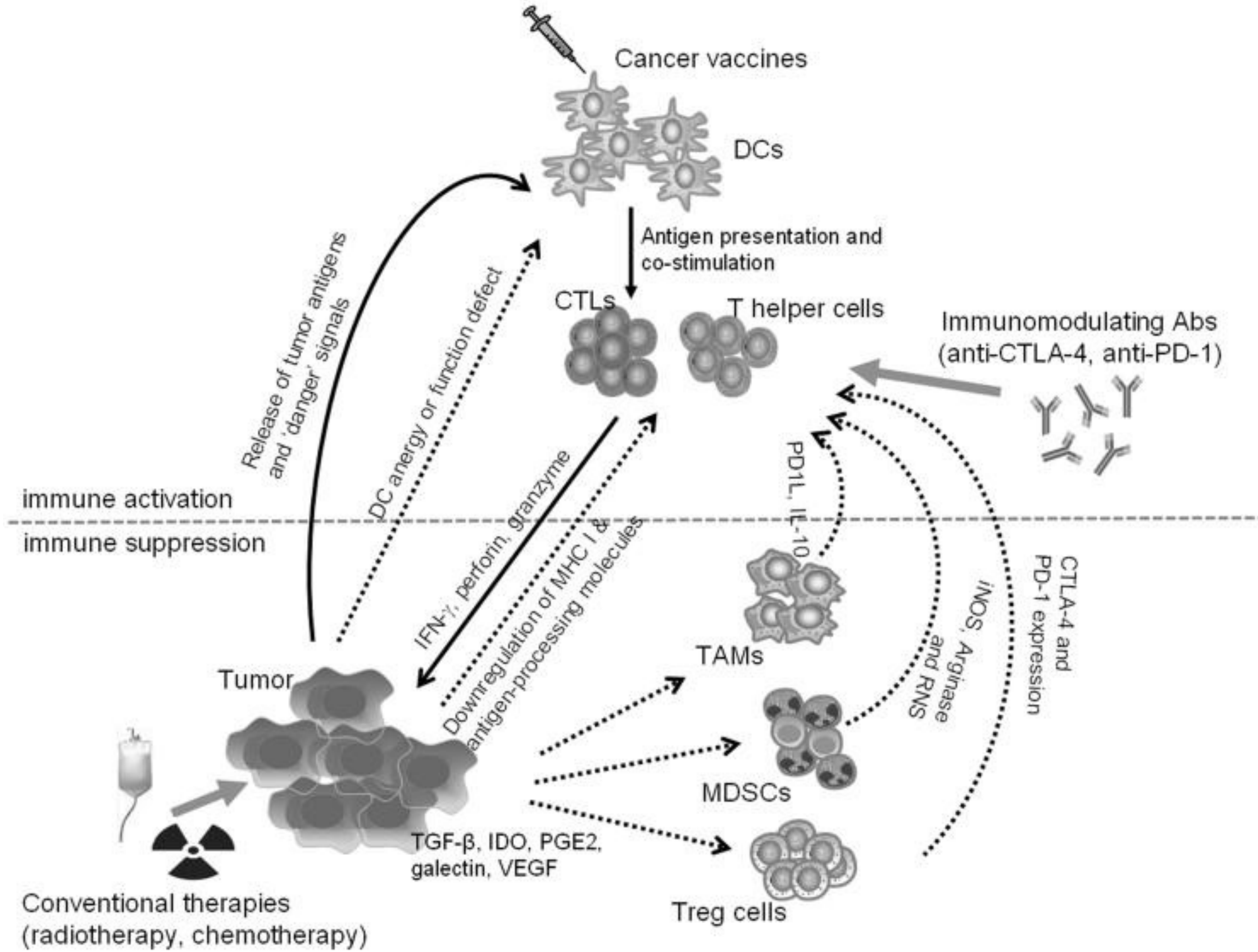
- Goal is to increase immune response against tumor antigens
- Hard to generate strong immune response due to similarity to the host tissue
- Tumors typically also have higher mutation rate and may develop several variants of tumor antigen

Therapeutic vaccines

- Autologous tumor cell vaccines
  - Isolate cells, lyse and inject with adjuvants
  - Presents entire spectrum of antigens
  - Transduction of stimulatory proteins such as IL-12, GM-CSF
  - Risk of autoimmunity
  - Need lots of tumor sample and may involve surgery

Potential tumour immune evasion mechanisms

- Physical barriers imposed by tumour architecture
- Induction of host myeloid and lymphoid suppressor cells
- Blocking T cell effector function
  - Expression of proteases that interfere with cytotoxicity
  - Expression of FAS-L
  - Downregulation of cytokine receptors
  - Resistance to apoptosis
- Evasion of T cell recognition
  - Loss or mutation in target antigen
  - Downregulation of MHC and other components important for antigen processing



Adv Cancer Res. 2013; 119: 421–475

Desirable properties of cancer vaccines

- **Targeting** antigen(s) preferentially or exclusively expressed on tumours but not on healthy tissue
- **Safety** — Delivery systems that are non-toxic to humans
- **Quantity** — Delivery systems capable of inducing sufficient numbers of effector T cells
- **Quality** — Delivery systems capable of inducing T cells of sufficient quality
  - Express T cell receptors with high avidity for target antigen
  - Have desired effector function (cytotoxicity, cytokine secretion)
  - Traffick to tumour sites
  - Duration of response sufficient to mediate tumour regression

Therapeutic vaccines

- Allogenic tumor cell vaccines
  - Uses few established human tumor cell lines
  - Limitless source and cost effective
  - Risk of autoimmunity
  - May not be effective at all

Sipuleucel-T (Provenge)

- Used clinically for prostate cancer since 2010
- Patients DCs are isolated
- Incubated with fusion protein (PA2024)
  - The antigen prostatic acid phosphatase (PAP), which is present in 95% of prostate cancer cells and
  - An immune signaling factor granulocyte-macrophage colony stimulating factor (GM-CSF) that helps the APCs to mature
- Injected back in the patients
- Improves life by 3 years

Stimuli-sensitive or Responsive Delivery Systems

- Magnetically controlled release systems
- Ultrasonically controlled systems
- Temperature sensitive hydrogels
- pH sensitive hydrogels
- Ligand (such as glucose) sensitive hydrogels
- Light sensitive hydrogels
- Other stimuli
  - Specific ion-sensitive hydrogels
  - Specific antigen-responsive hydrogels

Environmentally-Responsive/Smart Systems

pH-responsive systems for Intracellular Drug Delivery

Polymer design inspired by the mode of action of viruses

pH-sensitive polymers, such as poly(propylacrylic acid), PPAA, that become hydrophobic at the endosomal pHs and disrupt the endosomal membrane.

Endosomal pH: 5.5-6.5

In another design, membrane-disruptive polymers have acid-cleavable bonds that degrade within the endosome, exposing the backbone which is membrane-disruptive, leading to the release of the drug to the cytosol

Environmentally-Responsive/Smart Systems

pH-Responsive Enteric Coatings for oral delivery

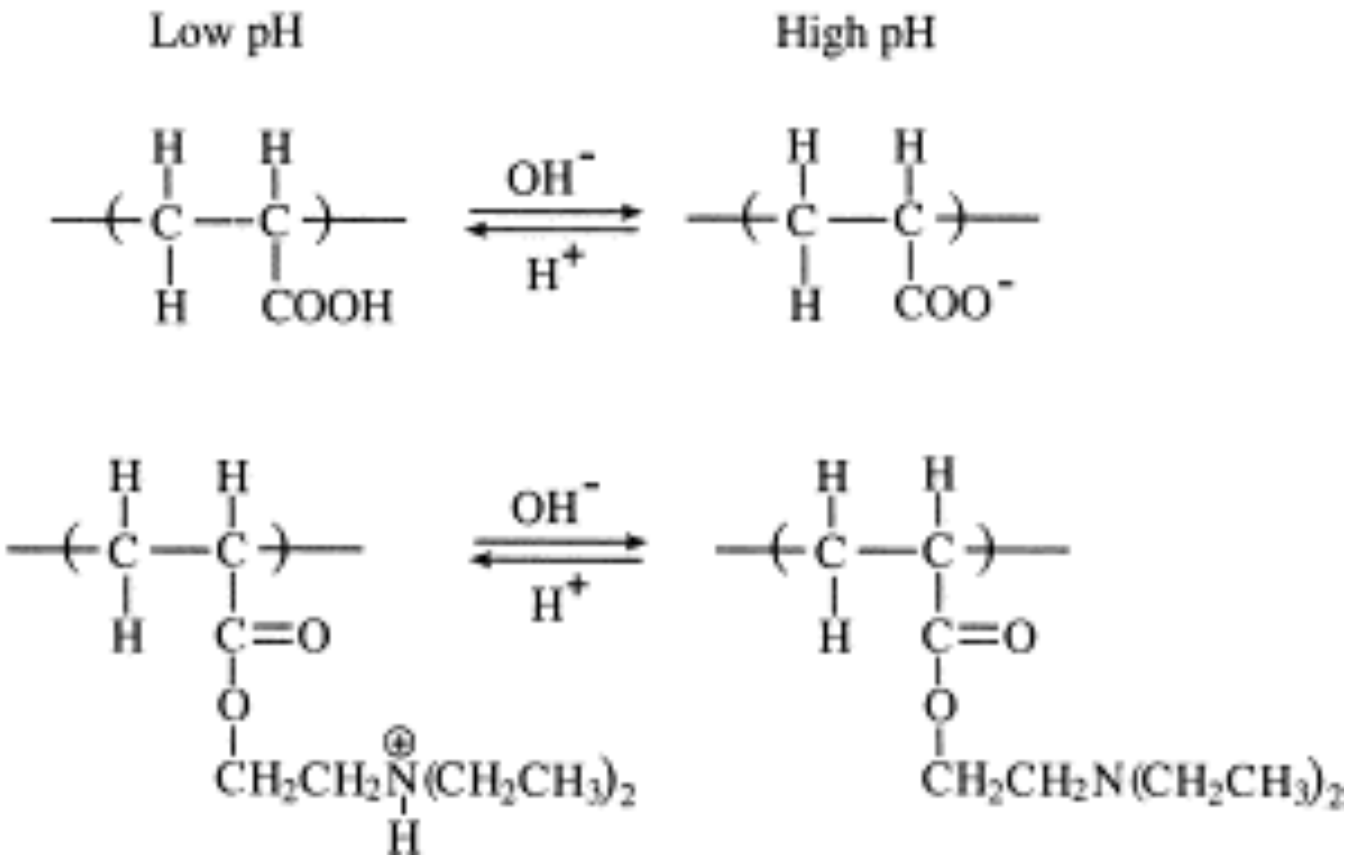
An enteric coating is a polymer that is insoluble at a low pH and soluble at physiologic pH.

The most common applications:

(a) to protect acid-sensitive drugs from gastric juices,
(b) to prevent the absorption of certain drugs in the stomach, and
(c) to deliver drugs to the small intestine if that is their prime absorption site.

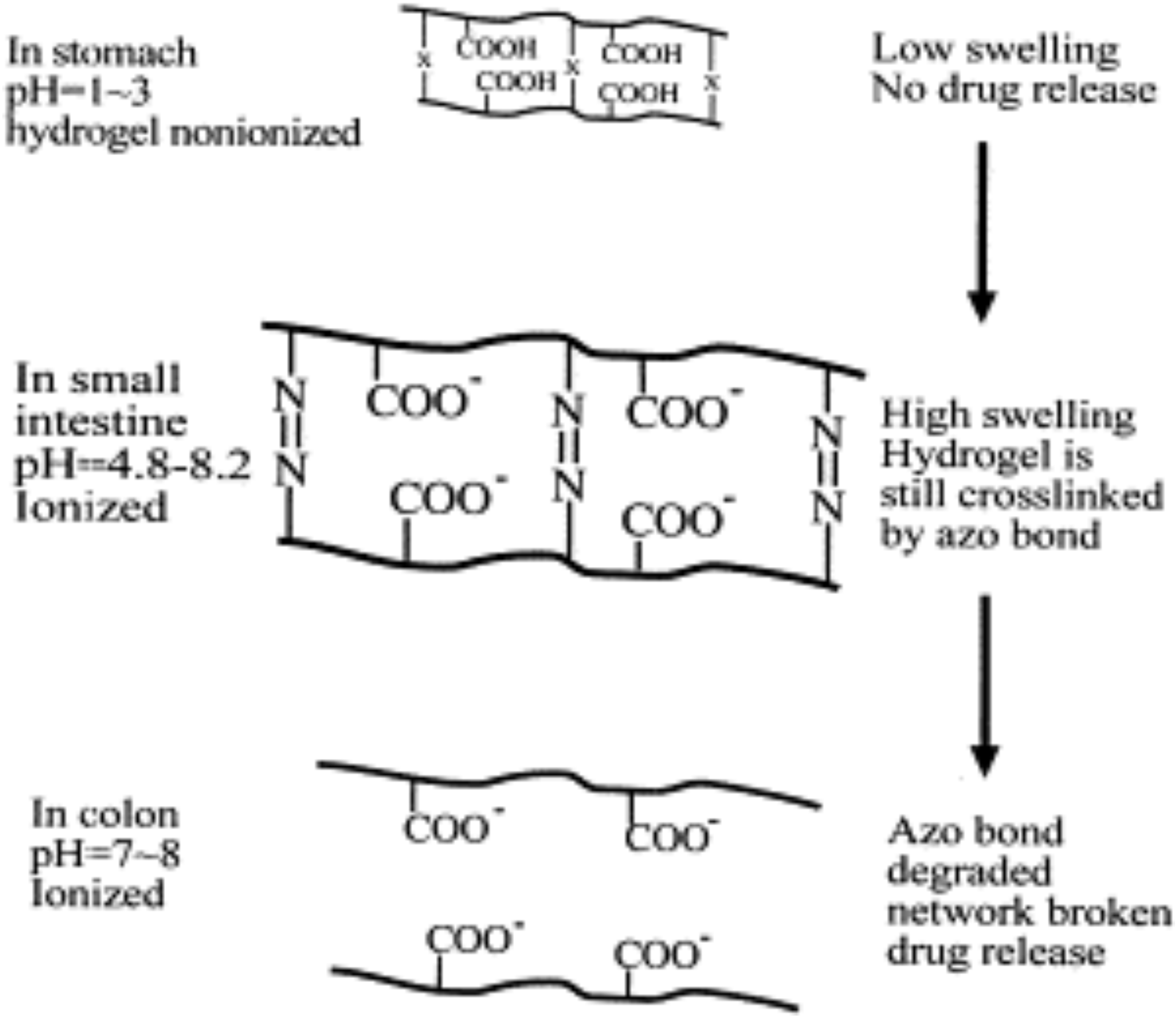
Modern enteric coatings are based on polycarboxylic acids that are hydrophobic and water-insoluble at a low pH become water soluble at a higher pH where the carboxyl groups ionize.

pH sensitive hydrogels



pH-dependent ionization of polyelectrolytes. Poly(acrylic acid) (top) and poly(*N,N*'-diethylaminoethyl methacrylate) (bottom).

pH sensitive, colon specific delivery



Azo reductase is produced by microbes present in colon

Temperature responsive hydrogels

- Most polymers increase their water-solubility as the temperature increases
- However, some polymers decrease their water-solubility as temperature increases (lower critical solution temperature (LCST))
- Hydrogels made of LCST polymers shrink as the temperature increases above the LCST
- Made of polymer chains that either possess moderately hydrophobic groups or contain a mixture of hydrophilic and hydrophobic segments
- At lower temperatures, hydrogen bonding between hydrophilic segments of the polymer chain and water molecules are dominat, leading to enhanced dissolution in water
- As the temperature increases, however, hydrophobic interactions among hydrophobic segments become strengthened, while hydrogen bonding becomes weaker. The net result is shrinking of the hydrogels due to inter-polymer chain association through hydrophobic interactions.

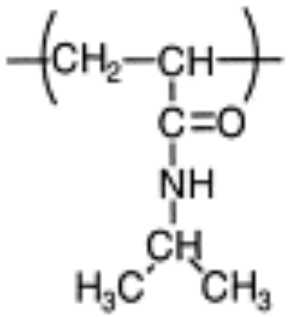
Temperature responsive hydrogels

- The LCST can be changed by adjusting the ratio of hydrophilic and hydrophobic segment of the polymer. More hydrophobic constituent lowers LCST

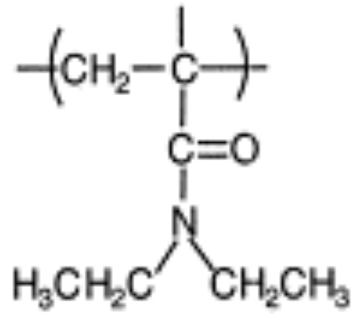
- **Poly(*N*-isopropylacrylamide) (PNIPAAm) and Poly(*N,N*-diethylacrylamide (PDEAAm) are the most extensively used because their LCST is in the range of 25-32°C**

- **Certain types of block copolymers made of poly(ethylene oxide) (PEO) and poly(propylene oxide) (PPO) also possess an inverse temperature sensitive property.**

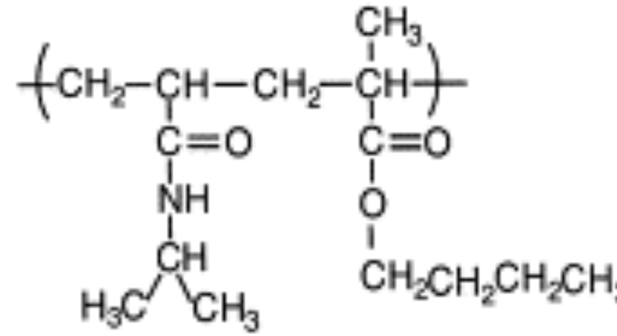
Temperature responsive hydrogels: Polymers



Poly(N-isopropylacrylamide) (PNIAAm)



Poly(N,N-diethylacrylamide) (PDEAAm)

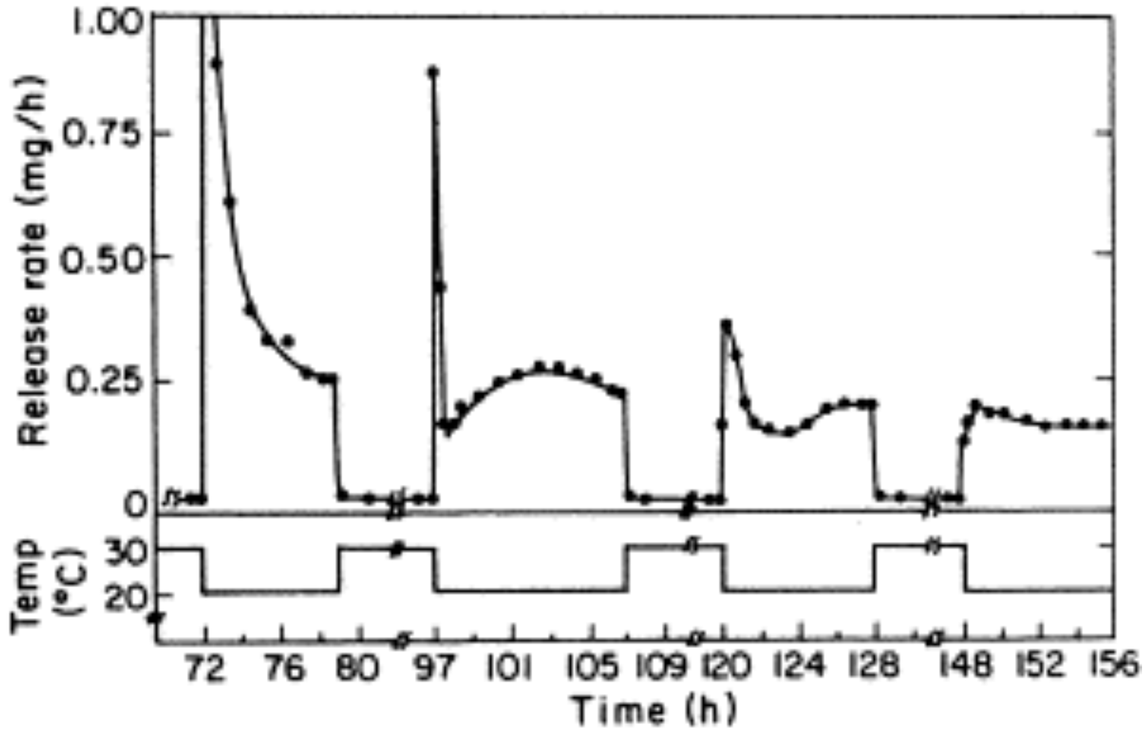


P(NIAAm-co-BMA)

Pluronic ® $HO-(CH_2CH_2O)_x-(CH_2CH(CH_3)O)_y-(CH_2CH_2O)_x-H$ PEO-PPO-PEO

Pluronic ® R $HO-(CH_2CH(CH_3)O)_x-(CH_2CH_2O)_y-(CH_2CH(CH_3)O)_x-H$ PPO-PEO-PPO

Temperature responsive hydrogels: Example



A pulsatile indomethacin release pattern was found to be regulated by temperature changes between 20°C and 30°C due to the reversible swelling properties of copolymers of *N*-isopropylacrylamide and butylmethacrylate

Temperature responsive Sol-Gel Hydrogels

- If the polymer chains in hydrogels are not covalently crosslinked, temperature-sensitive hydrogels may undergo solution-gelation phase transitions, instead of swelling-shrinking transitions

- Thermally reversible gels with inverse temperature dependence gels at higher temperatures

- A large number of PEO-PPO block copolymers are commercially available under the names of Pluronics® (or Poloxamers®) and Tetronics®.

Temperature responsive hydrogels: Example

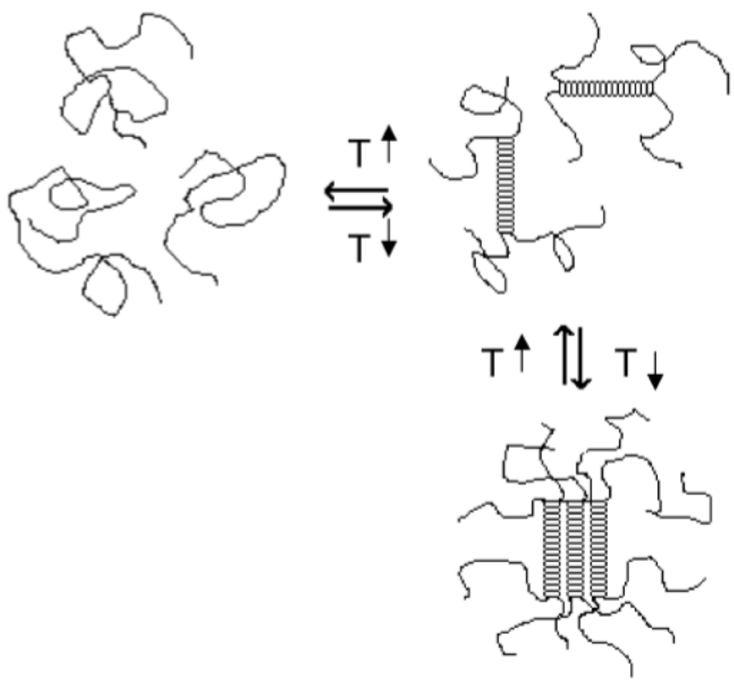
ELSEVIER
Advanced Drug Delivery Reviews 54 (2002) 37–51
Advanced DRUG DELIVERY Reviews
www.elsevier.com/locate/drugdeliv

Thermosensitive sol–gel reversible hydrogels

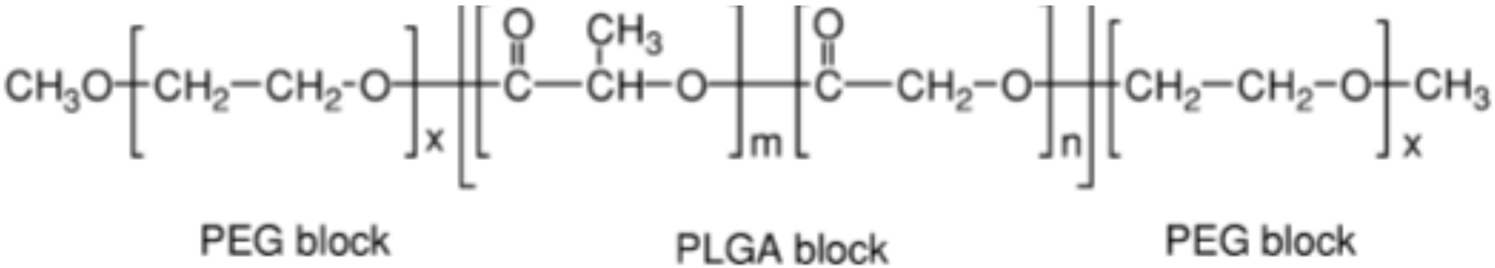
Byeongmoon Jeong[a], Sung Wan Kim[b], You Han Bae[b,*]

[a] Pacific Northwest National Laboratory (PNNL), 902 Battelle Blvd. P.O. Box 999, K2-44, Richland, WA 99352, USA
[b] Center for Controlled Chemical Delivery, Department of Pharmaceutics and Pharmaceutical Chemistry, University of Utah, Salt Lake City, UT 84112, USA

Temperature responsive natural polymer gels

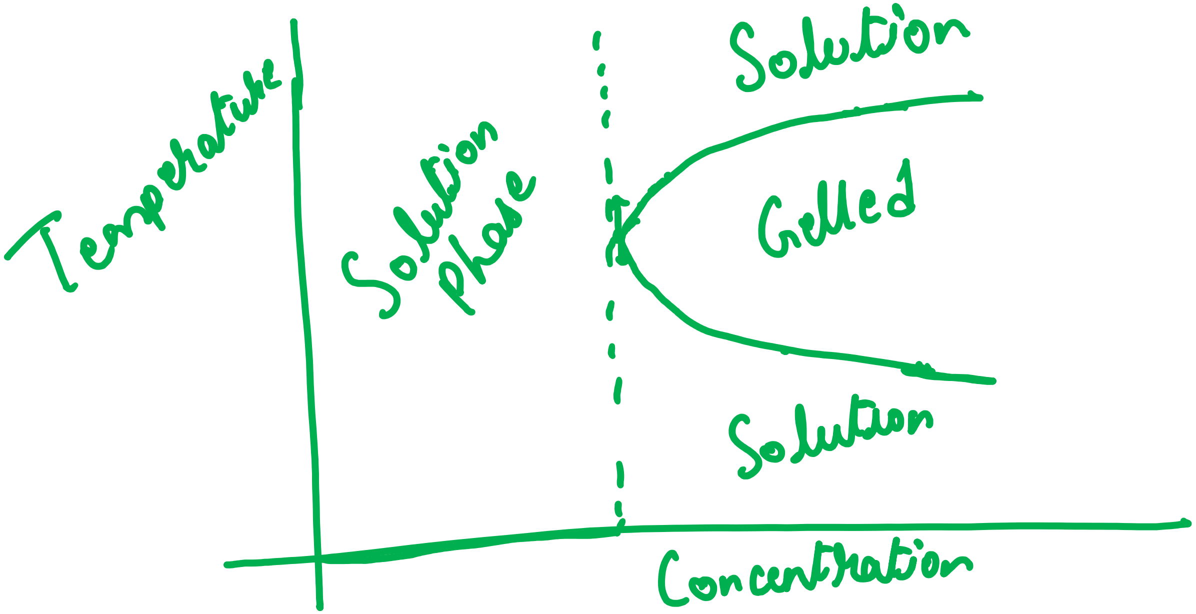


Thermo-reversible Hydrogels

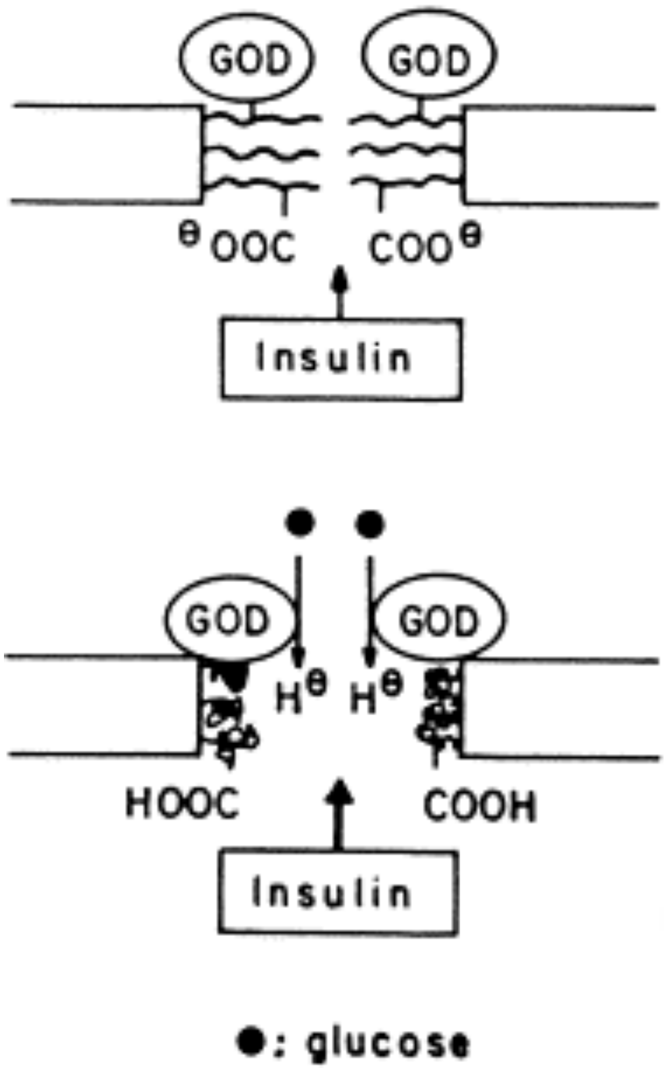


Temperature vs concentration phase diagram?

Sol – Gel
Solution – gelation



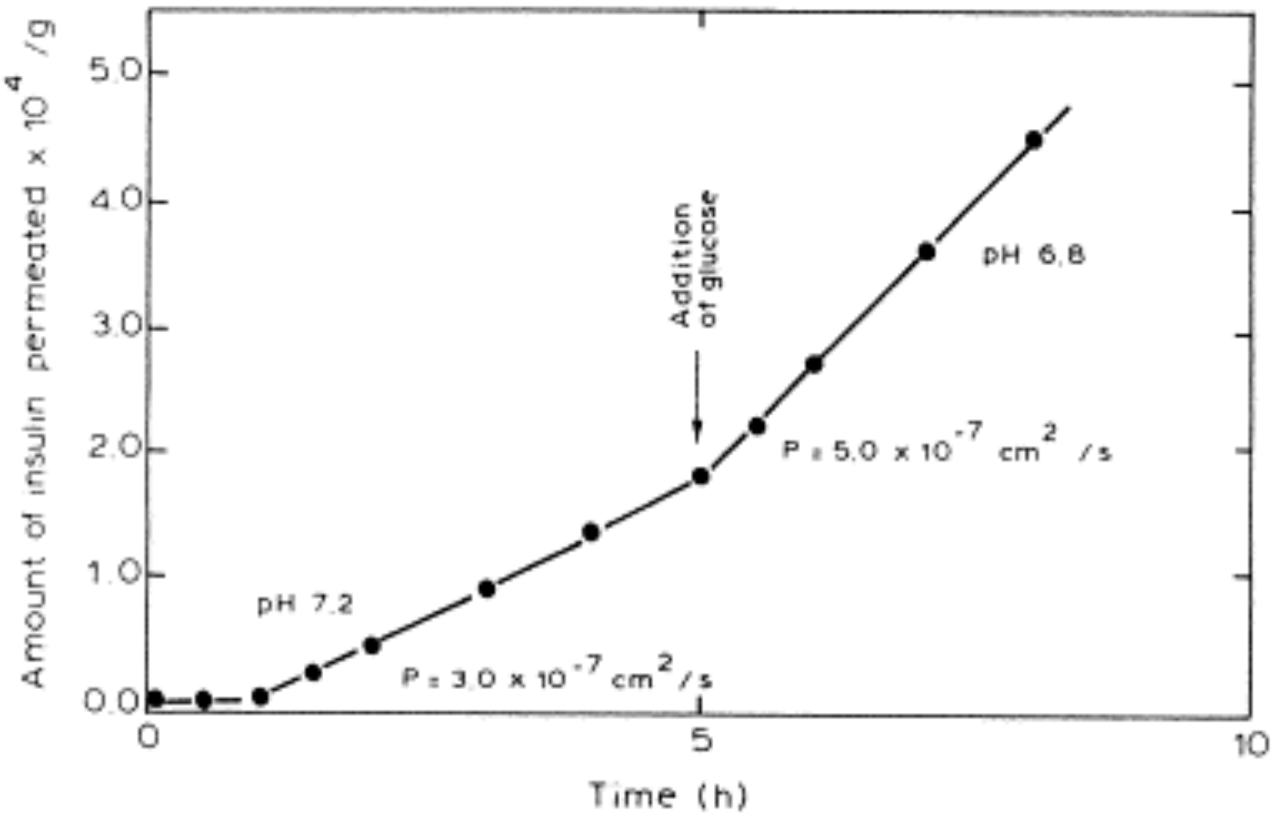
Glucose sensitive gels



Principle of controlled-release system of insulin: glucose oxidase. Above: in the absence of glucose, the chains of poly(acrylic acid) grafts are rod-like, lowering the porosity of the membrane and suppressing insulin permeation. Below: in the presence of glucose, gluconic acid produced by glucose oxidase protonates the poly(acrylic acid), making the graft chains coil-like and opening the pores to enhance insulin permeation

$$\text{Glucose} \xrightarrow{\text{Glucose oxidase}} \text{Gluconic acid} + H_2O_2$$

Glucose sensitive gels



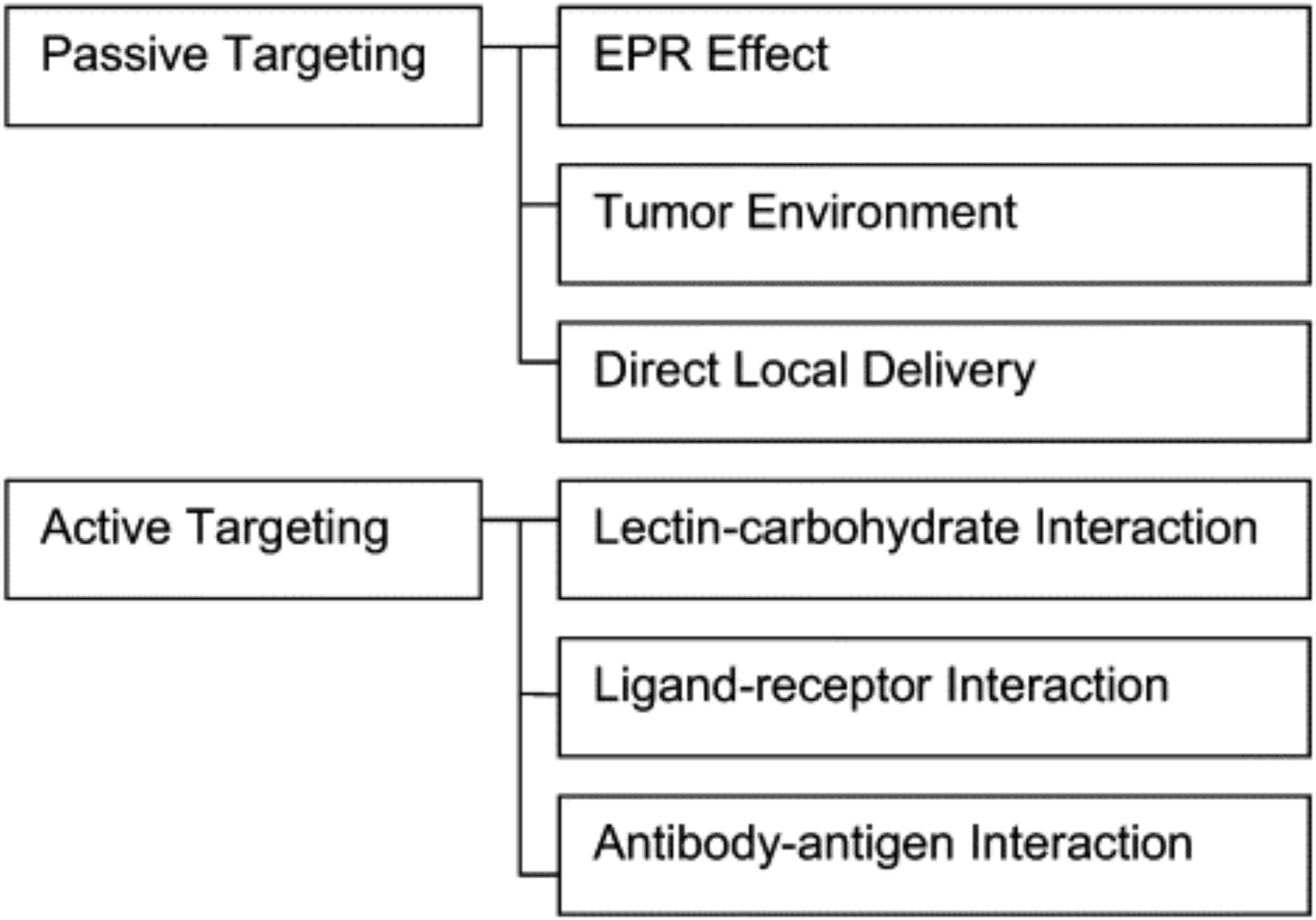
Permeation of insulin through poly(acrylic acid) grafted membrane in 0.1 M Tris-HCl-buffered solution. The concentration of added glucose is 0.2 M

Bio-inspired shielding strategies for particles and drugs

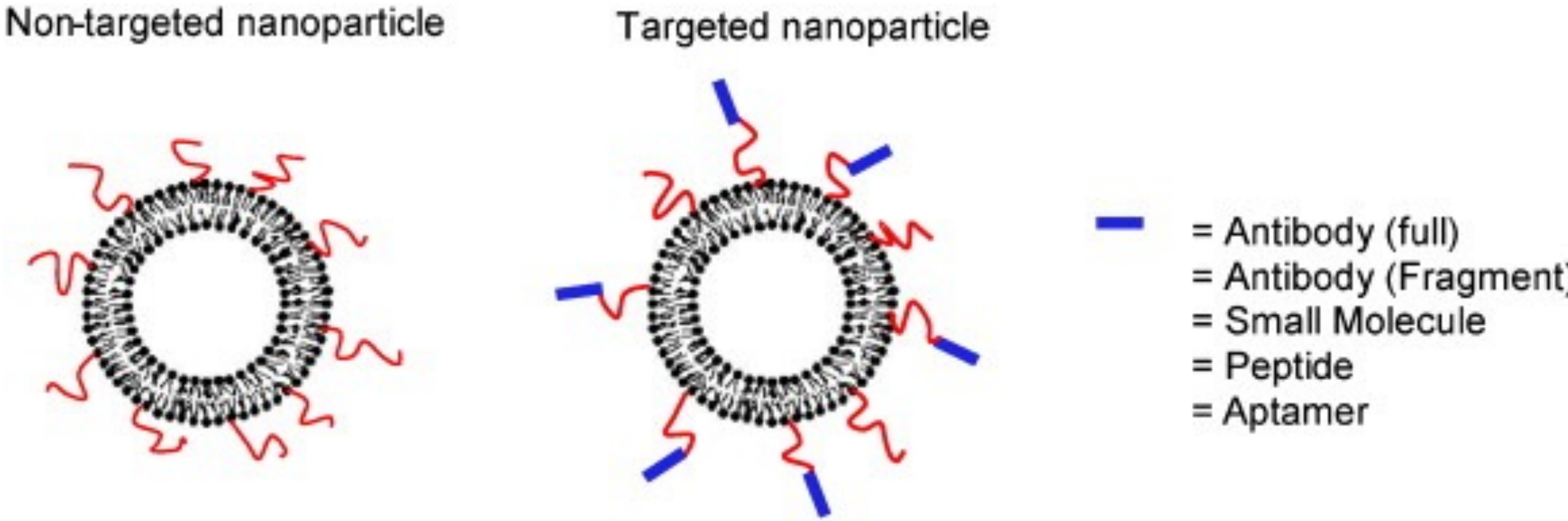
- Coating with Lipids
- Polymers
- Carbohydrates
- Proteins

- Give paper: **Bioinspired Shielding Strategies for Nanoparticle Drug Delivery Applications**

Targeted Drug Delivery System

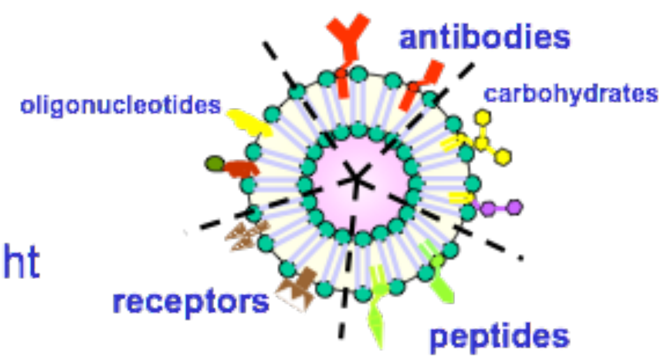


Targeted Drug Delivery System



Targeting moieties:

- Antibodies
- Proteins
- Lipoproteins
- Hormones
- Charged molecules
- Polysaccharides
- Low-molecular-weight ligands



www.certh.gr/dat/E2353434/file.pdf

Tumor targeting

- Tumors upregulate their receptors to satisfy their need for fast growth and generation of new blood vessels
- Some molecules routinely upregulated in many tumor types are
  - Transferrin
  - EGF receptor
  - Folic acid
  - VEGF-A

Cons for targeted delivery systems

- Important to note that very few targeted systems are being used in clinics (e.g. trastuzumab emtansine (antibody-drug conjugate), for HER-2-positive breast cancer)

- Still debatable whether adding ligands for systemic administration has any therapeutic benefit

- Several issues such change in pharmacokinetics, masking by protein adsorption, ligand denaturation and accessibility of target

Clinically approved drug-delivery systems

Type of drug delivery system	Clinically approved drugs
Nanoparticles	Abraxane (Paclitaxel), Doxil (Doxorubicin), DaunoXome (Daunorubicin), Marqibo (Vincristine), MEPACT (Mifamurtide), Onivyde MM-398 (Irinotecan), ADYNOVATE (antihemophilic factor (recombinant) PEGylated), Estrasorb (estradiol), AmBisome (amphotericin B), Depocyte (cytarabine), Visudyne (Verteporfin)
Microparticle-based depots	Zmax (Azithromycin), Decapeptyl/Trelstar (Triptorelin), Vivitrol (Naltrexone), Arestin (Minocycline), Risperdal/Consta (Risperidone), Sandostatin LAR Depot (Octreotide), Nutropin Depot (Somatropin), Lupron Depot (Leuprolide), DepoCyt (Cytarabine), DepoDur (Morphine), Bydureon (Exenatide), Somatuline LA (Lanreotide), Zoladex (Goselerin), Suprefact Depot (Buselerin), Signifor (Pasireotide)
Transdermal materials and devices	Transderm-Scop (Scopolamine), Nitro-Dur (Nitroglycerin), Catapres-TTS (Clonidine), Estraderm (Estradiol), Duragesic (Fentanyl), Androderm (Testosterone), Combipatch (Estradiol with norethindrone), Lidoderm (Lidocaine), Climara Pro (Estradiol with levonorgestrel), Oxytrol (Oxybutynin), Synera (Lidocaine and tetracaine), Daytrana (Methylphenidate), Emsam (Selegiline), Neupro (Rotigotine), Exelon (Rivastigmine), Sancuso (Granisetron), Butrans (Buprenorphine), Ortho Evra (Estradiol and norelgestromin), Qutenza (Capsaicin), Flector (Diclofenac epolamine), NicoDerm/Habitrol/ProStep (Nicotine), Retin-A (Tretinoin), IONSYS (Fentanyl), SonoPrep (Lidocaine via ultrasound), Iontocaine (Lidocaine and epinephrine via iontophoresis), LidoSite (Lidocaine and epinephrine via iontophoresis)
Oral	Concerta (Methylphenidate), Ditropan XL (Oxybutynin), Teczem (Enalapril Diltiazem), Dilacor XR (Diltiazem), Covera-HS (Verapamil), DynaCirc CR (Isradipine), Minipress XL (Prazosin), Procardia XL (Nifedipine), Fortamet (Metformin), Altoprev (Lovastatin), Glucotrol XL (Glipizide), Invega (Paliperidone), Tegretol-XL (Carbamazepine), Allegra D (Pseudoephedrine and Fexofenadine), Efidac/24 (Pseudoephedrine and Brompheniramine or Chlorphenir-amine), Volmax (Albuterol), Orenitram (Treprostinil), Sudafed 24 h (Pseudoephedrine), Exalgo (Hydromorphone), Vesanoid (Tretinoin), Syndros (Dronabinol), Venclexta (venetoclax), Farydak (panobinostat), Renagel (Sevelamer)
Pulmonary	Tudorza/Pressair (Aclidinium), Proventil HFA (Albuterol), Ventolin HFA (Albuterol), ProAir HFA (Albuterol), Combivent Respimat (Albuterol and ipratropium), DuoNeb (Albuterol and ipratropium), Brovana (Arformoterol), QVAR (Beclomethasone), Pulmicort Flexhaler (Budesonide), Symbicort (Budesonide and Formoterol), Alvesco (Ciclesonide), Breo/Ellipta (Fluticasone and vilanterol), Flovent HFA (Fluticasone), Flovent/Diskus (Fluticasone), Foradil/Aerolizer (Formoterol), Perforomist (Formoterol), Arcapta Neohaler (Indacaterol), Atrovent HFA (Ipratropium), Xopenex HFA (Levalbuterol), Asmanex/Twisthaler (Mometasone), Dulera (Mometasone and Formoterol), Serevent/Diskus (Salmeterol), ADVAIR Diskus (Salmeterol Fluticasone), ADVAIR HFA (Salmeterol Fluticasone), Spiriva/Handihaler (Tiotropium), Cayston (Aztreonam), Ventavis (Iloprost), Tyvaso (Treprostinil), TOBI Podhaler (Tobramycin), Afrezza (human insulin)
Implants	Vitrasert (Ganciclovir), Retisert (Fluocinolone), Ozurdex (Dexamethasone), Zoladex (Goserelin), Gliadel (Prolifeprosan and Carmustine), Vantas/Supprelin LA (Histrelin), Viadur (Leuprolide), Nexplanon (Etonogestrel), NuvaRing (Etonogestrel and ethinyl estradiol), Mirena/Norplant (Levonorgestrel), Paragard (Copper)

Adv. Mater. **2018**, 1705328

Steps towards approval

- **STEP ONE** in the marketing process is to make absolutely sure that the product that you wish to market is feasible

- **STEP TWO** is to determine how regulatory agencies (ICMR) may classify your device - which one of the classes the device may fall into. Classification identifies the level of regulatory control that is necessary to assure the safety and effectiveness of a medical device.

- **STEP THREE** is the development of data and/or information necessary to submit a marketing application, and to obtain clearance to market

GMP and QS Regulations

- Good Manufacturing Practice (GMP) and QS (Quality Systems) Regulations
- These regulations require that domestic or foreign manufacturers have a quality system for the design, manufacture, packaging, labeling, storage, installation, and servicing of finished medical devices intended for clinical trials and commercial distribution
- Anything used in humans should follow these guidelines

Investigation Phases in Clinical Trials

- Phase 1 Feasibility Study
  - Small numbers to assess product safety
  - Generally localized to one or two centers
- Phase 2 Study
  - Proper/safe dosing; potential efficacy
- Phase 3 Pivotal Study
  - Well controlled clinical trail supports safety/efficacy; leads to FDA application for premarket approval
  - Multi-center involvement
  - 90% of Phase III trials fail primarily due to lack of efficacy or due to unforeseen toxicity in specific human populations.

Overall process for approval

- New drugs need a full and comprehensive evaluation pathway (requires approval at each step)
  - Small animal trials (Animal Ethics committee)
  - Large animal trials (Large Animal/Primate Ethics committee)
  - Extensive toxicity screening
  - Human and follow-up clinical trials
- Repurposed drugs and drug delivery vehicles packed with pre-approved drugs and/or pre-evaluated excipients (e.g. PLGA and variants, phospholipids etc) have a much easier pathway

Interested Researchers: Drug Delivery Lab

- rachit@iisc.ac.in
- Bioengineering, TSH Building
- Positions:
  - Undergraduate semester projects
  - M. Tech projects
  - Ph.D.